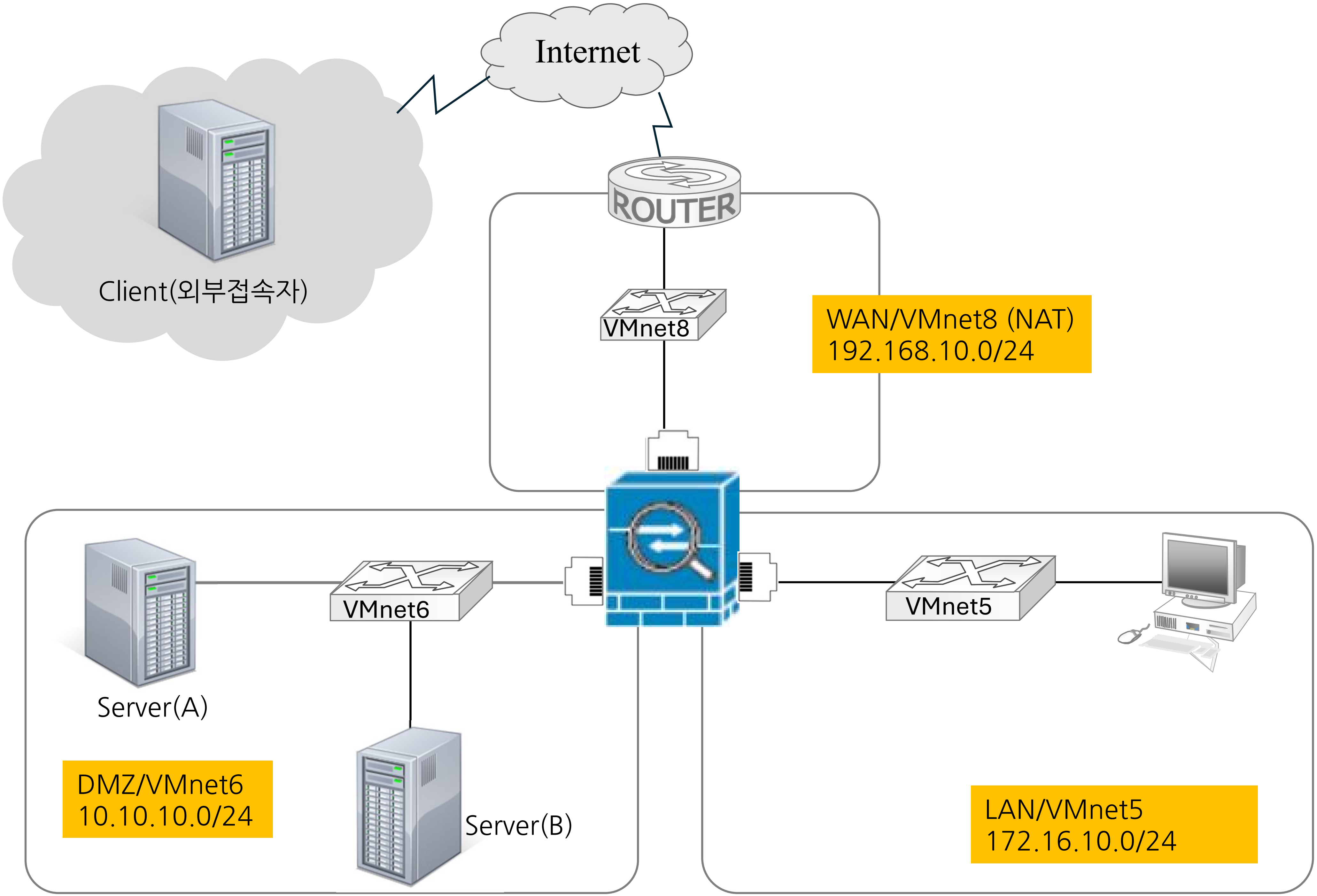
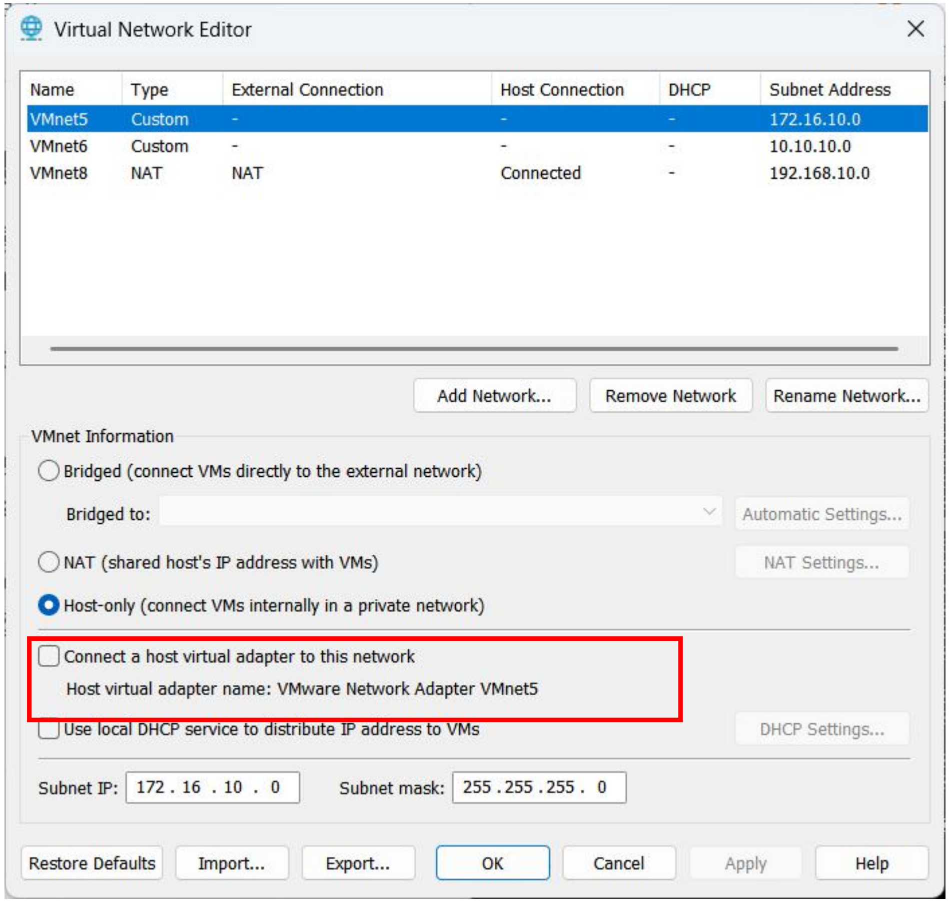
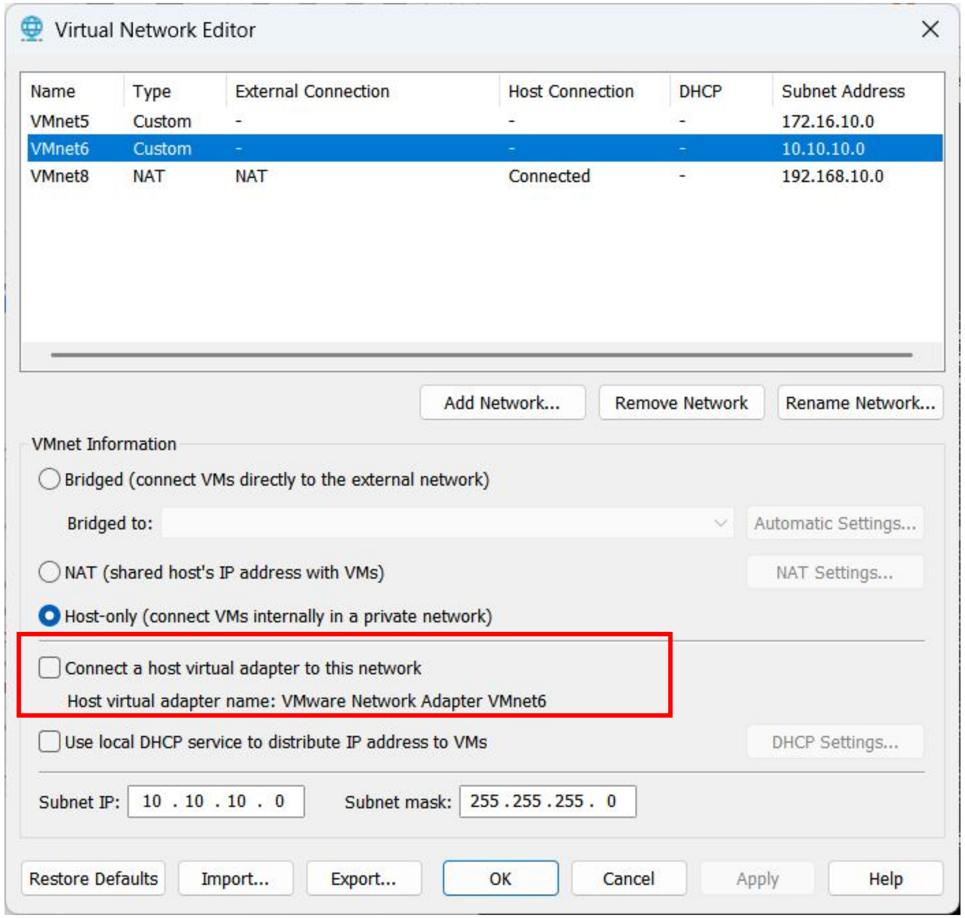
pfSense UTM 설치& 운영

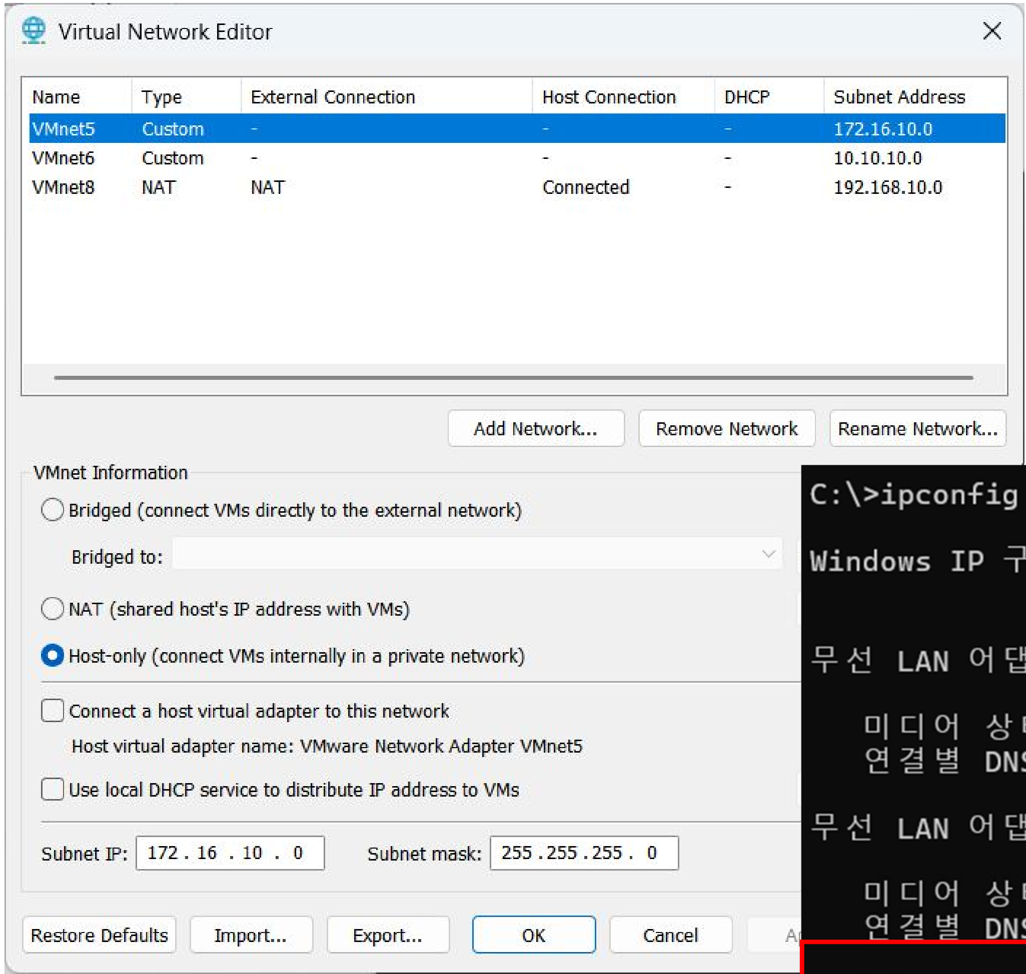

Internet
Client(외부접속자)
ROUTER
VMnet8
WAN/VMnet8 (NAT)
192.168.10.0/24
VMnet6
VMnet5
Server(A)
DMZ/VMnet6
10.10.10.0/24
Server(B)
LAN/VMnet5
172.16.10.0/24

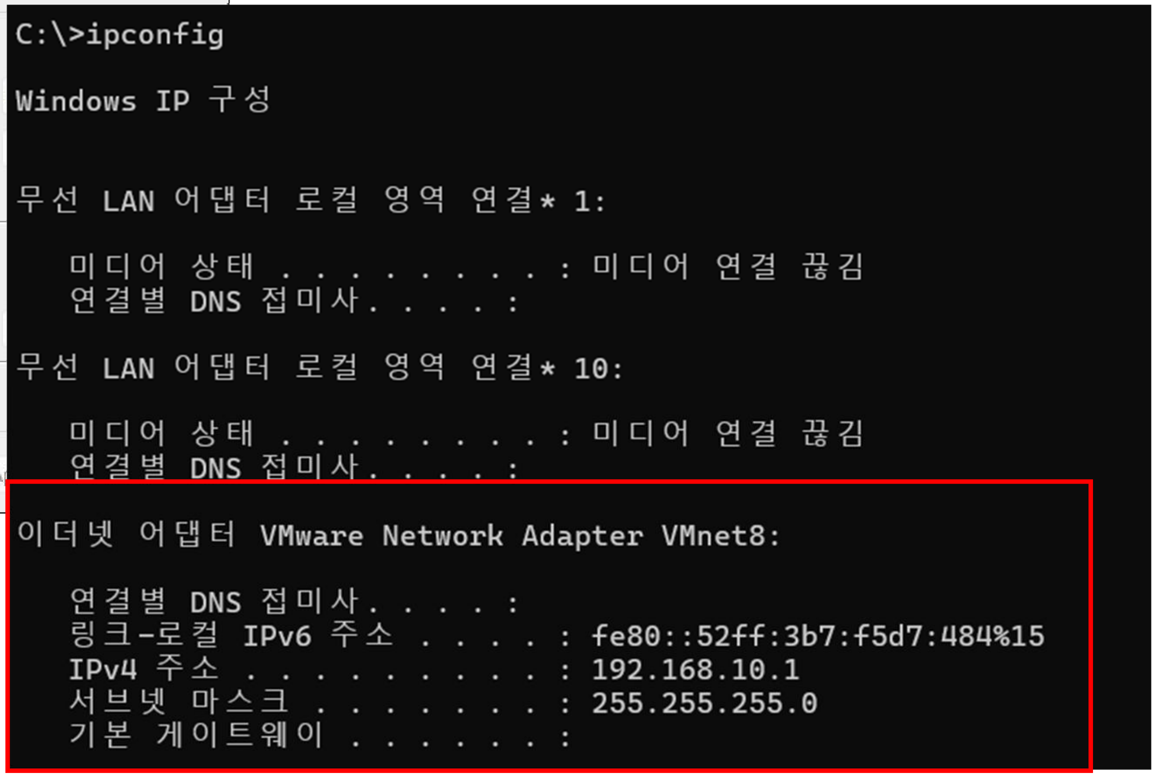
Edit > Virtual Network Editor

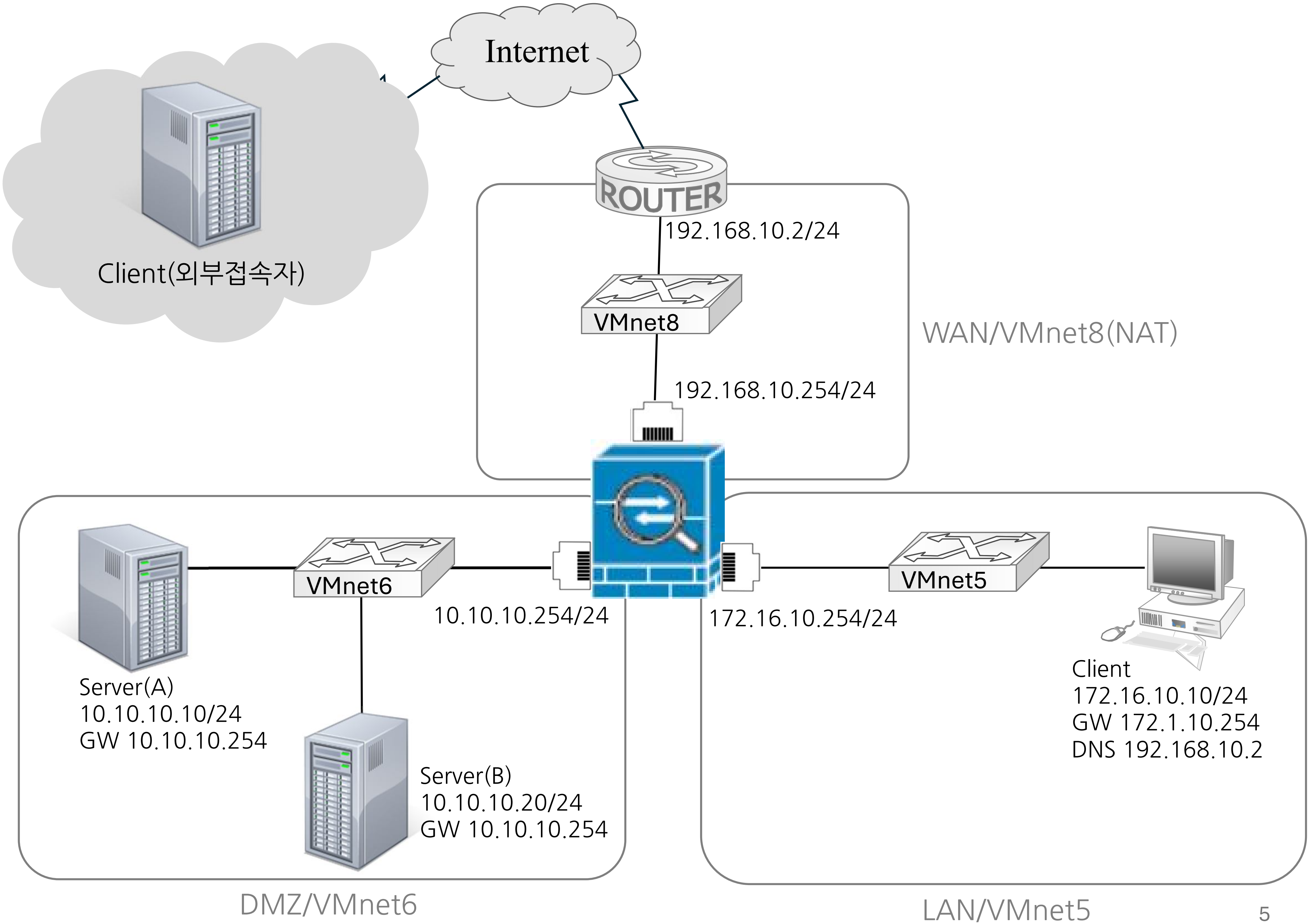




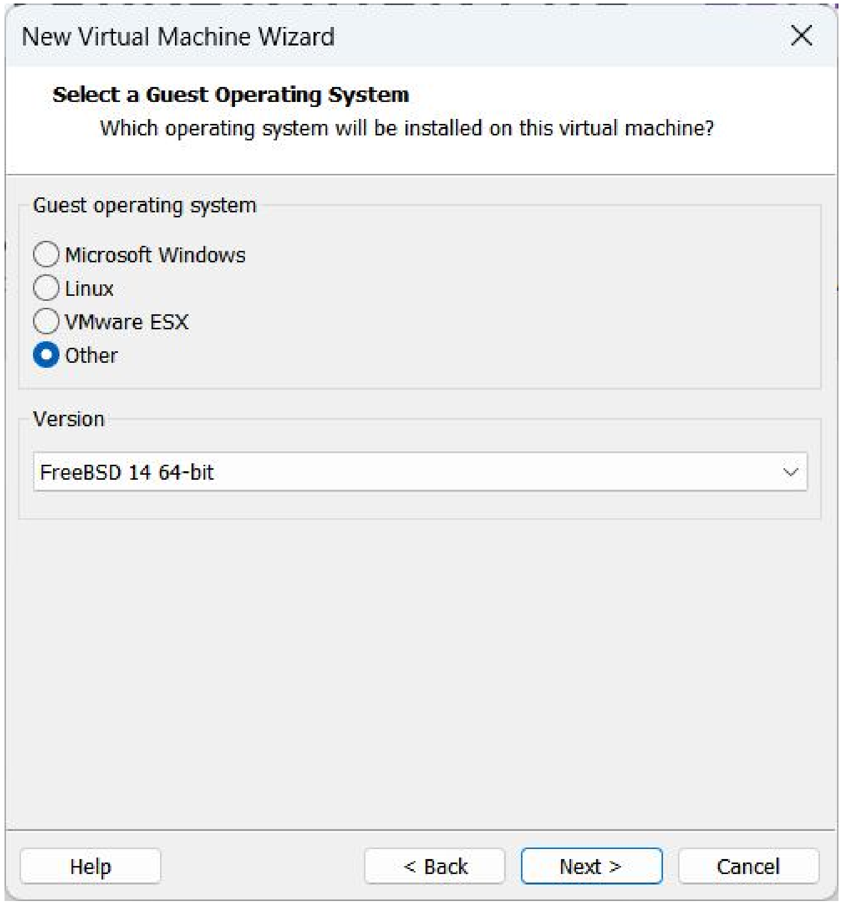
Edit > Virtual Network Editor

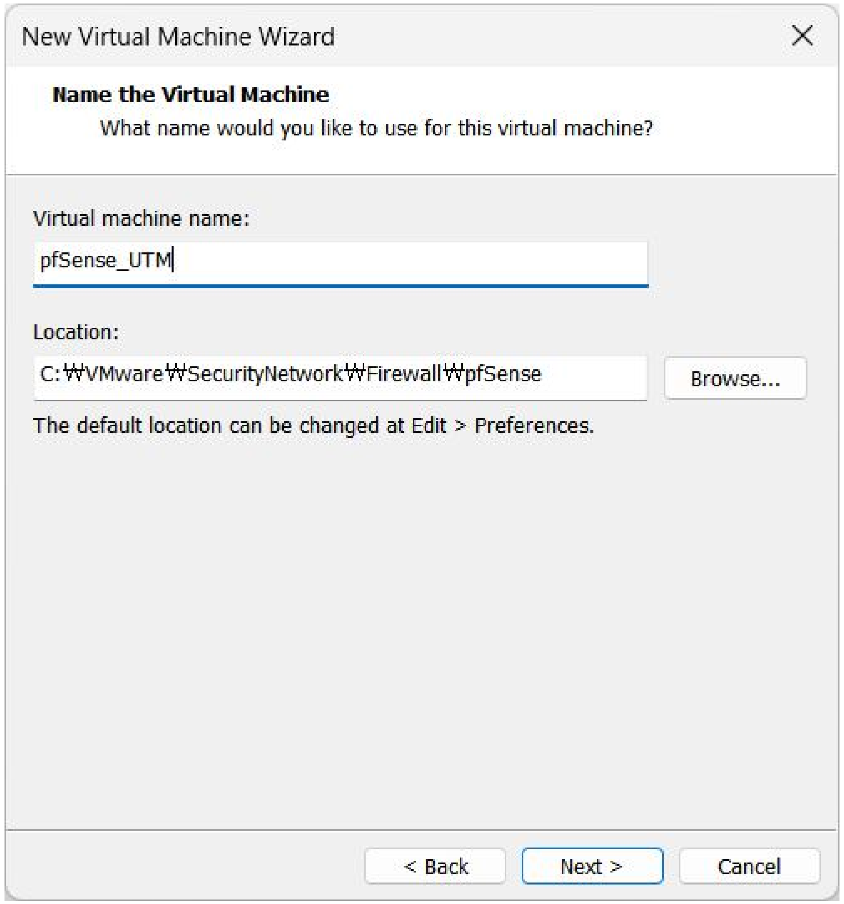





Internet
Client(외부접속자)
ROUTER
192.168.10.2/24
VMnet8
WAN/VMnet8(NAT)
192.168.10.254/24
VMnet6
10.10.10.254/24
VMnet5
172.16.10.254/24
Server(A)
10.10.10.10/24
GW 10.10.10.254
Server(B)
10.10.10.20/24
GW 10.10.10.254
Client
172.16.10.10/24
GW 172.1.10.254
DNS 192.168.10.2
DMZ/VMnet6
LAN/VMnet5

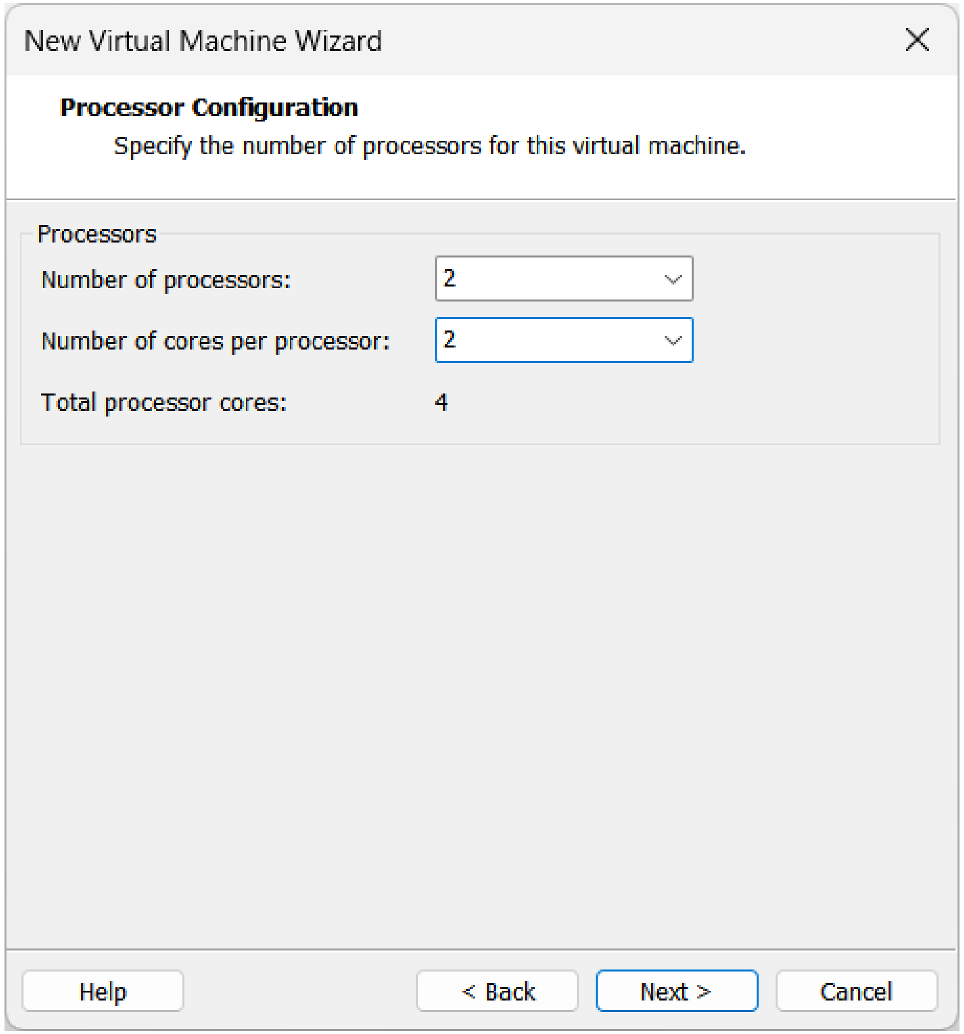
1) Virtual machine 생성

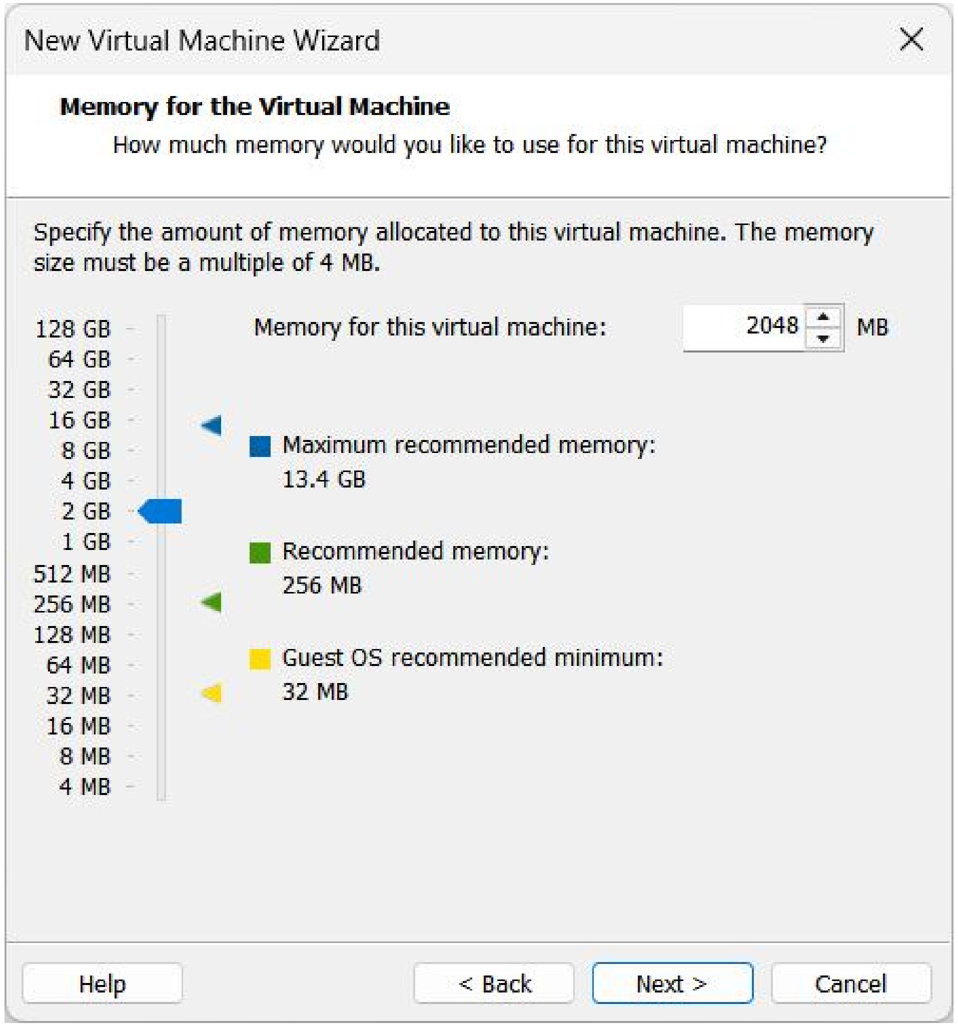




① Version FreeBSD 14 64-bit

Virtual machine 생성

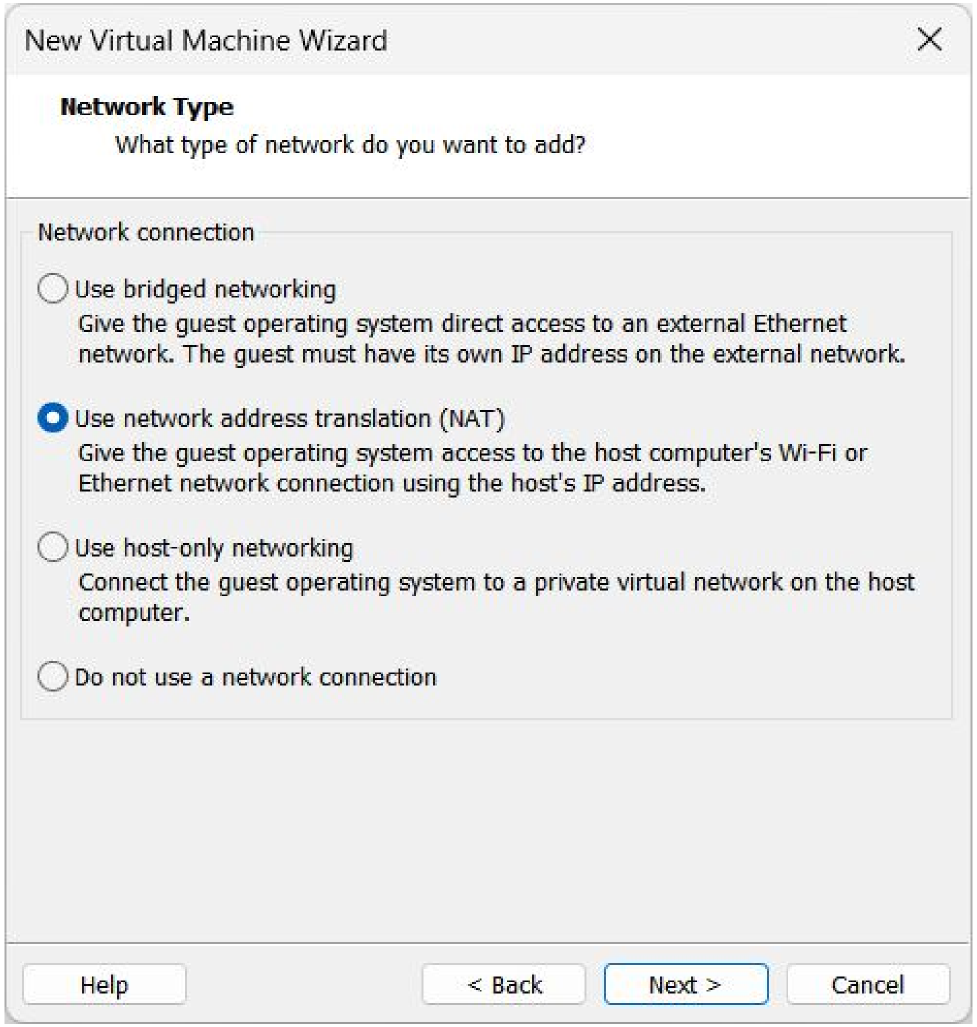




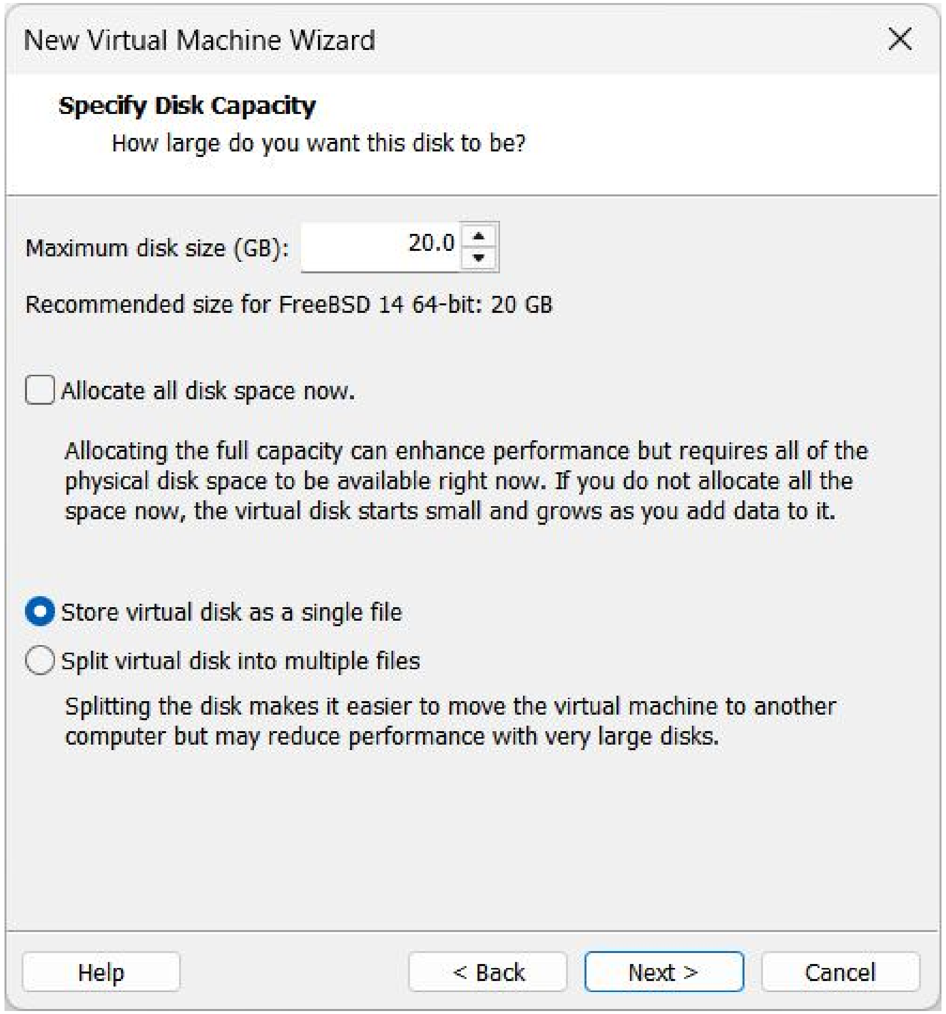
② Processor cores : 4

③ RAM : 2GB

Virtual machine 생성

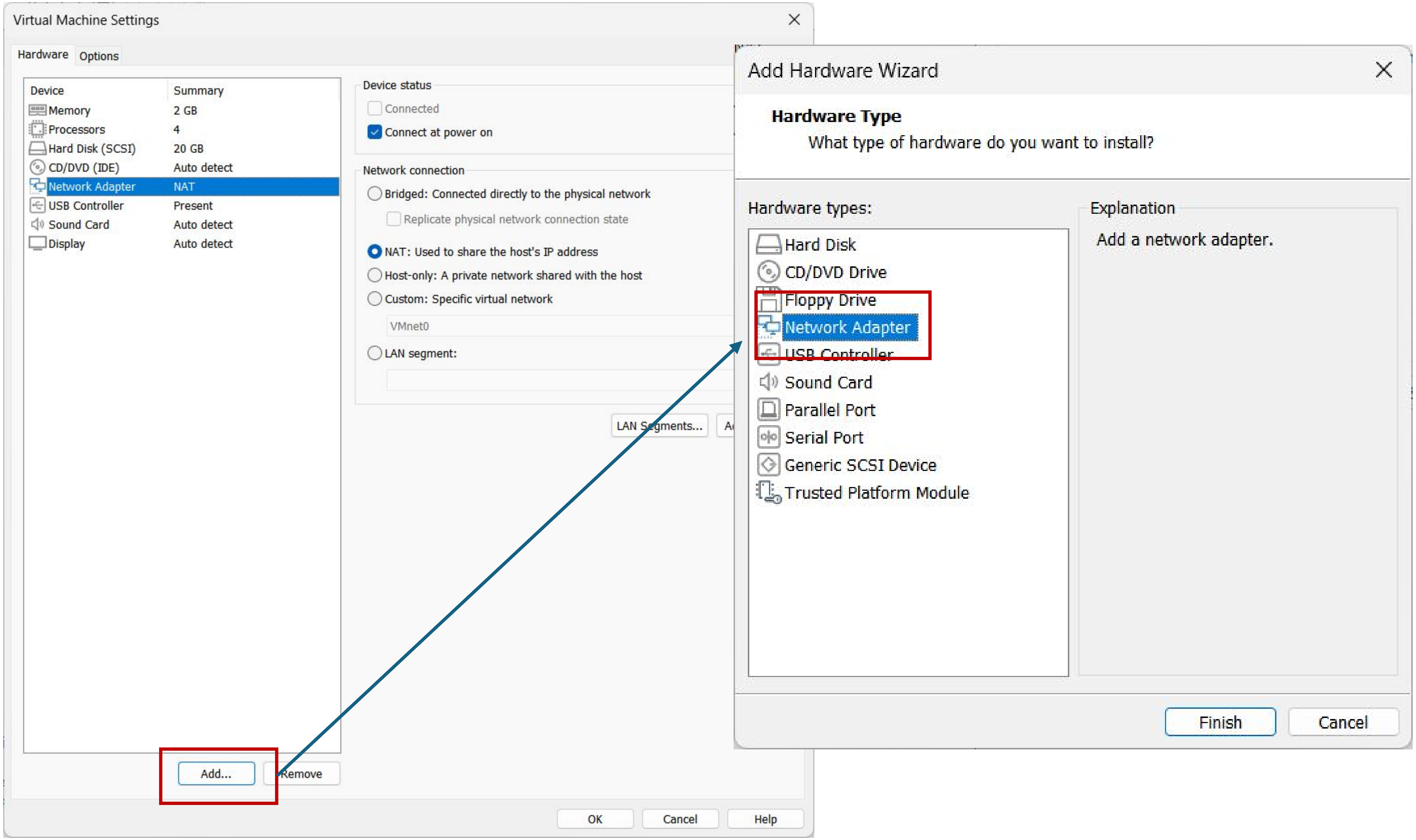


④ Network Type : NAT

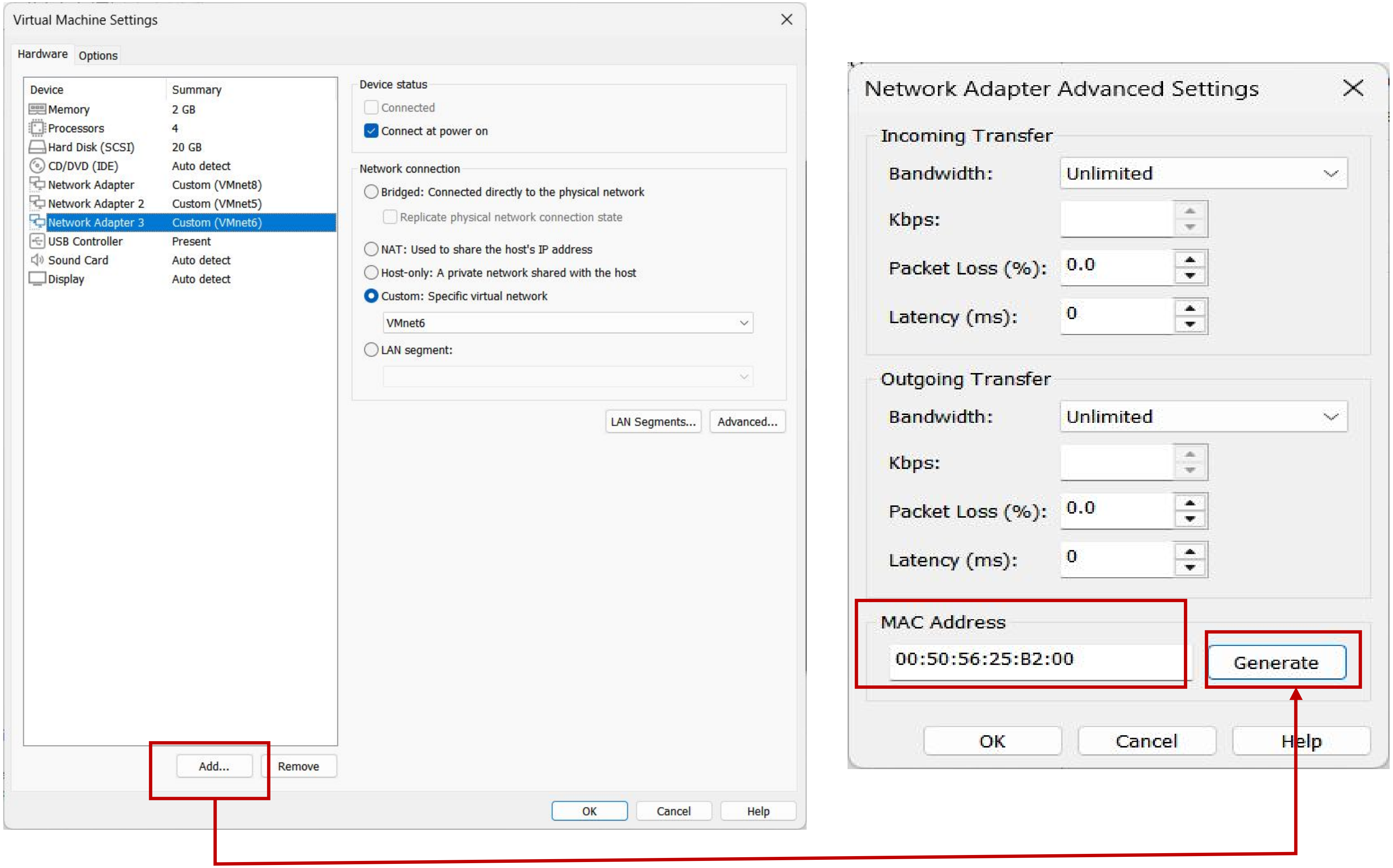


⑤ HardDisk : 20GB

LAN Card 추가 후 MAC 주소 생성과 확인

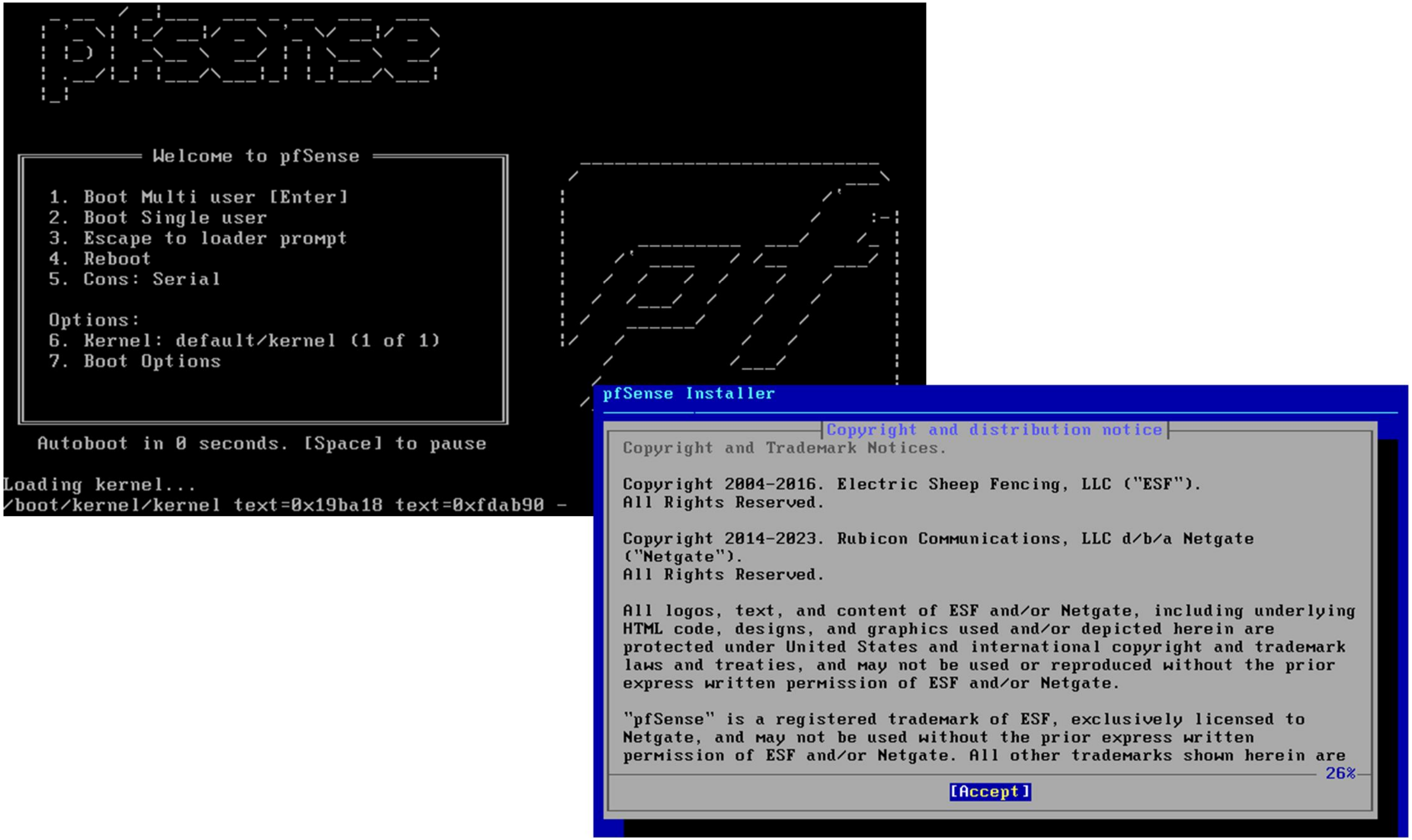


LAN Card 추가 후 MAC 주소 생성과 확인

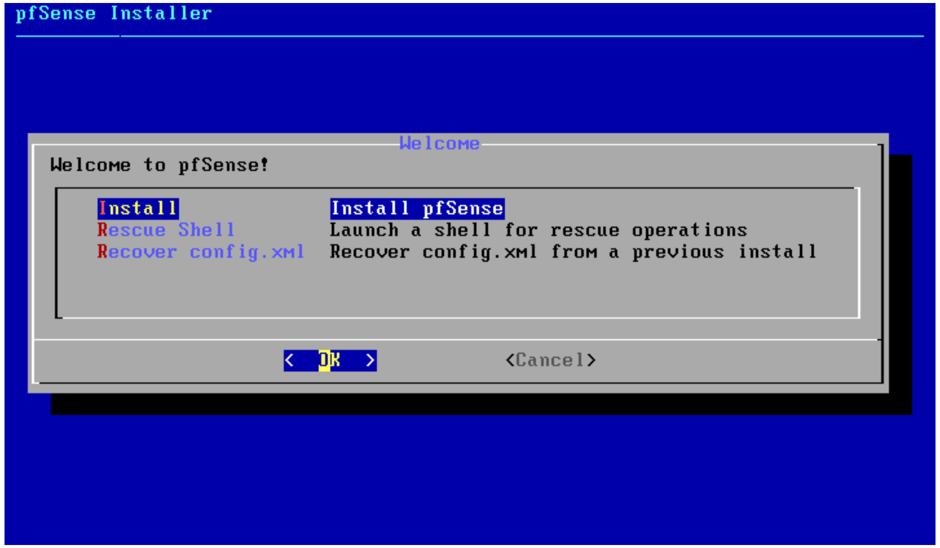


⑥LAN card 추가 후 생성된 LAN card의 MAC 주소 기록

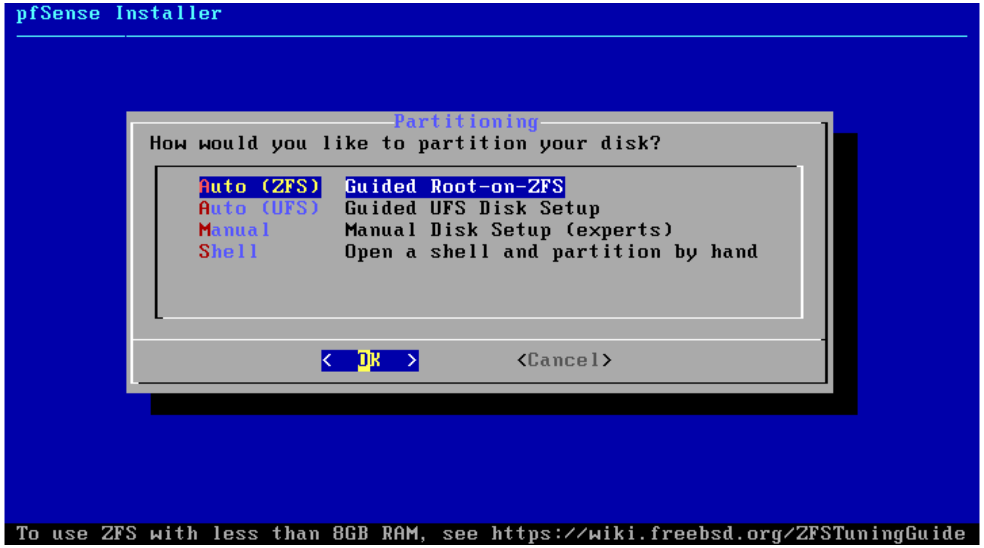
2) pfSense IOS 설치



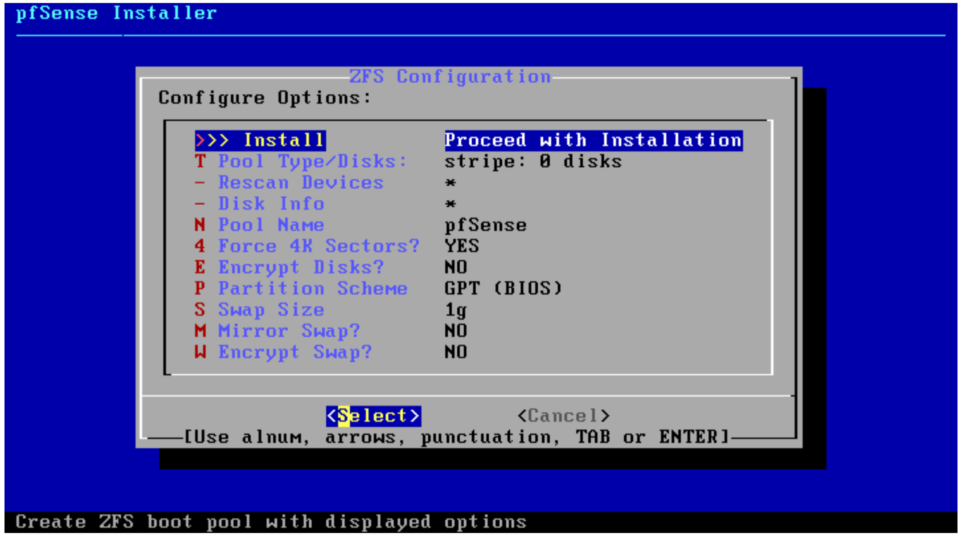
① Accept



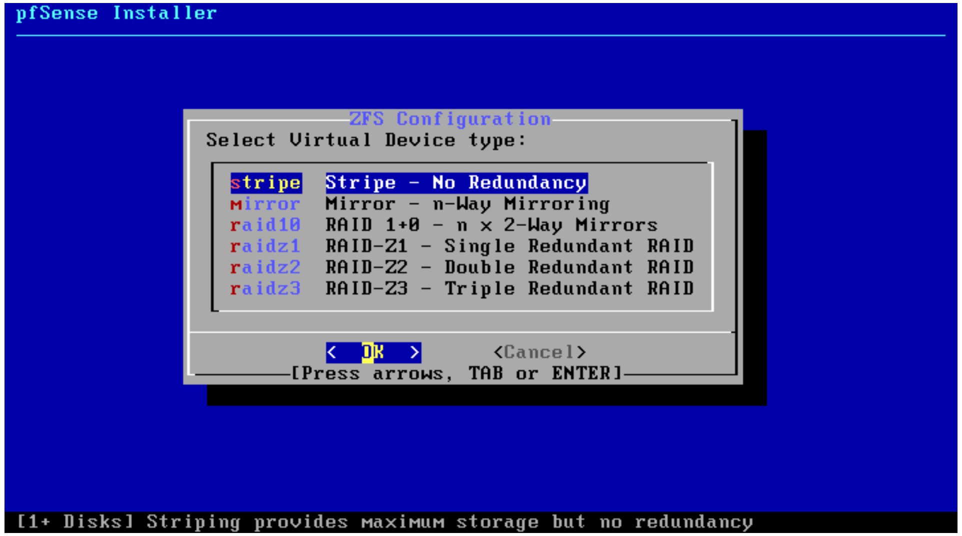
② Install pfsense



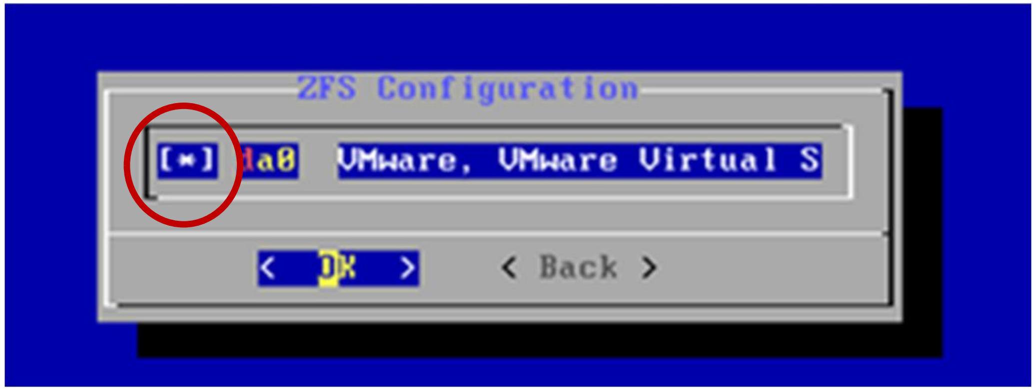
③ Auto(ZFS)



④ Install

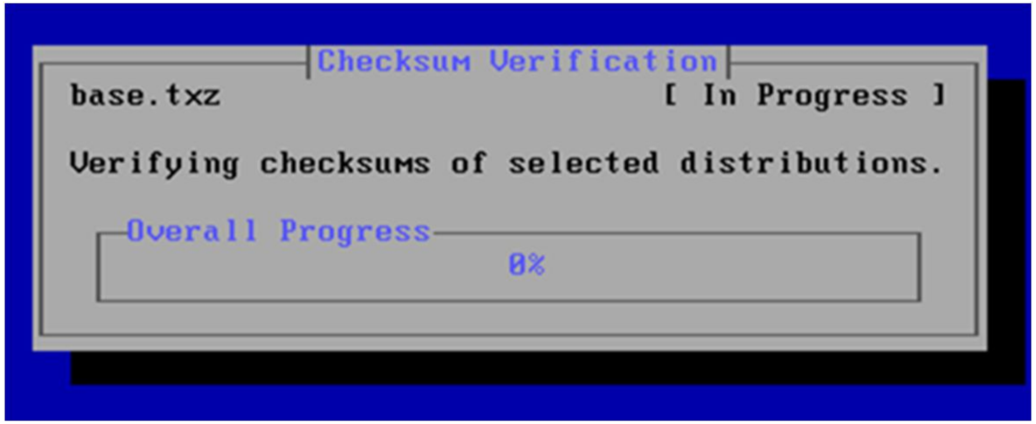


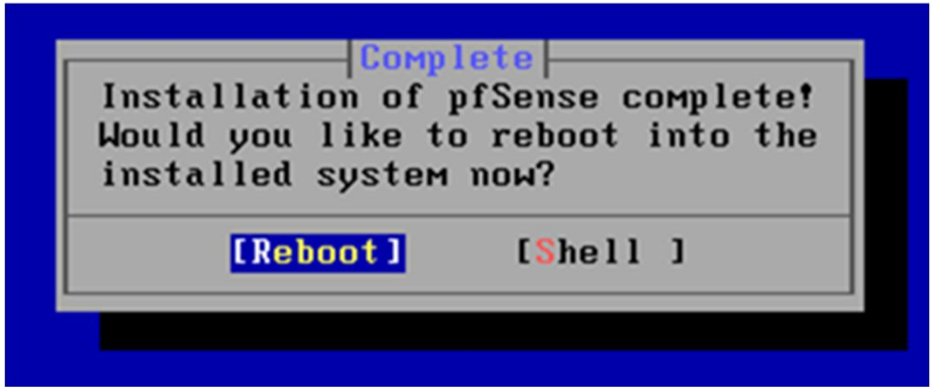
⑤ stripe



⑥ * 체크 후 OK

⑦ Yes





⑧ Reboot

3) pfSense 네트워크 인터페이스 확인

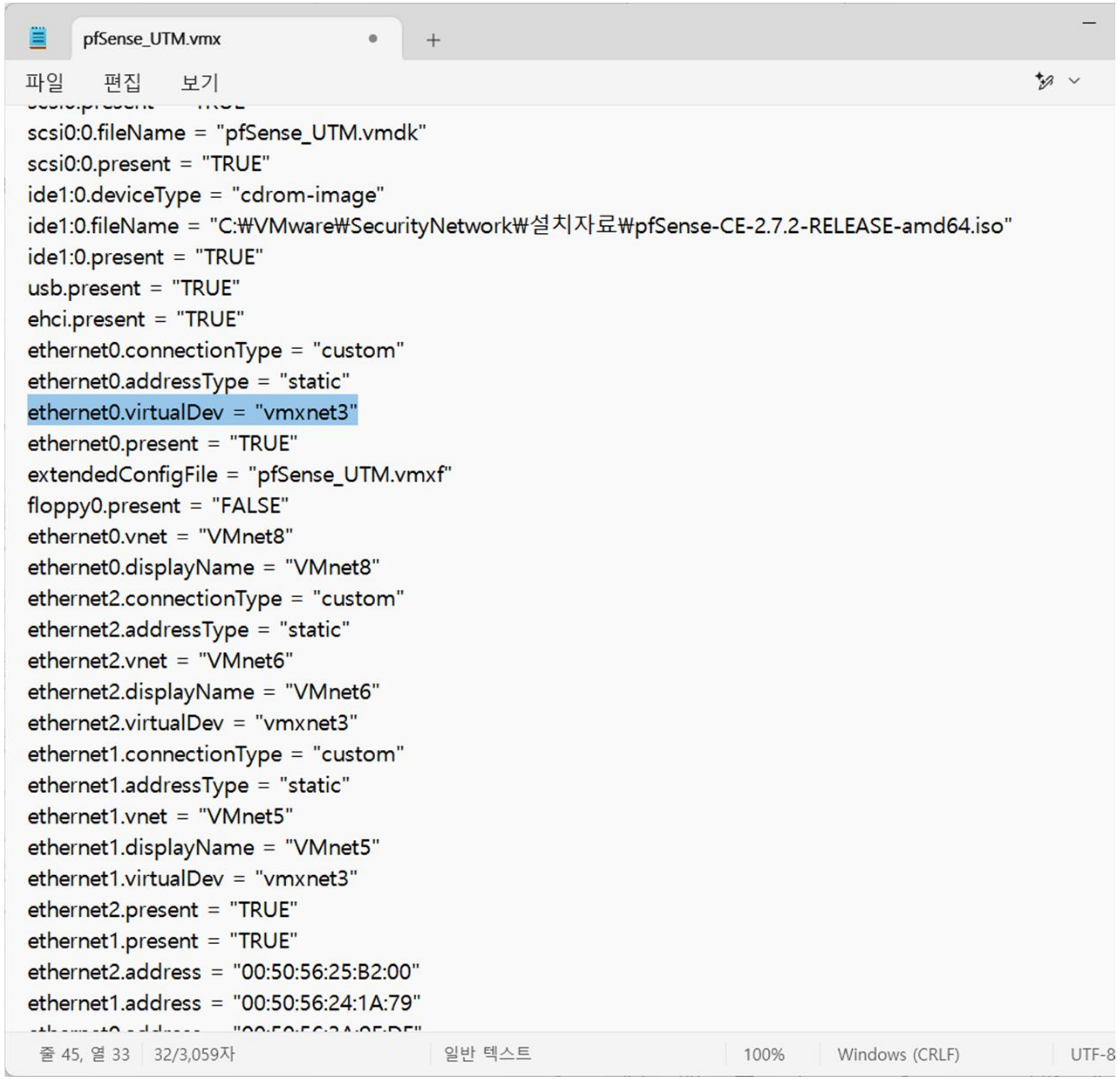
```
done.
Starting CRON... done.
pfSense 2.7.2-RELEASE amd64 20231206-2010
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: abfef3ef2b25954e3af8

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

 WAN (wan)       -> em0        ->
 LAN (lan)       -> em1        -> v4: 192.168.1.1/24

 0) Logout (SSH only)                  9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart webConfigurator
 3) Reset webConfigurator password    12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option: 1
```

```
Valid interfaces are:

em0     00:50:56:3a:9f:de   (up) Intel(R) Legacy PRO/1000 MT 82545EM (Copper)
em1     00:50:56:24:1a:79   (up) Intel(R) Legacy PRO/1000 MT 82545EM (Copper)
em2     00:50:56:25:b2:00 (down) Intel(R) Legacy PRO/1000 MT 82545EM (Copper)

Do VLANs need to be set up first?
If VLANs will not be used, or only for optional interfaces, it is typical to
say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now [y|n]?
```

기록해 놓은 랜카드 MAC주소와 인터페이스 이름 확인

4) pfSense 칩셉 변경 (e1000 ➔ vmxnet3)

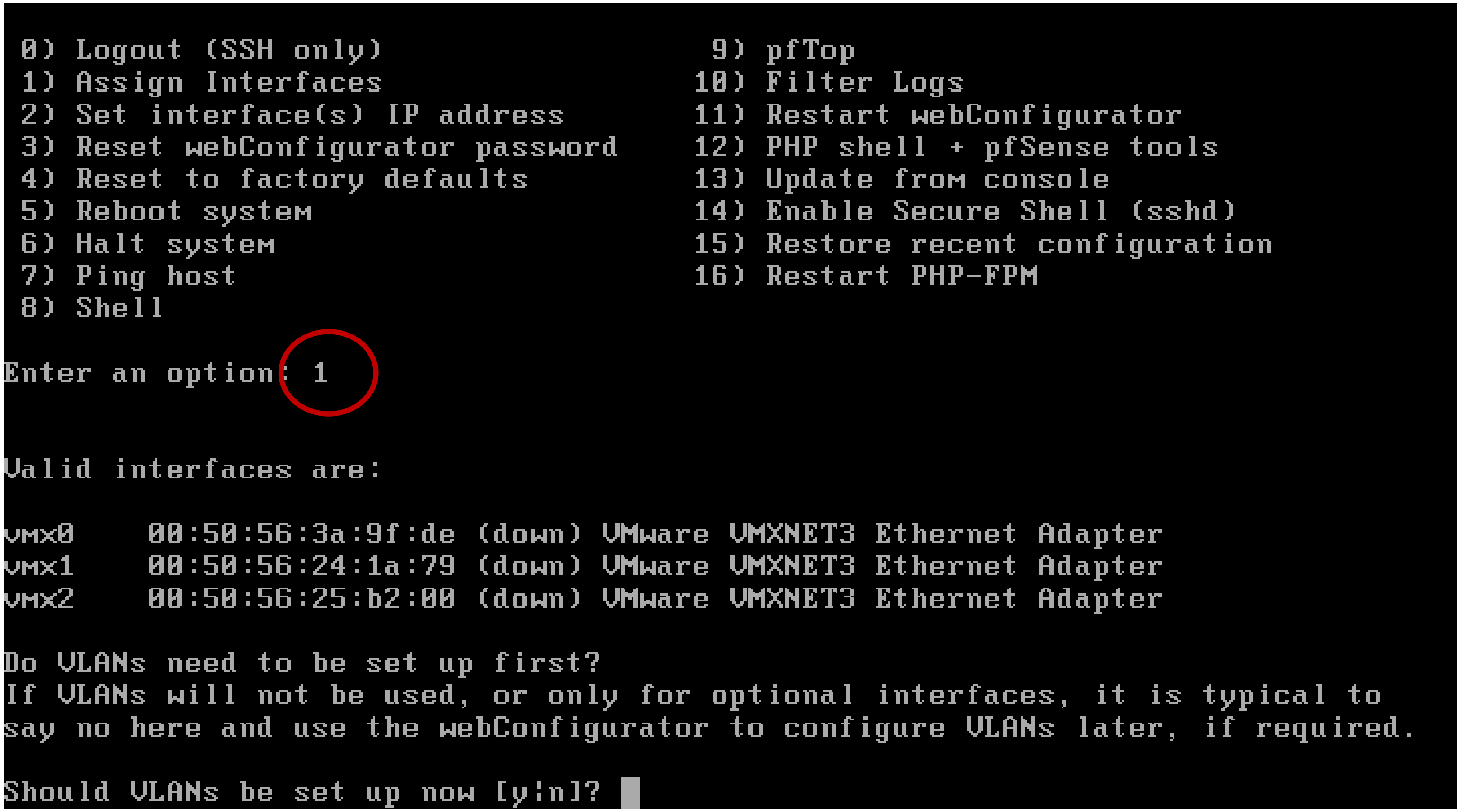
```
pfSense_UTM.vmx
파일 편집 보기
scsi0:0.fileName = "pfSense_UTM.vmdk"
scsi0:0.present = "TRUE"
ide1:0.deviceType = "cdrom-image"
ide1:0.fileName = "C:\VMware\SecurityNetwork\설치자료\pfSe
ide1:0.present = "TRUE"
usb.present = "TRUE"
ehci.present = "TRUE"
ethernet0.connectionType = "custom"
ethernet0.addressType = "static"
ethernet0.virtualDev = "e1000"
ethernet0.present = "TRUE"
extendedConfigFile = "pfSense_UTM.vmxf"
floppy0.present = "FALSE"
ethernet0.vnet = "VMnet8"
ethernet0.displayName = "VMnet8"
ethernet2.connectionType = "custom"
ethernet2.addressType = "static"
ethernet2.vnet = "VMnet6"
ethernet2.displayName = "VMnet6"
ethernet2.virtualDev = "e1000"
ethernet1.connectionType = "custom"
ethernet1.addressType = "static"
ethernet1.vnet = "VMnet5"
ethernet1.displayName = "VMnet5"
ethernet1.virtualDev = "e1000"
ethernet2.present = "TRUE"
ethernet1.present = "TRUE"
ethernet2.address = "00:50:56:25:B2:00"
ethernet1.address = "00:50:56:24:1A:79"
```



```
pfSense_UTM.vmx
파일 편집 보기
scsi0:0.fileName = "pfSense_UTM.vmdk"
scsi0:0.present = "TRUE"
ide1:0.deviceType = "cdrom-image"
ide1:0.fileName = "C:\VMware\SecurityNetwork\설치자료\pfSense-CE-2.7.2-RELEASE-amd64.iso"
ide1:0.present = "TRUE"
usb.present = "TRUE"
ehci.present = "TRUE"
ethernet0.connectionType = "custom"
ethernet0.addressType = "static"
ethernet0.virtualDev = "vmxnet3"
ethernet0.present = "TRUE"
extendedConfigFile = "pfSense_UTM.vmxf"
floppy0.present = "FALSE"
ethernet0.vnet = "VMnet8"
ethernet0.displayName = "VMnet8"
ethernet2.connectionType = "custom"
ethernet2.addressType = "static"
ethernet2.vnet = "VMnet6"
ethernet2.displayName = "VMnet6"
ethernet2.virtualDev = "vmxnet3"
ethernet1.connectionType = "custom"
ethernet1.addressType = "static"
ethernet1.vnet = "VMnet5"
ethernet1.displayName = "VMnet5"
ethernet1.virtualDev = "vmxnet3"
ethernet2.present = "TRUE"
ethernet1.present = "TRUE"
ethernet2.address = "00:50:56:25:B2:00"
ethernet1.address = "00:50:56:24:1A:79"
줄 45, 열 33   32/3,059자   일반 텍스트   100%   Windows (CRLF)   UTF-8
```

\SecurityNetwork\Firewall\pfSense\pfSense_UTM.vmx

5) 네트워크 인터페이스 기능 지정



n ➔ VLAN 기능 비활성화

네트워크 인터페이스 기능 지정

```
Should VLANs be set up now [y|n]? n

If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(vmx0 vmx1 vmx2 or a): vmx0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(vmx1 vmx2 a or nothing if finished): vmx1

Optional interface 1 description found: OPT1
Enter the Optional 1 interface name or 'a' for auto-detection
(vmx2 a or nothing if finished): vmx2

The interfaces will be assigned as follows:

WAN  -> vmx0
LAN  -> vmx1
OPT1 -> vmx2

Do you want to proceed [y|n]? y
```

- WAN 인터페이스 지정 : VMX0
- LAN 인터페이스 지정 : VM1
- Optional 인터페이스 지정 : OPT1

6) WAN 네트워크 인터페이스 설정과 IP 주소 지정

```
1 - WAN (vmx0 - dhcp)
2 - LAN (vmx1 - static)
3 - OPT1 (vmx2)

Enter the number of the interface you wish to configure: 1

Configure IPv4 address WAN interface via DHCP? (y/n) n

Enter the new WAN IPv4 address.  Press <ENTER> for none:
> 192.168.10.254

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new WAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new WAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
> 192.168.10.2

Configure IPv6 address WAN interface via DHCP6? (y/n)
```

- 설정할 인터페이스 번호 지정 : 1
- WAN IP address : 192.168.10.254
- 서브넷마스크 : 24
- 게이트웨이 주소 : 192.168.10.2
- 그 외 IPv4/IPv6 DHCP 기능 비활성화

WAN 네트워크 인터페이스 설정과 IP 주소 지정

```
For a WAN, enter the new WAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address WAN interface via DHCP6? (y/n) n

Enter the new WAN IPv6 address.  Press <ENTER> for none:
>

Do you want to enable the DHCP server on WAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Please wait while the changes are saved to WAN...
 Reloading filter...
 Reloading routing configuration...
 DHCPD...

The IPv4 WAN address has been set to 192.168.10.254/24
You can now access the webConfigurator by opening the following URL in your web
browser:
                http://192.168.10.254/

Press <ENTER> to continue.
```

LAN/DMZ 네트워크 인터페이스 설정과 IP 주소 지정

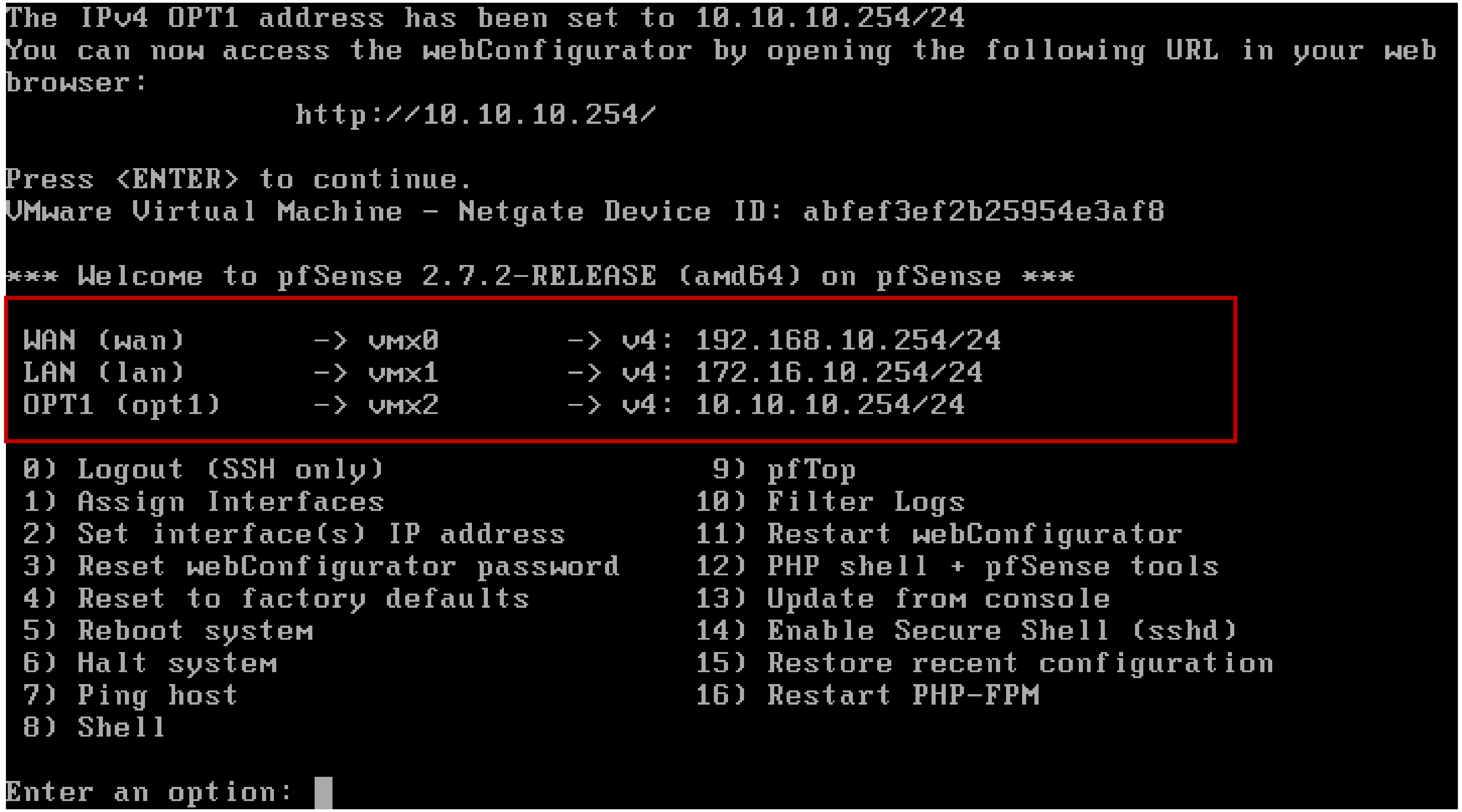
```
Enter the number of the interface you wish to configure: 3

Configure IPv4 address OPT1 interface via DHCP? (y/n) n

Enter the new OPT1 IPv4 address.  Press <ENTER> for none:
> 10.10.10.254

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new OPT1 IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new OPT1 IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address OPT1 interface via DHCP6? (y/n) n

Enter the new OPT1 IPv6 address.  Press <ENTER> for none:
>

Do you want to enable the DHCP server on OPT1? (y/n) n
```

- 설정할 인터페이스 번호 지정 : 3
- WAN IP address : 10.10.10.254
- 서브넷마스크 : 24
- 그 외 IPv4/IPv6 DHCP 기능 비활성화

LAN/DMZ 네트워크 인터페이스 설정과 IP 주소 지정

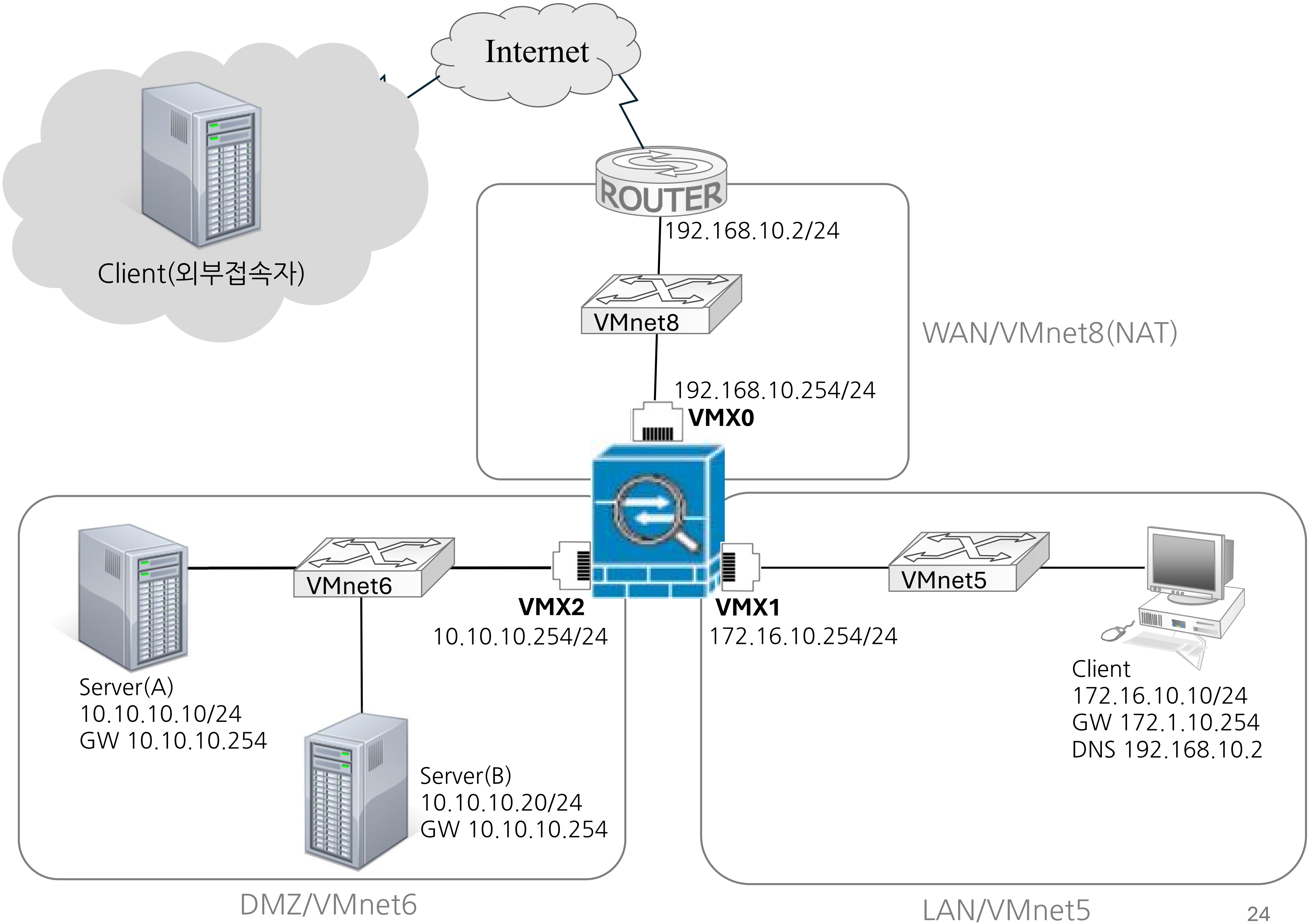
```
For a WAN, enter the new OPT1 IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address OPT1 interface via DHCP6? (y/n) n

Enter the new OPT1 IPv6 address.  Press <ENTER> for none:
>

Do you want to enable the DHCP server on OPT1? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Please wait while the changes are saved to OPT1...
 Reloading filter...
 Reloading routing configuration...
 DHCPD...

The IPv4 OPT1 address has been set to 10.10.10.254/24
You can now access the webConfigurator by opening the following URL in your web
browser:
                http://10.10.10.254/

Press <ENTER> to continue.
```

7) 네트워크 인터페이스 IP 주소 확인



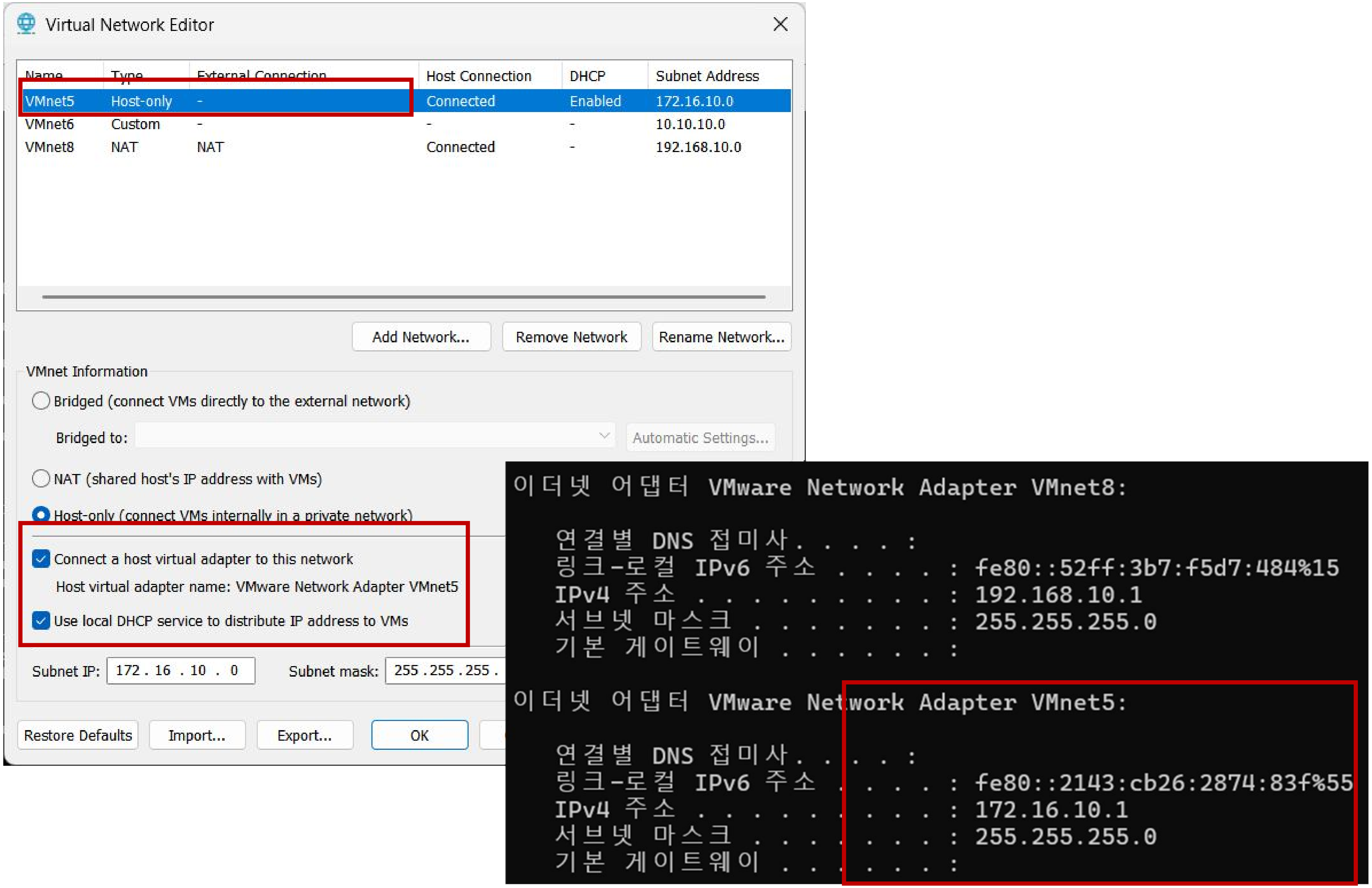
8) pfSense disable(비활성화)

pfctl -d



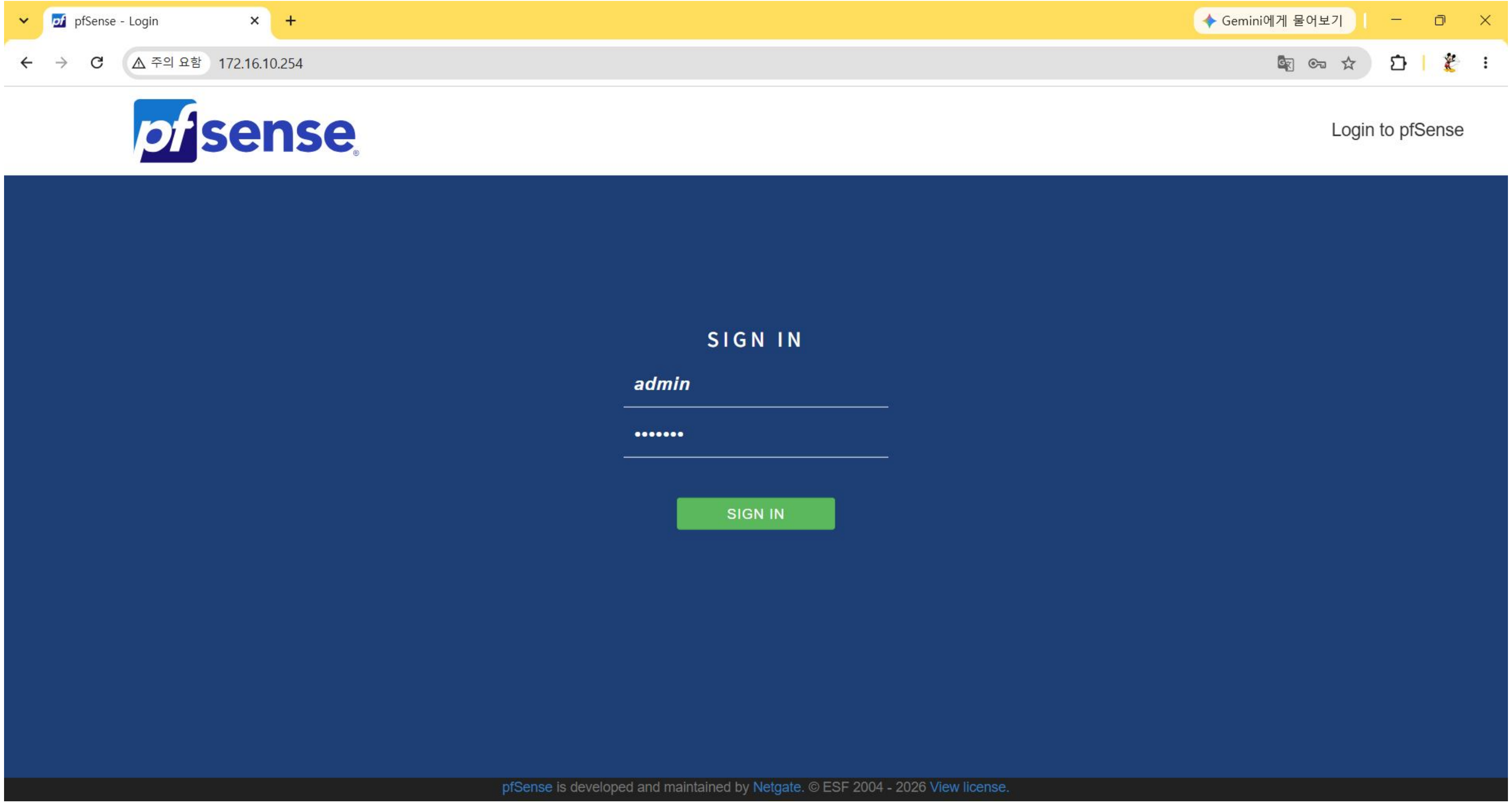
Internet
Client(외부접속자)
ROUTER
192.168.10.2/24
VMnet8
WAN/VMnet8(NAT)
192.168.10.254/24
VMX0
VMnet6
VMX2
10.10.10.254/24
VMX1
172.16.10.254/24
VMnet5
Client
172.16.10.10/24
GW 172.1.10.254
DNS 192.168.10.2
Server(A)
10.10.10.10/24
GW 10.10.10.254
Server(B)
10.10.10.20/24
GW 10.10.10.254
DMZ/VMnet6
LAN/VMnet5

9) 방화벽 설정을 위한 네트워크 환경 변경



10) Web pfSendse 접속

http://172.16.10.254



admin/pfsense

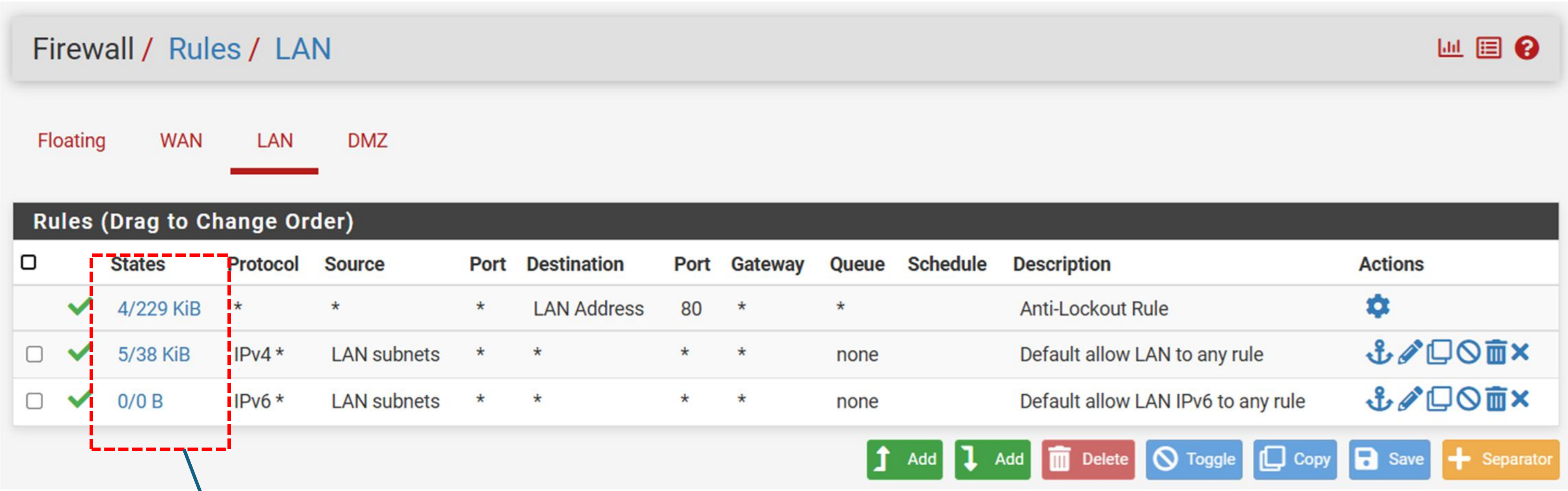
pfSens 명령어

- pfctl -d : 방화벽 rule 비활성화
- pfctl -e : 방화벽 rule 활성화
- pfctl -f /tmp/rules.debug : 방화벽 rule 강제 재로드
- pfctl -s state : 현재 활성화된 방화벽 상태 테이블 확인
- pfctl -F state : 상태 테이블 초기화

- ifconfig : 네트워크 인터페이스 정보 및 할당된 IP 주소 확인
- netstat -rn : 라우팅 테이블 확인
- exit : shell 상태에서 메뉴로 전환

Default 접근제어 Rule

Default LAN Rules



현재세션수/누적트래픽양

*세션 수 : 실시간으로 접속을 요청한 세션수

*누적트래픽 양 : 해당 규칙이 적용되어 통과한 누적 트래픽 데이터 총량

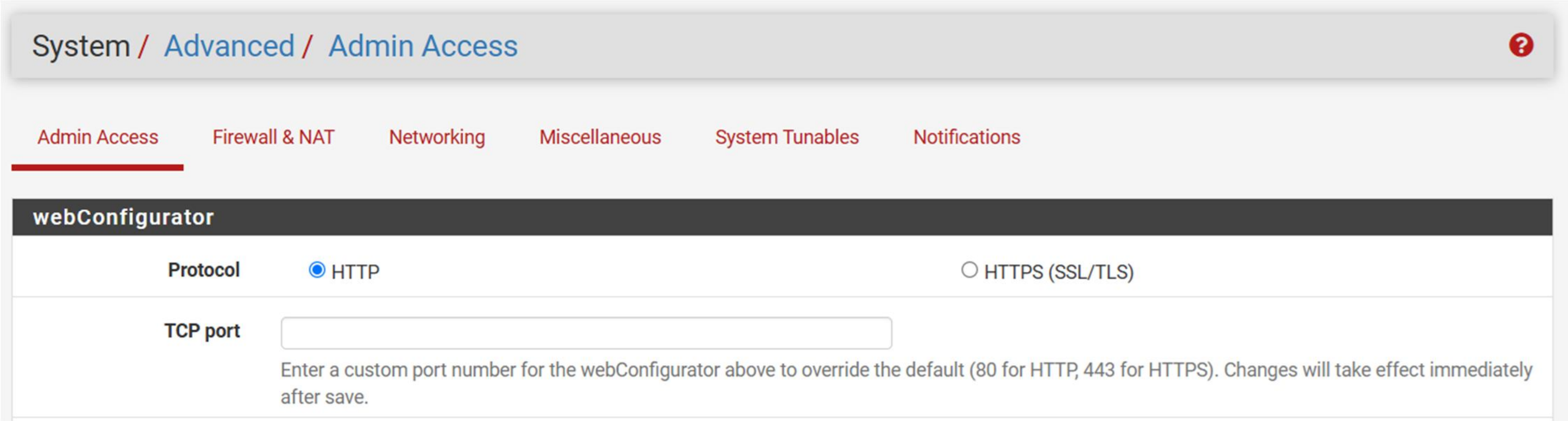
(KiB, MiB, GiB 단위, 통신을 많이 할수록 증가)

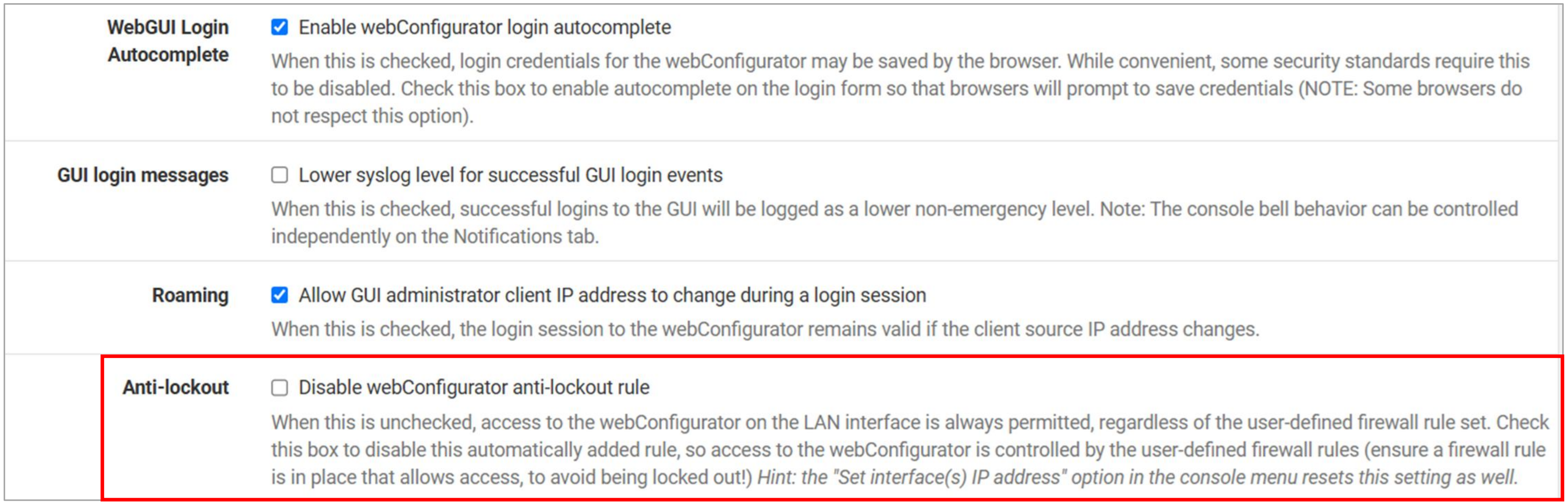
Anti-Lockout Rule

- 접속 차단 방지 정책
- 관리자가 LAN 상에서 pfSense로 Web 및 SSH 접속을 무조건 허용하는 규칙
- 모든 트래픽 다찬 규칙을 넣더라도 WebUI접속은 항상 유지하게 함
- 보안상 비활성화가 필요한 경우
- System > Advanced > Admin Access > Anti-Lockout 항목 체크
- (Disable the webConfigurator anti-lockout rule)

* 해당 룰을 비활성화기 전 에 Web또는 ssh 접속을 반드시 허용하는 수동 방화벽 규칙을 만들어 둬야 함

Anti-Lockout Rule 비활성화 방법



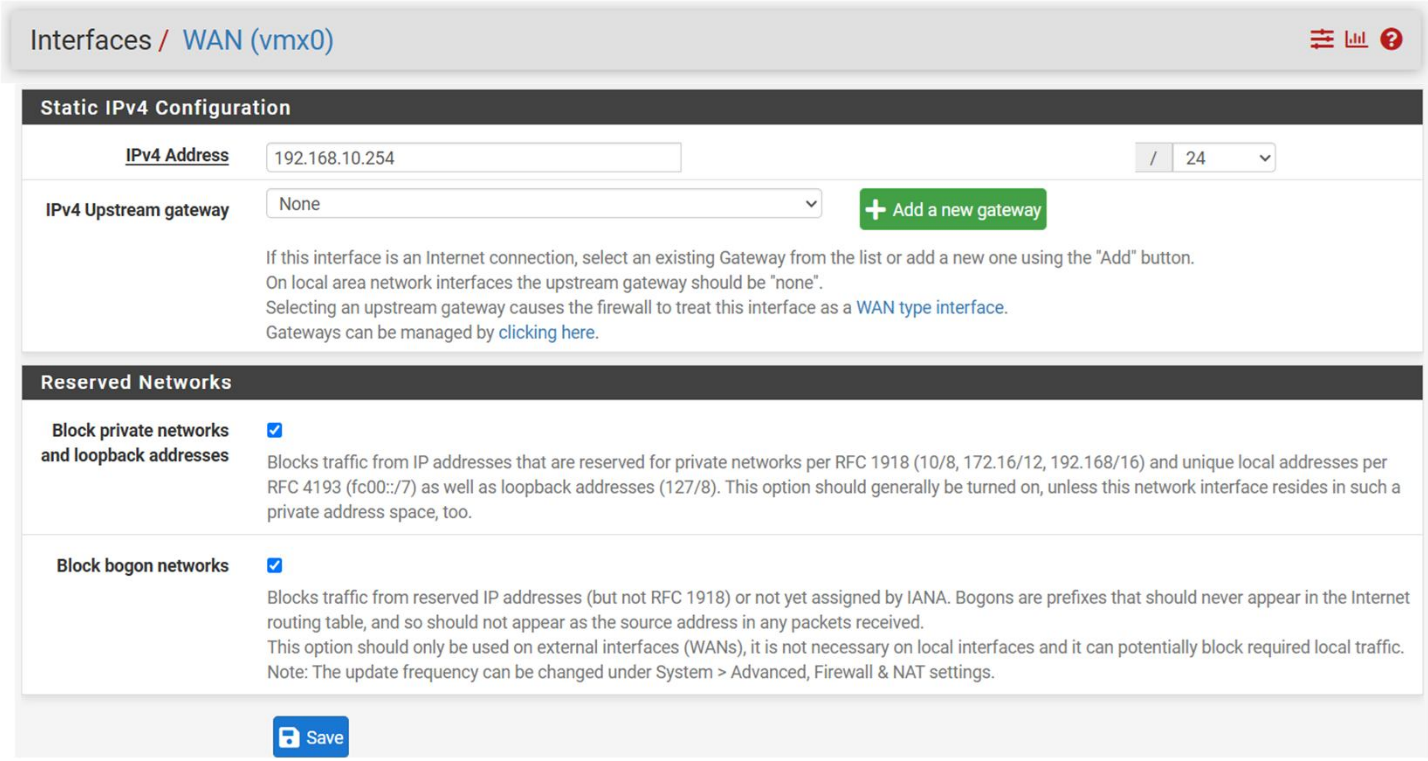


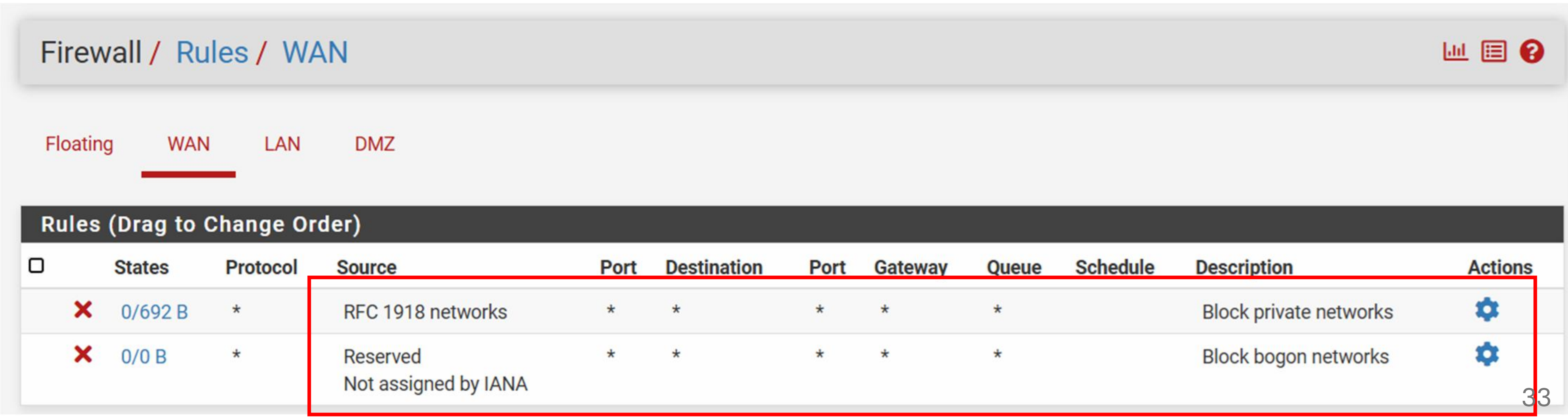
WebUI 접속 허용 수동 방화벽 설정

- Action : Pass
- Interface : LAN
- Address Family : IPv4
- Protocol : TCP
- Source : Single host or alias (192.168.1.2)
- Destination : LAN address
- Destiantion Port Range : From HTTP / To HTTP

- 기본의 모든 허용 규칙 보다 무조건 위쪽에 위치해야 함

Default WAN Rules





Block private networks and loopback addresses

- RFC 1918(사설IP)와 루프백수조(127.0.0.0/8)
- 외부망(인터넷)에서 사설IP나 loopback IP를 송신지IP로 들어오는 것은
- 보안상 위험한 트래픽으로 간주하여 원천 차단한다는 규칙
- 해당 rule을 활성화 시 IP Spoofing이나 잘못된 라우팅 정책으로 내부로 들어오는 트래픽들을 차단 가능

Block bogon networks

- 인터넷에서 비정상 IP주소(bogon, 보곤) 접근을 차단
- Bogon 주소는 인터넷망에서 절대 사용될 수 없는 주소

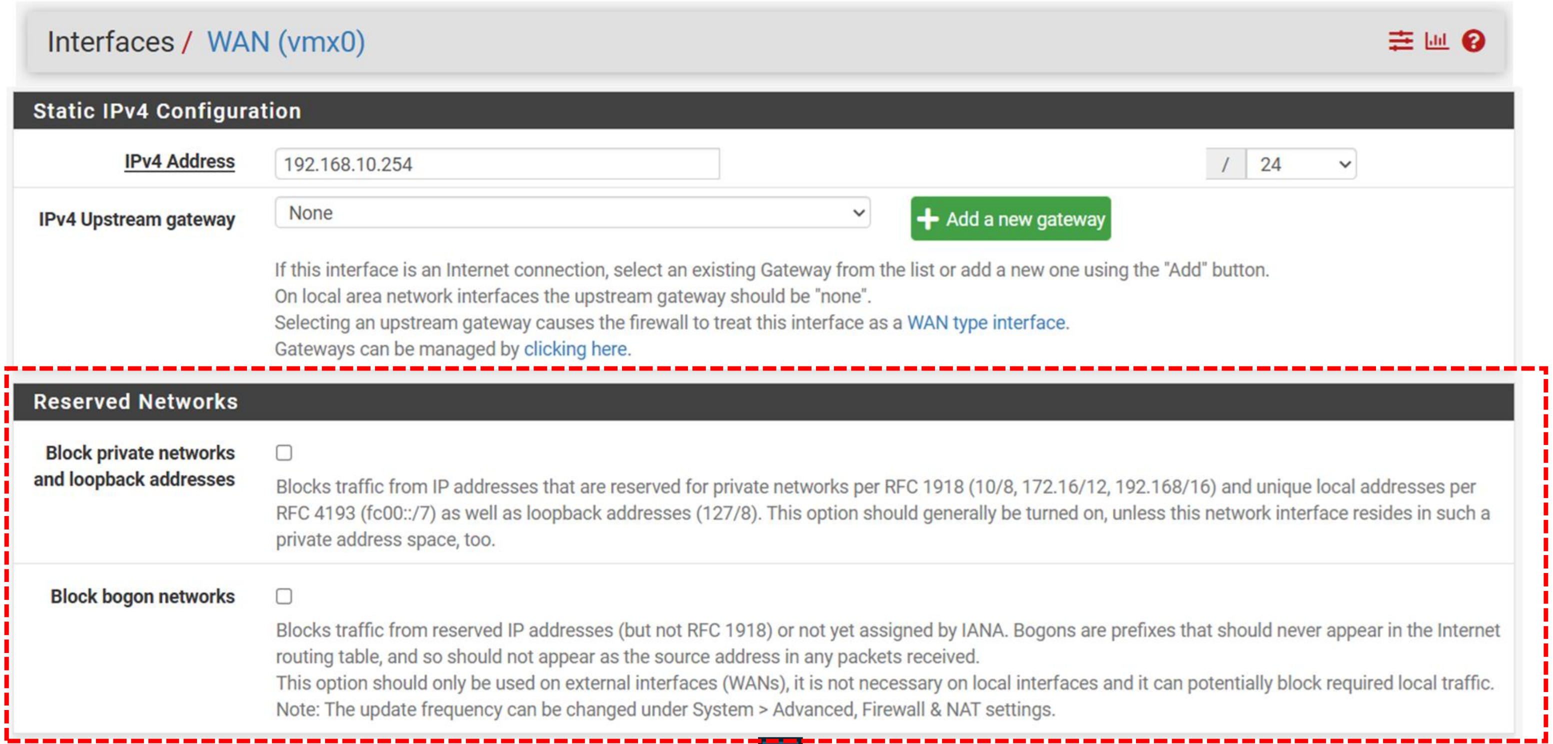
  - 미할당 주소 : IANA에서 통신사나 기업에 아직 분배하지 않은 공인 IP 대역

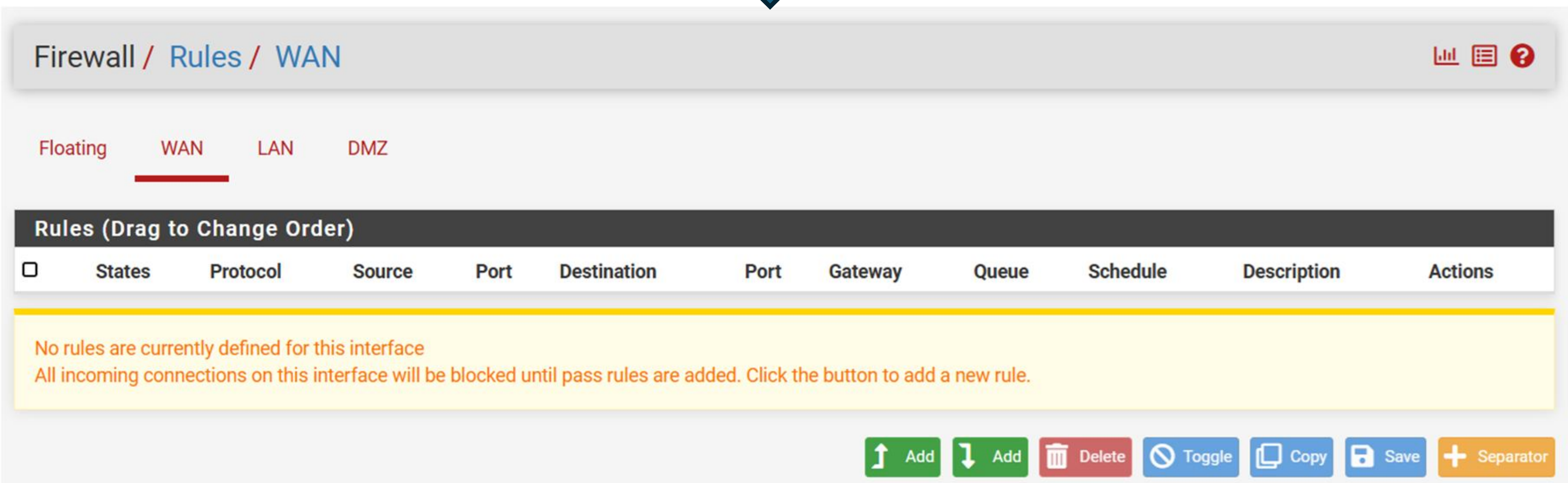
  - 예약된 주소 : 테스트용, 연구용, 또는 특수목적(사설IP)으로 할당된 주소

  - 삭제된 주소 : 과거에 사용되었으나 현재는 회수되어 비어 있는 IP 대역

- 해당 rule 활성화 시 다양한 네트워크 공격을 차단할 수 있음(IP spoofing, DDoS 공격 등)
- **Block private networks and loopback addresses 활성화는 단지 사설IP와 루프백 주소에만 초점을 맞춘 반면 Block bogon networks은 존재하지 않은 유령IP가지 포함하는 넓은 개념의 필터링**

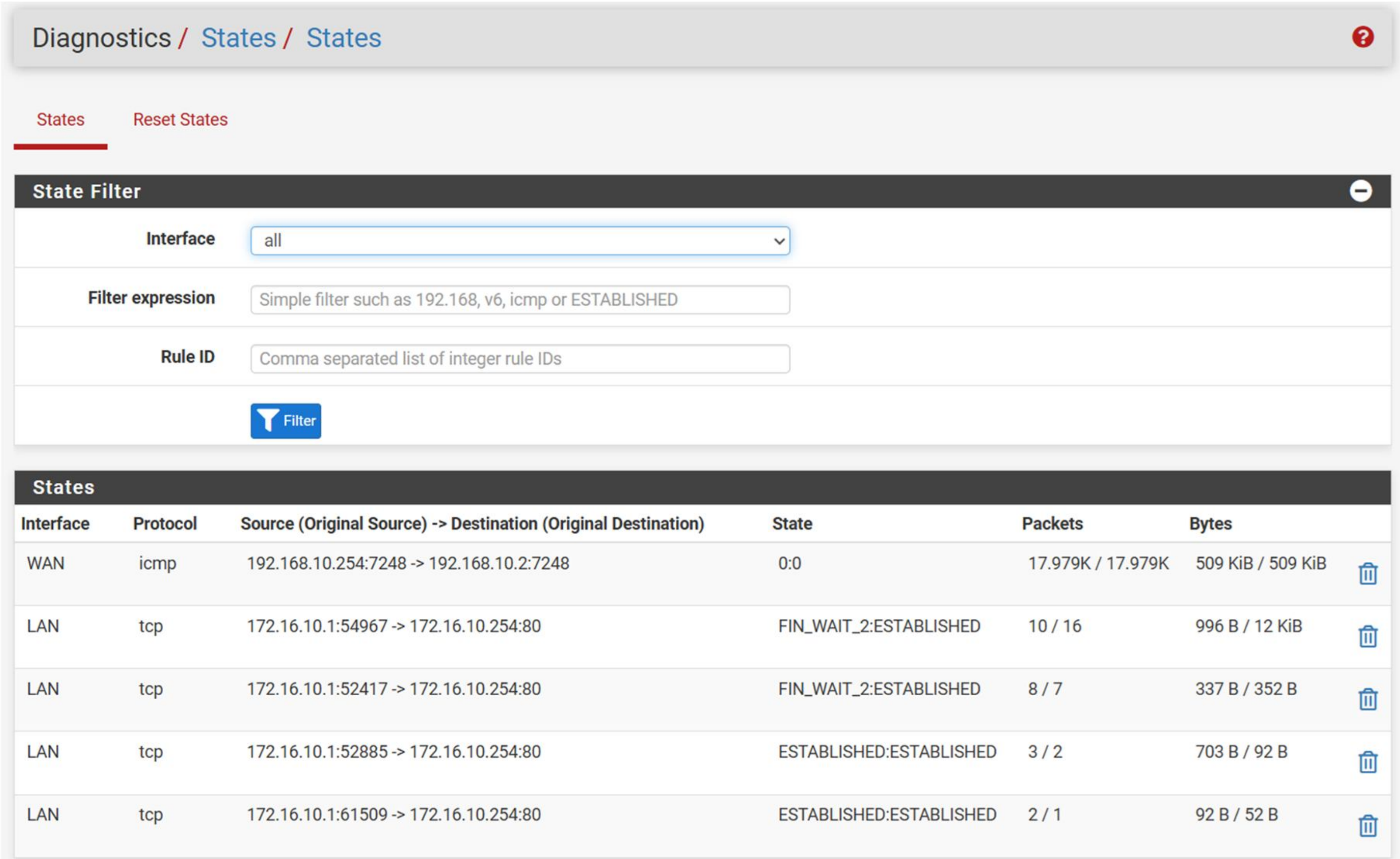
Block private networks /bogon network 비활성화





State Table(상태 테이블 또는 세션 테이블)

연결된 세션 관계를 테이블에 기록해 놓고 접근 룰이 없어서 해당 세션관계의 트래픽들을 통과 시킴

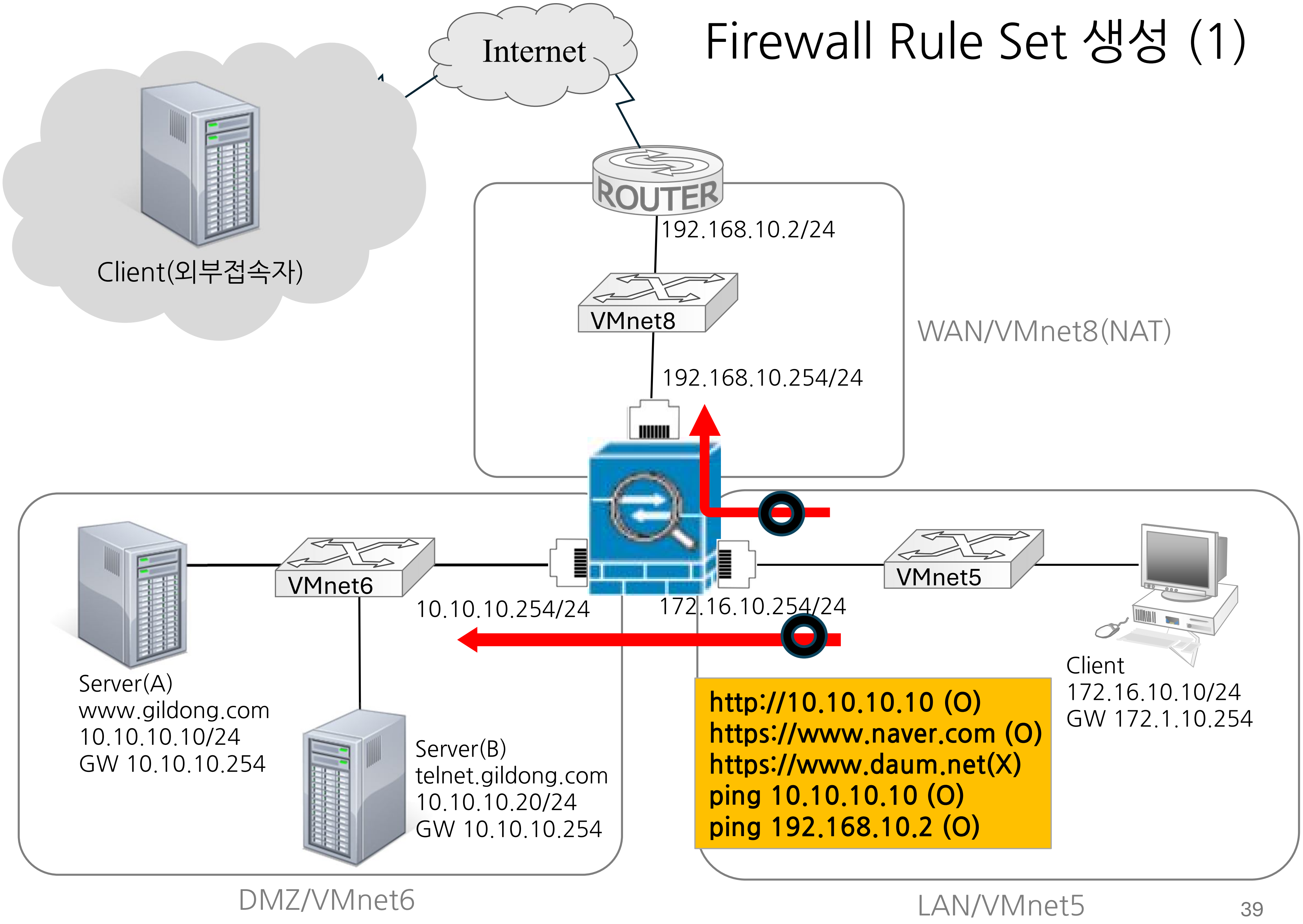


Interface	Protocol	Source (Original Source) -> Destination (Original Destination)	State	Packets	Bytes
WAN	icmp	192.168.10.254:7248 -> 192.168.10.2:7248	0:0	17.979K / 17.979K	509 KiB / 509 KiB
LAN	tcp	172.16.10.1:54967 -> 172.16.10.254:80	FIN_WAIT_2:ESTABLISHED	10 / 16	996 B / 12 KiB
LAN	tcp	172.16.10.1:52417 -> 172.16.10.254:80	FIN_WAIT_2:ESTABLISHED	8 / 7	337 B / 352 B
LAN	tcp	172.16.10.1:52885 -> 172.16.10.254:80	ESTABLISHED:ESTABLISHED	3 / 2	703 B / 92 B
LAN	tcp	172.16.10.1:61509 -> 172.16.10.254:80	ESTABLISHED:ESTABLISHED	2 / 1	92 B / 52 B

송신지 정보→ 수신지정보 연결상태정보 전송량

Firewall Rule Set 생성

Firewall Rule Set 생성 (1)



http://10.10.10.10 (O)
https://www.naver.com (O)
https://www.daum.net(X)
ping 10.10.10.10 (O)
ping 192.168.10.2 (O)

Firewall Rule Set 생성 (1)

1) WAN/DMZ 망으로 ping 테스트 허용

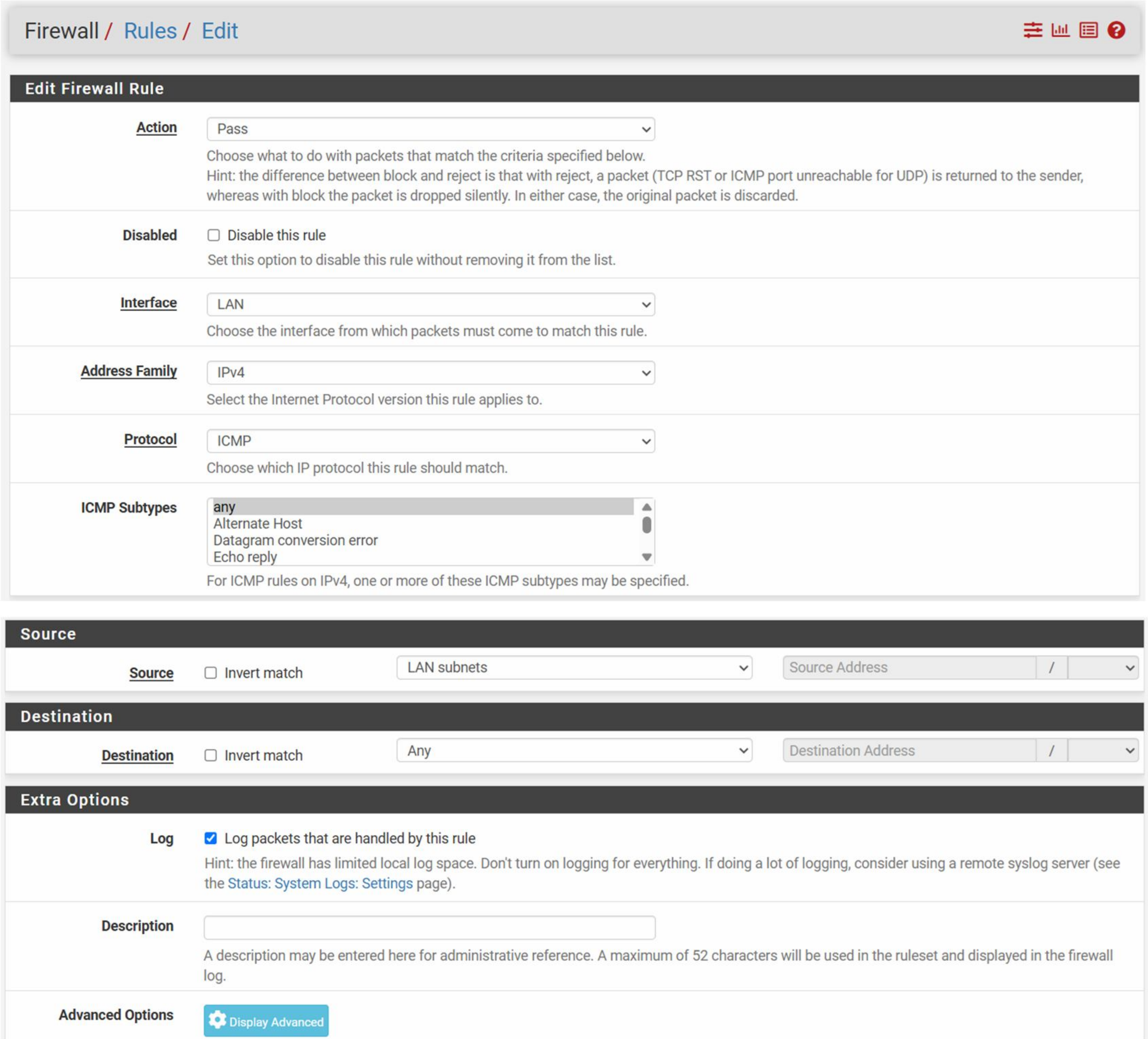
2) DMZ 모든 서버들로 접속 허용

3) 외부망은 www.naver.com 외 모든 사이트 접속 차단

*** Rule 생성 전 접근 가능 여부 테스트**
https://www.naver.com (X)
https://www.daum.net(X)
http://10.10.10.10 (X)
http://10.10.10.20 (X)
ping 10.10.10.10 (X)
ping 192.168.10.2 (X)

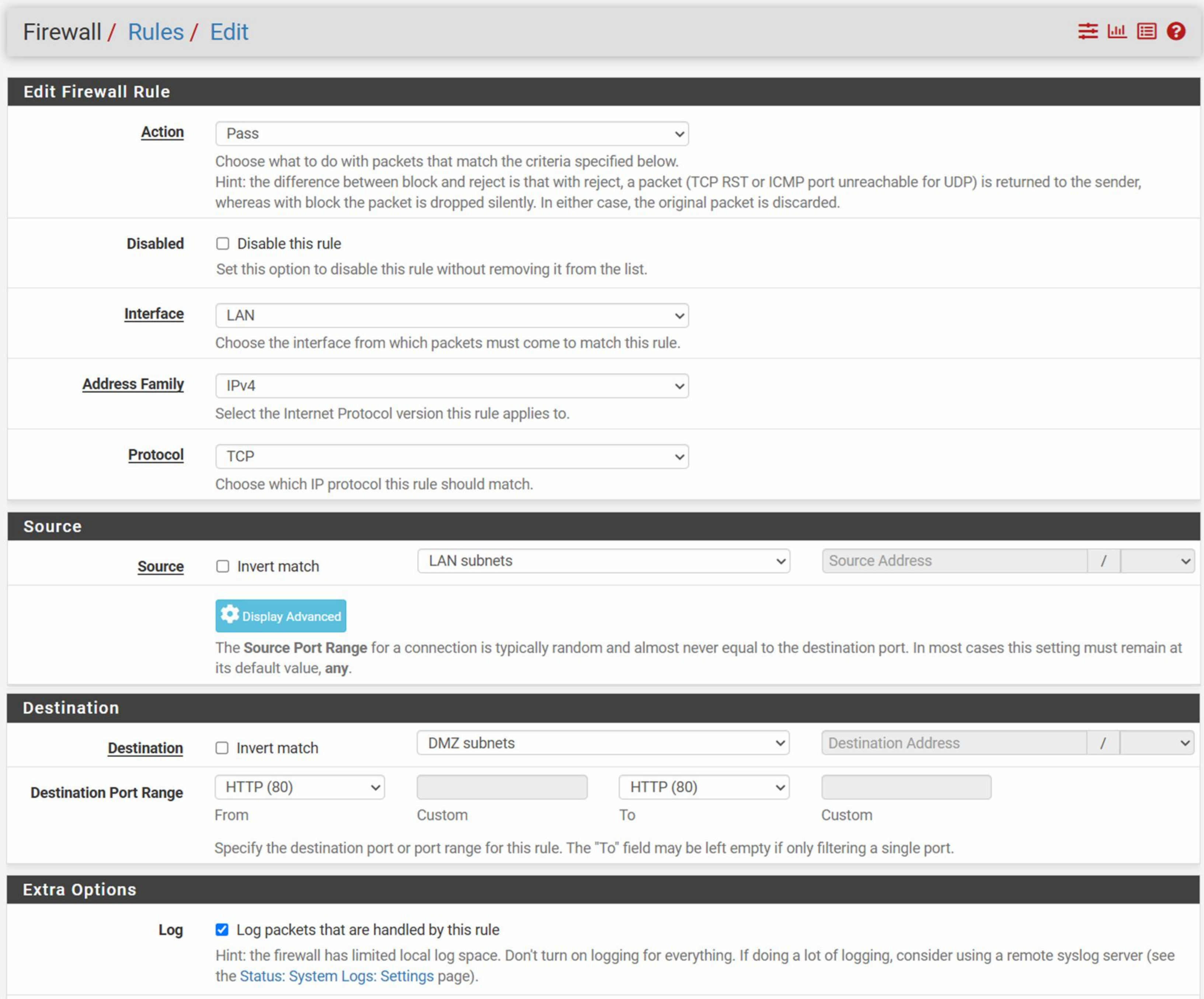
1) WAN/DMZ 망으로 ping 테스트는 허용

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	ICMP
ICMP Subtypes	Any
Source	LAN subnets
Destination	Any
Log	Log packets that are handled by the rule


Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
ICMP
Choose which IP protocol this rule should match.
ICMP Subtypes
any
Alternate Host
Datagram conversion error
Echo reply
For ICMP rules on IPv4, one or more of these ICMP subtypes may be specified.
Source
Source
Invert match
LAN subnets
Source Address
/
Destination
Destination
Invert match
Any
Destination Address
/
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).
Description
A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.
Advanced Options
Display Advanced

2) DMZ 모든 서버들로 접속 허용

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	TCP
Source	LAN Subnets
Destination	DMZ Subnets
Destination Port Range	HTTP
Log	Log packets that are handled by the rule


Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
LAN subnets
Source Address
/
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
DMZ subnets
Destination Address
/
Destination Port Range
HTTP (80)
HTTP (80)
From
Custom
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

3) 외부망은 www.naver.com 외 모든 사이트 접속 차단

① DNS 서버 접근 허용 룰 생성

② https://www.naver.com 접근 허용 룰 생성

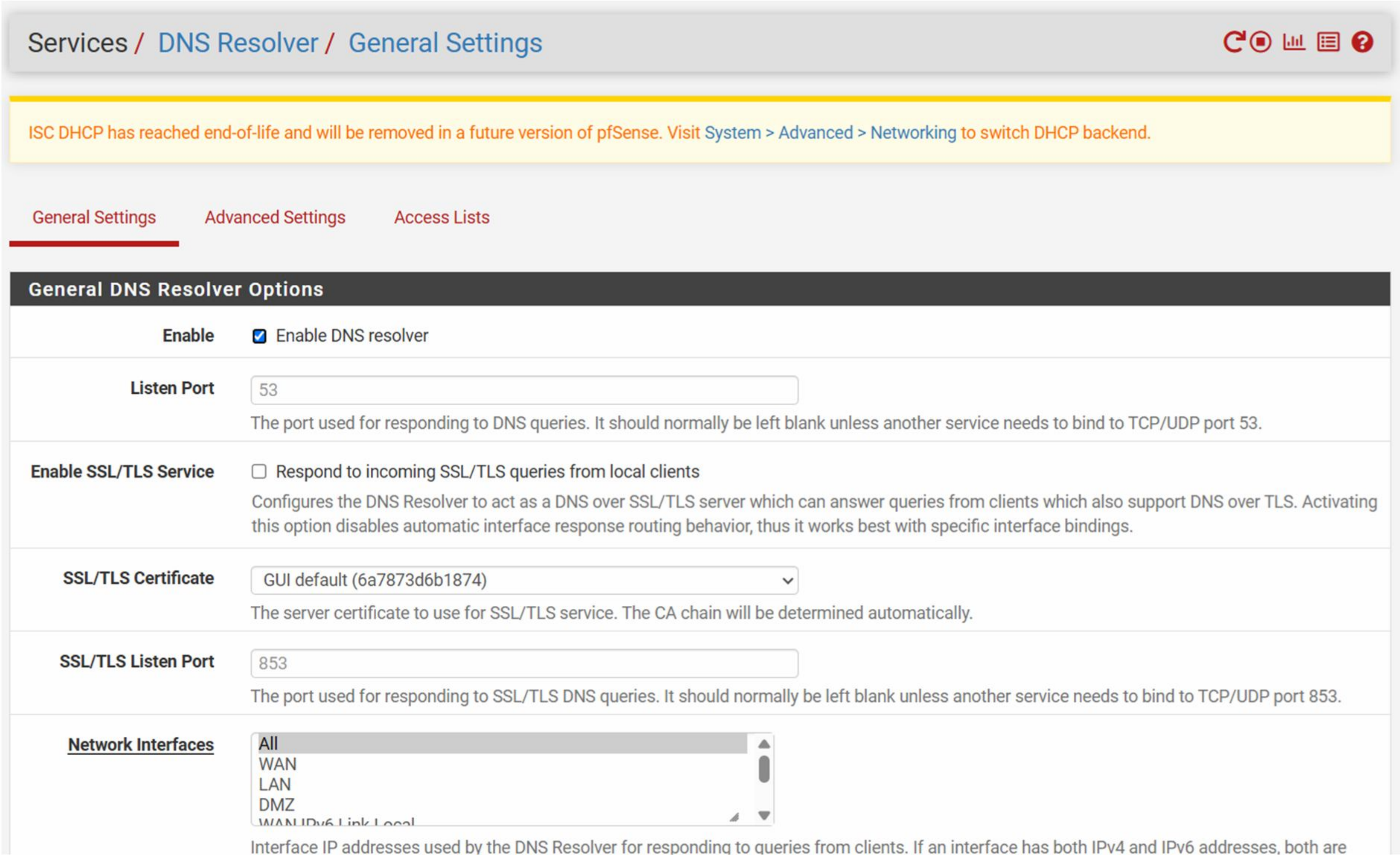
③ 그 외 외부 사이트 접속 차단 룰 생성

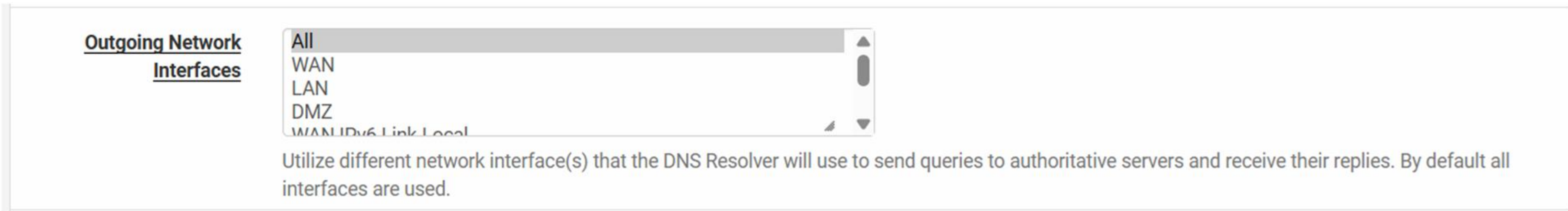
① DNS 서버 접근 허용 룰 생성

1단계. 내부 DNS 서버 사용 설정

Service > DNS resolver > General Settings > General DNS Resolver Options

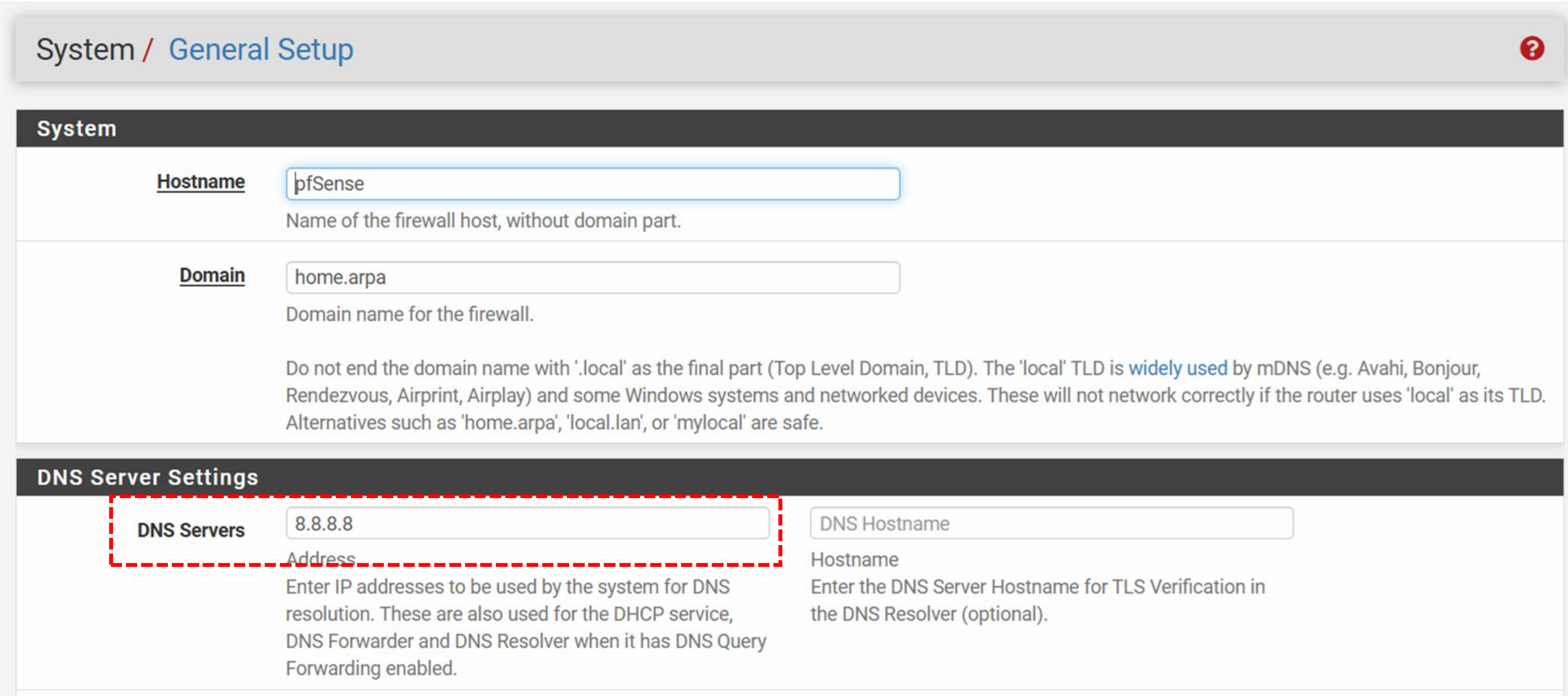
설정 필드	필드값
Enable	Enable DNS resolver
Network Interface	All
Outgoing Network Interfaces	All


Services / DNS Resolver / General Settings
ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend.
General Settings
Advanced Settings
Access Lists
General DNS Resolver Options
Enable
Enable DNS resolver
Listen Port
53
The port used for responding to DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 53.
Enable SSL/TLS Service
Respond to incoming SSL/TLS queries from local clients
Configures the DNS Resolver to act as a DNS over SSL/TLS server which can answer queries from clients which also support DNS over TLS. Activating this option disables automatic interface response routing behavior, thus it works best with specific interface bindings.
SSL/TLS Certificate
GUI default (6a7873d6b1874)
The server certificate to use for SSL/TLS service. The CA chain will be determined automatically.
SSL/TLS Listen Port
853
The port used for responding to SSL/TLS DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 853.
Network Interfaces
All
WAN
LAN
DMZ
Interface IP addresses used by the DNS Resolver for responding to queries from clients. If an interface has both IPv4 and IPv6 addresses, both are


Outgoing Network Interfaces
All
WAN
LAN
DMZ
Utilize different network interface(s) that the DNS Resolver will use to send queries to authoritative servers and receive their replies. By default all interfaces are used.

2단계. 외부 DNS 서버 사용 설정

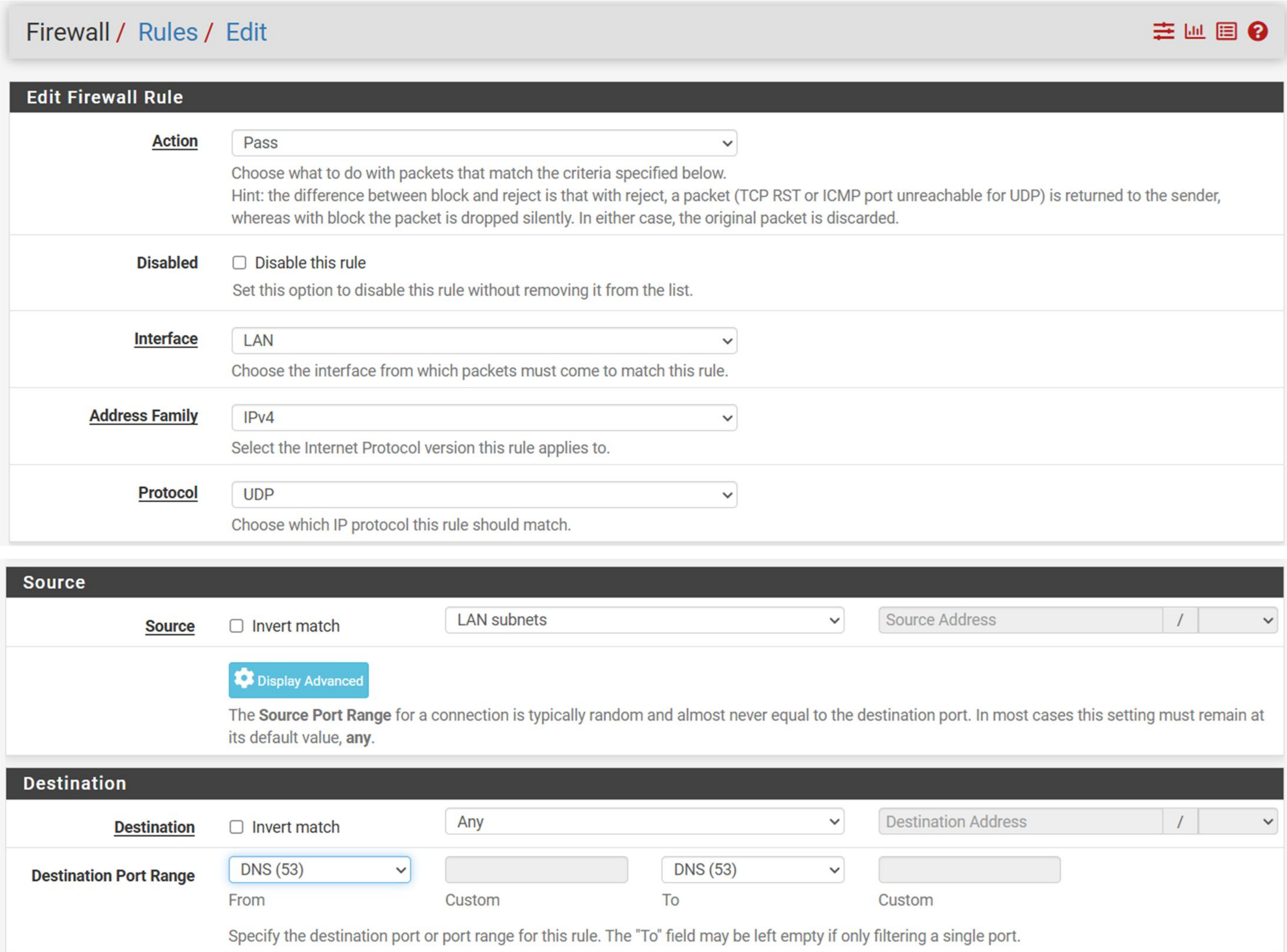
System > General Setup > DNS Server Settings > DNS servers



3단계. DNS 서버 접속 허용 룰 생성

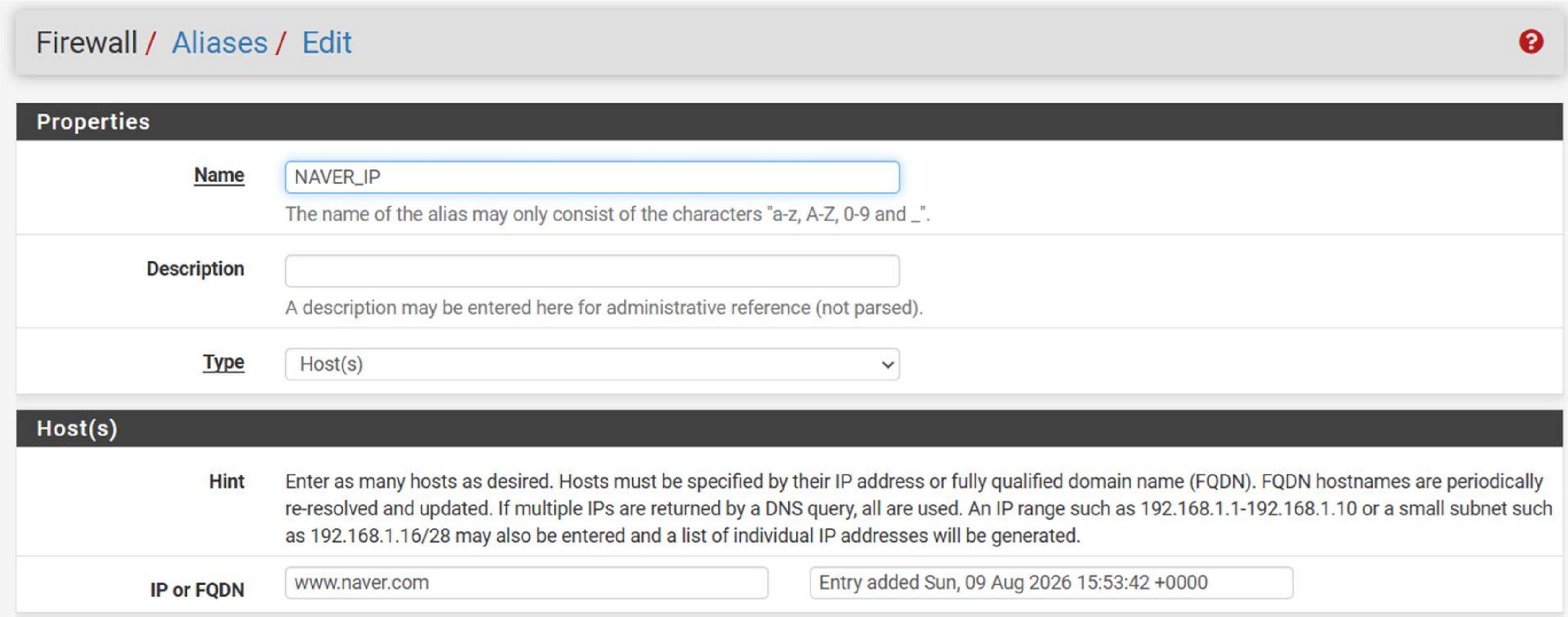
Firewall > Rules > LAN

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	UDP
Source	LAN Subnets
Destination	Any
Destination Port Range	DNS(53)
Log	Log packets that are handled by the rule


Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
UDP
Choose which IP protocol this rule should match.
Source
Source
Invert match
LAN subnets
Source Address
/
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
Any
Destination Address
/
Destination Port Range
DNS (53)
From
Custom
DNS (53)
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

② https://www.naver.com 접근 허용 룰 생성

[Alias 지정] Firewall > Aliases

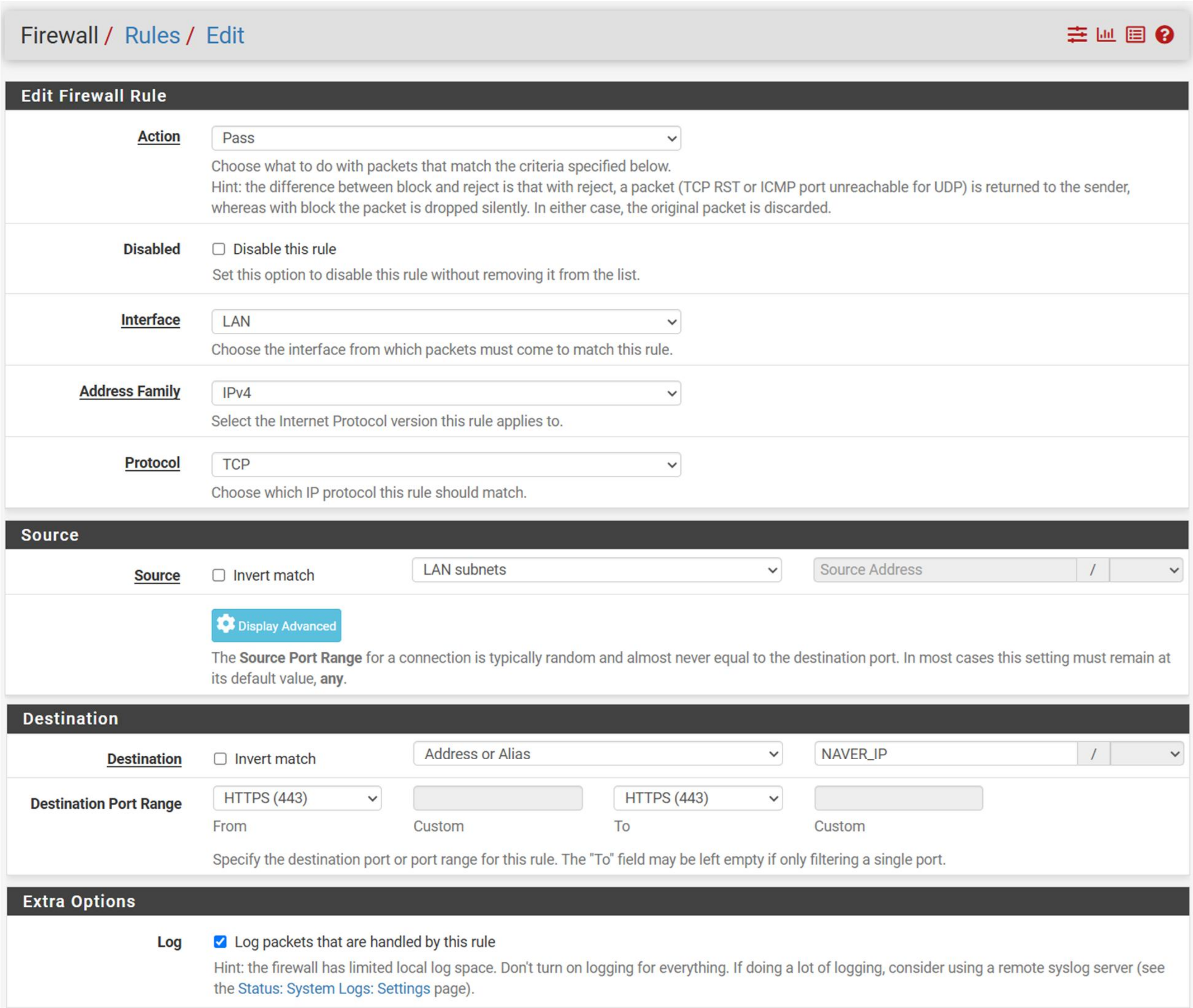


설정 필드	필드값
Properties	Name : NAVER_IP Type : Host(s)
Hosts	IP or FQDN : www.naver.com

4단계. https://www.naver.com 접속 허용 룰 생성

Firewall > Rules > LAN

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	TCP
Source	LAN Subnets
Destination	Address or Alias : NAVER_IP
Destination Port Range	HTTPS(443)
Log	Log packets that are handled by the rule

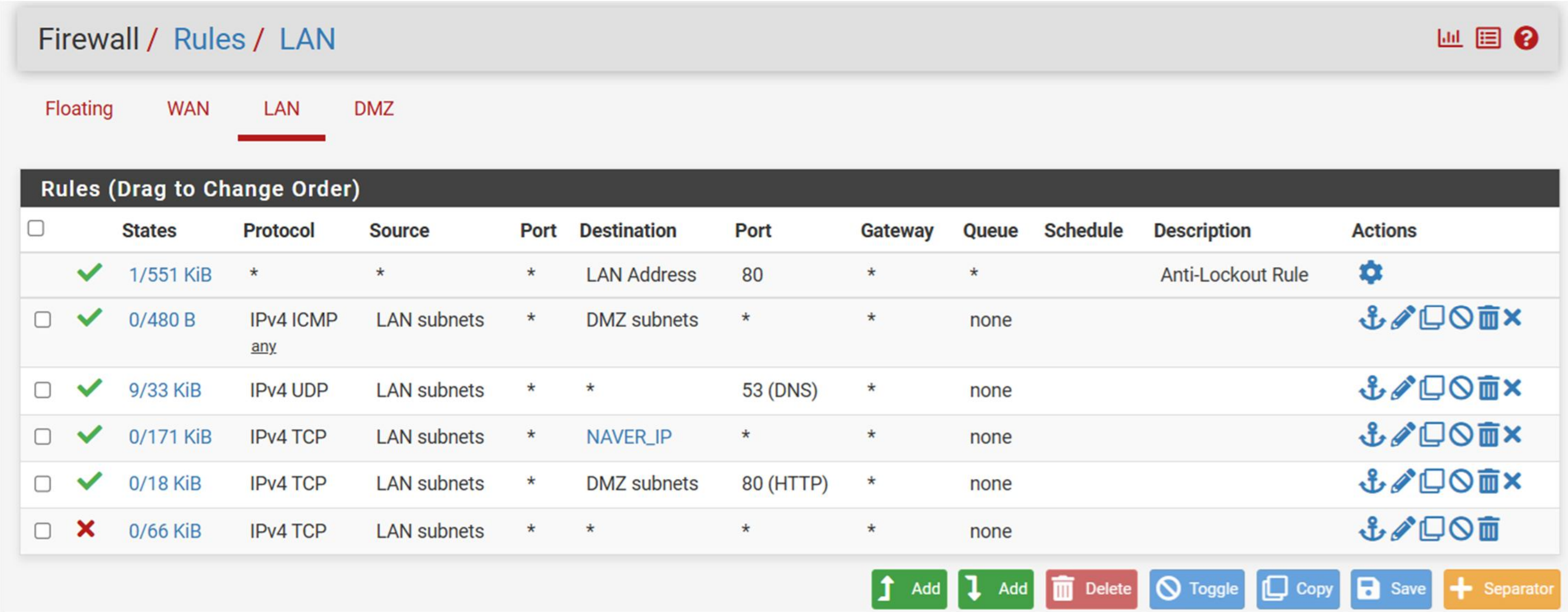

Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
LAN subnets
Source Address
/
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
Address or Alias
NAVER_IP
/
Destination Port Range
HTTPS (443)
From
Custom
HTTPS (443)
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

③ 그 외 외부 사이트 접속 차단 룰 생성

Firewall > Rules > LAN

설정 필드	필드값
Action	Block
Interface	LAN
Address Family	IPv4
Protocol	Any
Source	Any
Destination	Any
Log	Log packets that are handled by the rule

LAN Interface에 적용된 Rule Set



Firewall / Rules / LAN

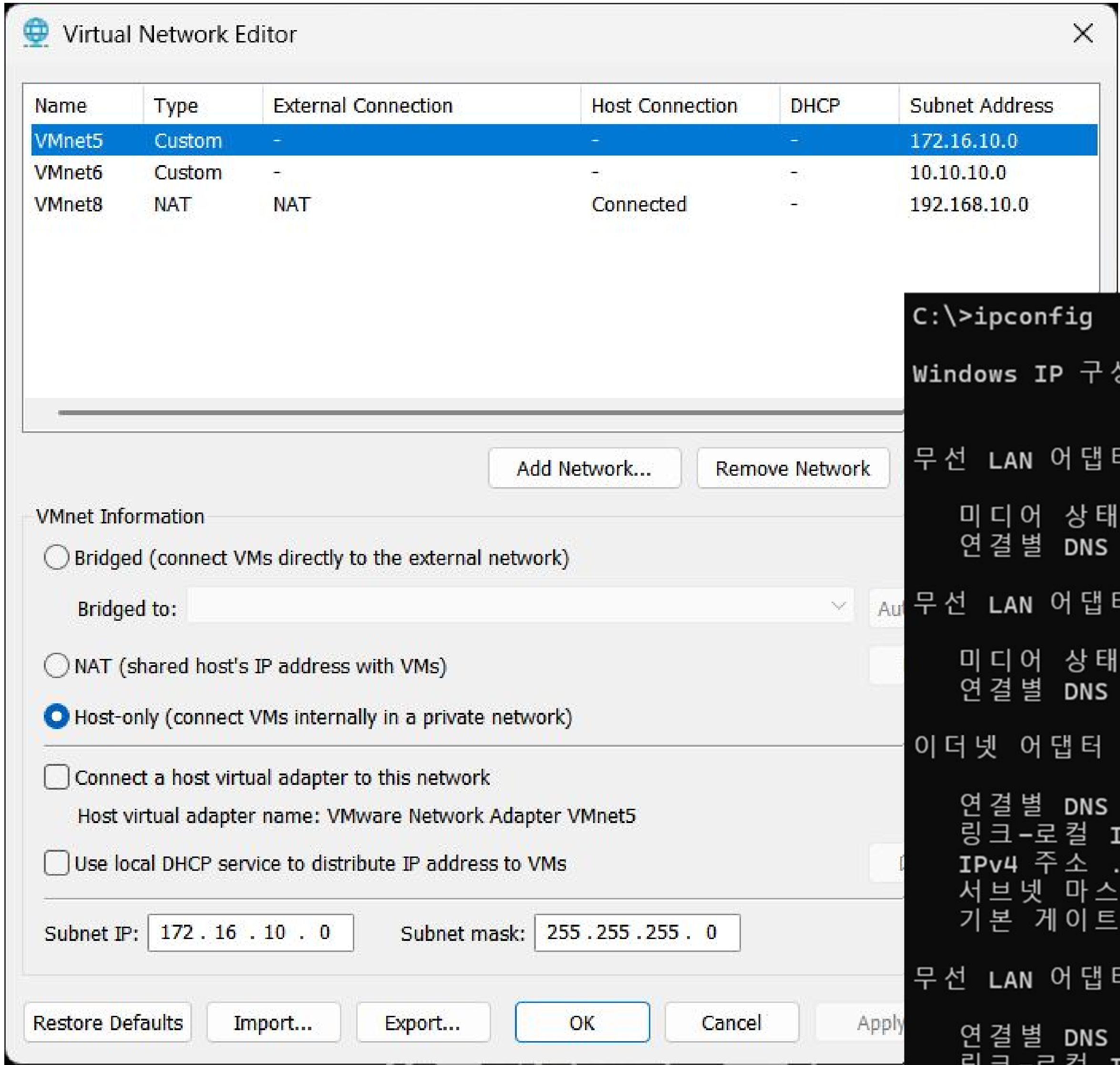
Floating WAN LAN DMZ

Rules (Drag to Change Order)

		States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	✔	1/551 KiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	✔	0/480 B	IPv4 ICMP any	LAN subnets	*	DMZ subnets	*	*	none			
☐	✔	9/33 KiB	IPv4 UDP	LAN subnets	*	*	53 (DNS)	*	none			
☐	✔	0/171 KiB	IPv4 TCP	LAN subnets	*	NAVER_IP	*	*	none			
☐	✔	0/18 KiB	IPv4 TCP	LAN subnets	*	DMZ subnets	80 (HTTP)	*	none			
☐	✘	0/66 KiB	IPv4 TCP	LAN subnets	*	*	*	*	none			

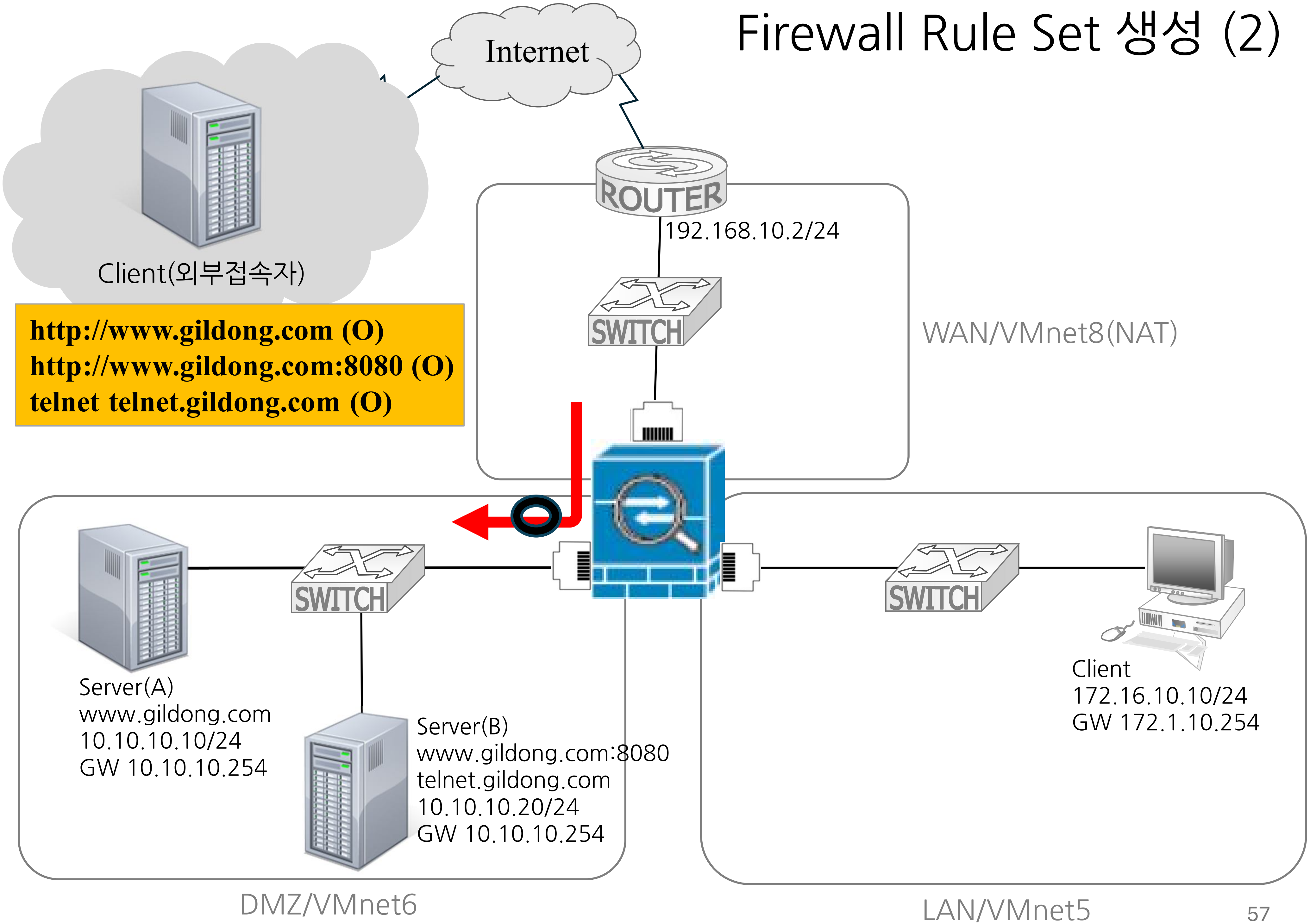
Add Add Delete Toggle Copy Save Separator

테스트 환경 구성 (VMWare> Edit > Virtual Network Editor)




```
C:\>ipconfig

Windows IP 구성


무선 LAN 어댑터 로컬 영역 연결* 1:

   미디어 상태 . . . . . . . . : 미디어 연결 끊김
   연결별 DNS 접미사. . . . :

무선 LAN 어댑터 로컬 영역 연결* 10:

   미디어 상태 . . . . . . . . : 미디어 연결 끊김
   연결별 DNS 접미사. . . . :

이더넷 어댑터 VMware Network Adapter VMnet8:

   연결별 DNS 접미사. . . . :
   링크-로컬 IPv6 주소 . . . . : fe80::52ff:3b7:f5d7:484%16
   IPv4 주소 . . . . . . . . . : 192.168.10.1
   서브넷 마스크 . . . . . . . : 255.255.255.0
   기본 게이트웨이 . . . . . . :

무선 LAN 어댑터 Wi-Fi:

   연결별 DNS 접미사. . . . :
   링크-로컬 IPv6 주소 . . . . : fe80::6d01:1c27:487b:8dcb%19
   IPv4 주소 . . . . . . . . . : 172.18.10.198
   서브넷 마스크 . . . . . . . : 255.255.252.0
   기본 게이트웨이 . . . . . . : 172.18.11.254
```

Firewall Rule Set 생성 (2)



Firewall Rule Set 생성 (2)

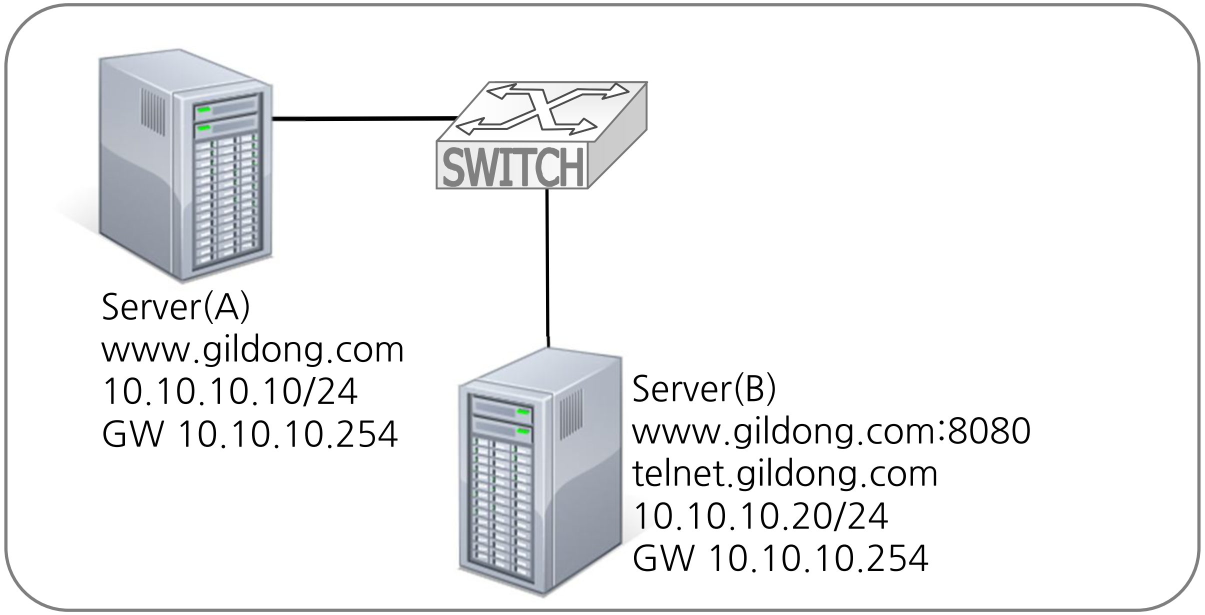
1) 외부 접속자는 DMZ 서버로 접근 가능

2) 외부 접속자는 Telnet 서버로 접근 가능

3) 그 외 모든 접근은 차단

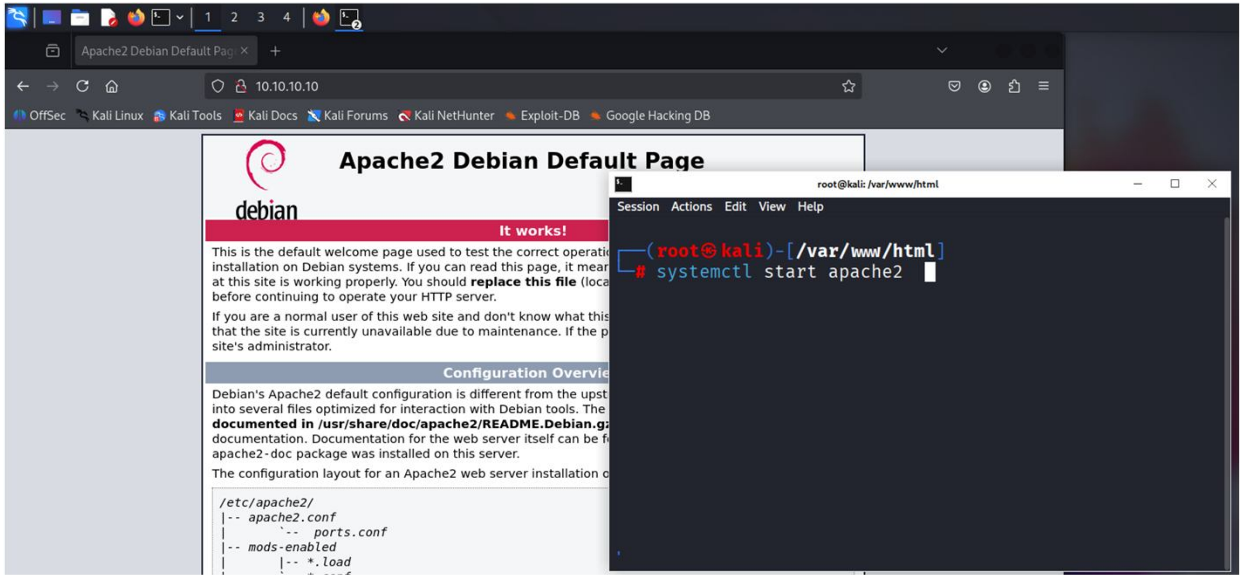
*** Rule 생성 전 접근 가능 여부 테스트**
http://www.gildong.com (X)
http://www.gildong.com:8080(X)
telnet telnet.gildong.com (X)

Firewall Rule Set 생성 (2)



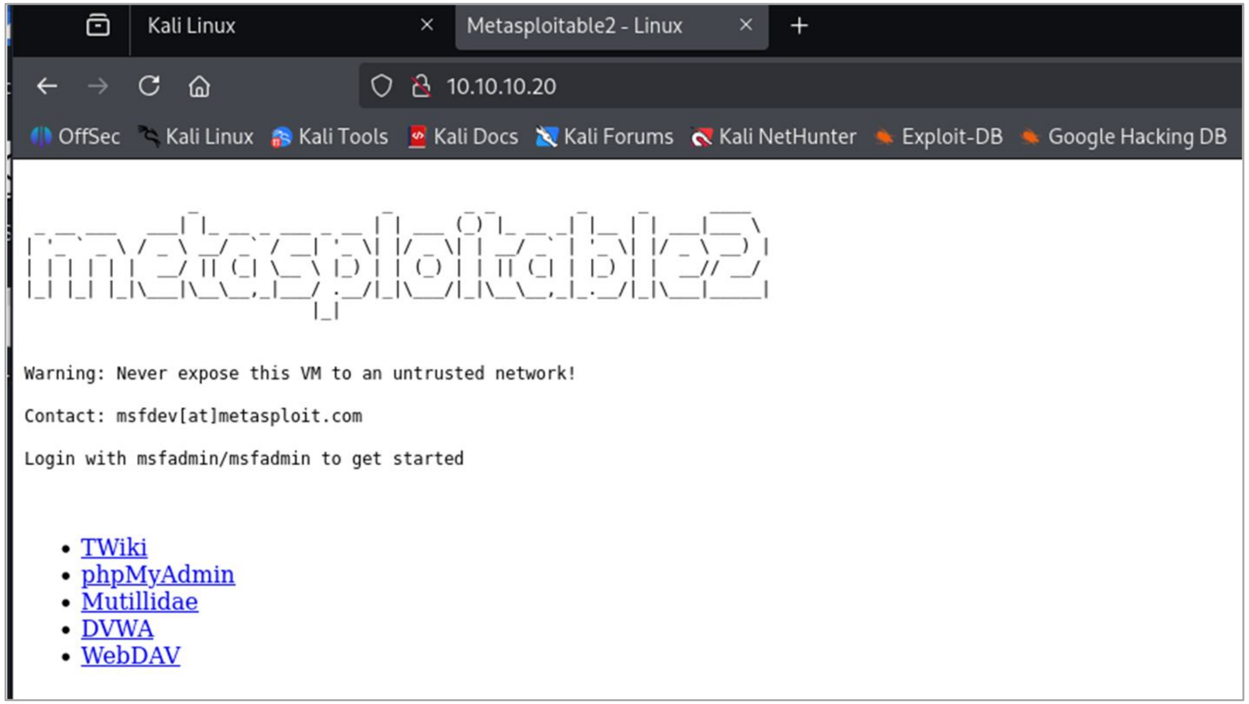
외부주소	내부주소
192.168.10.254:80	10.10.10.10:80
192.168.10.254:8080	10.10.10.20:8080
192.168.10.254:23	10.10.10.20:23

[DMZ Zone] Rule 생성 전 테스트 (웹서버 접속)



http://10.10.10.10

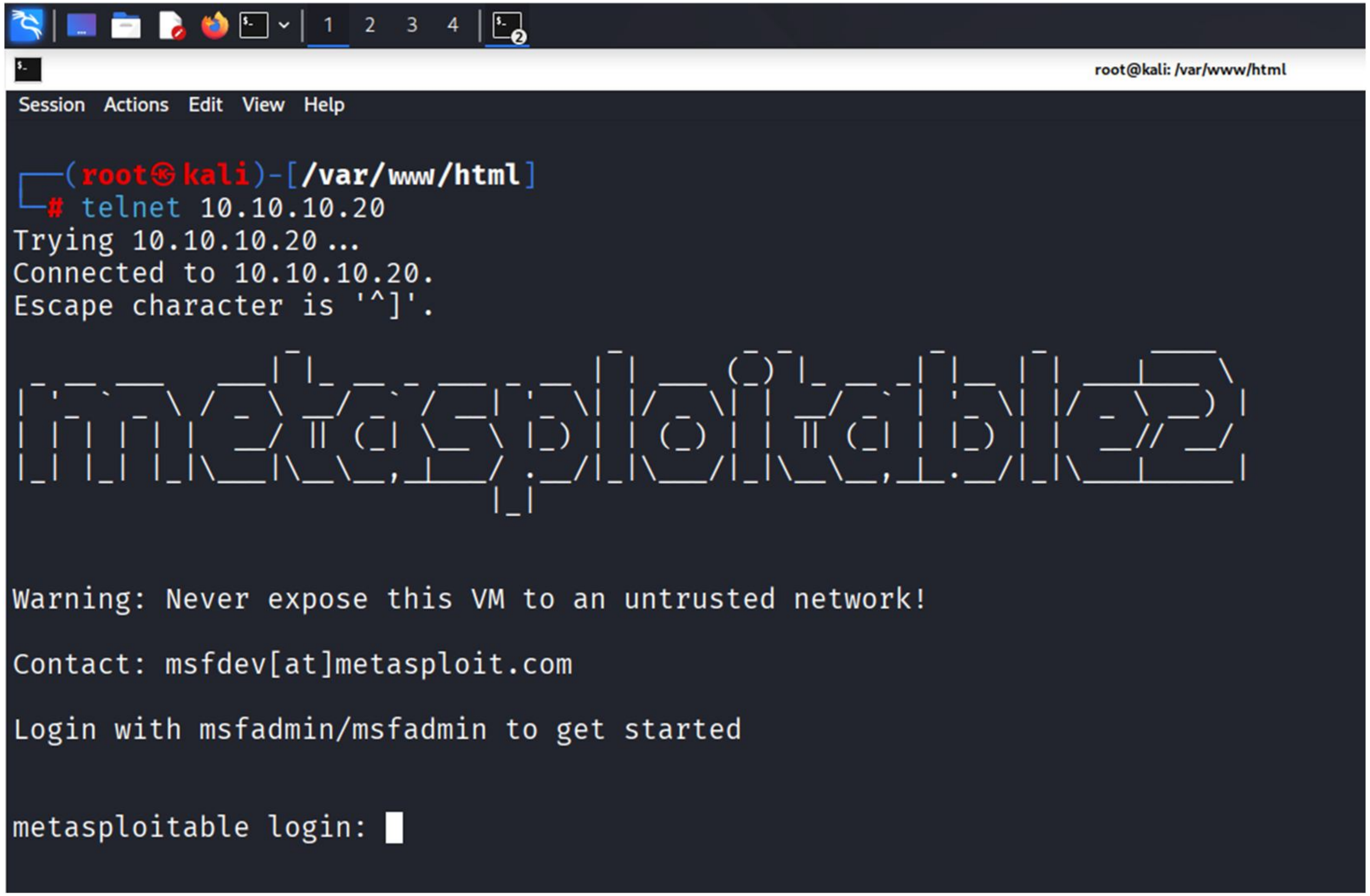
[WebServer 활성화]
sudo su - root
systemctl start apache2



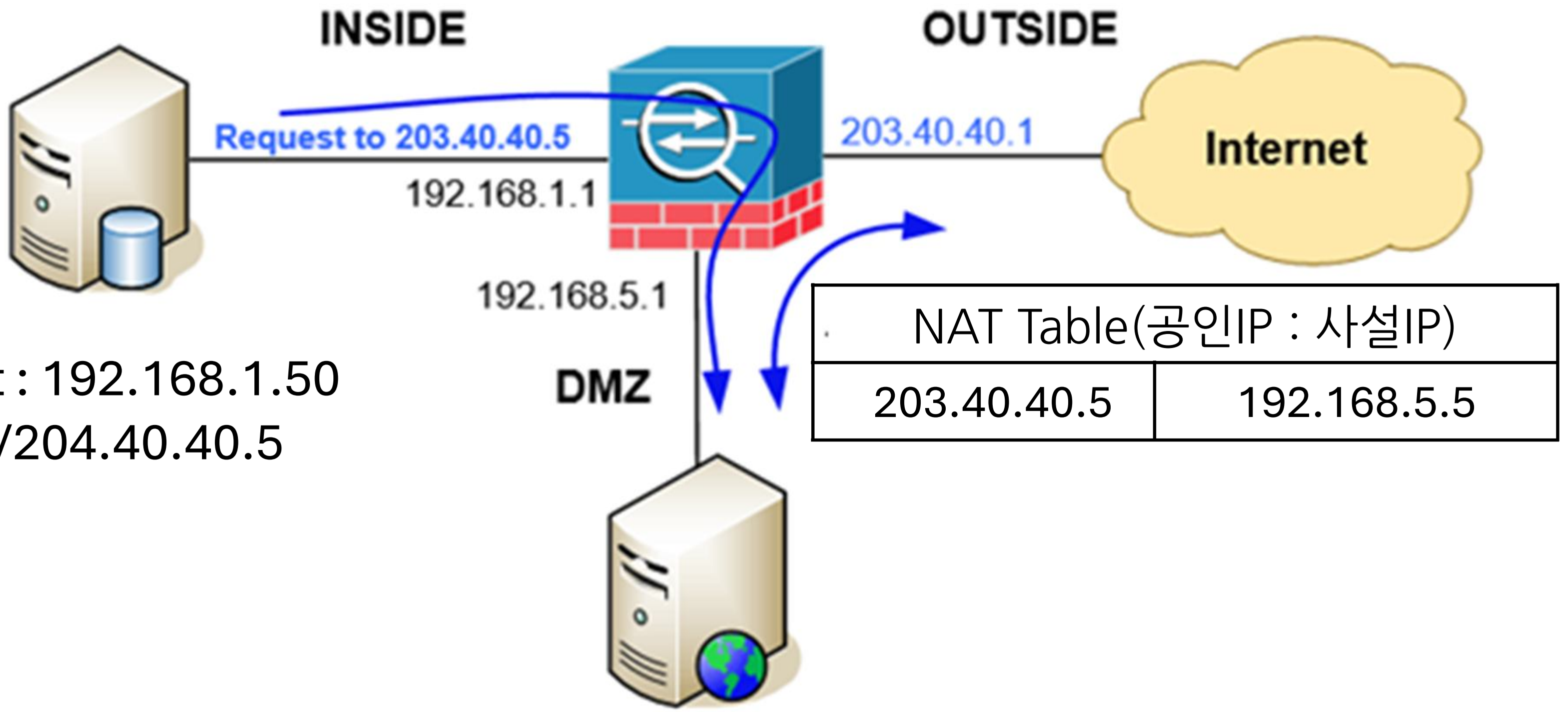
http://10.10.10.20

[DMZ Zone] ule 생성 전 테스트 (텔넷 서버 접속)

telnet 10.10.10.20



1단계. NAT Reflection 설정



Client : 192.168.1.50
http://204.40.40.5

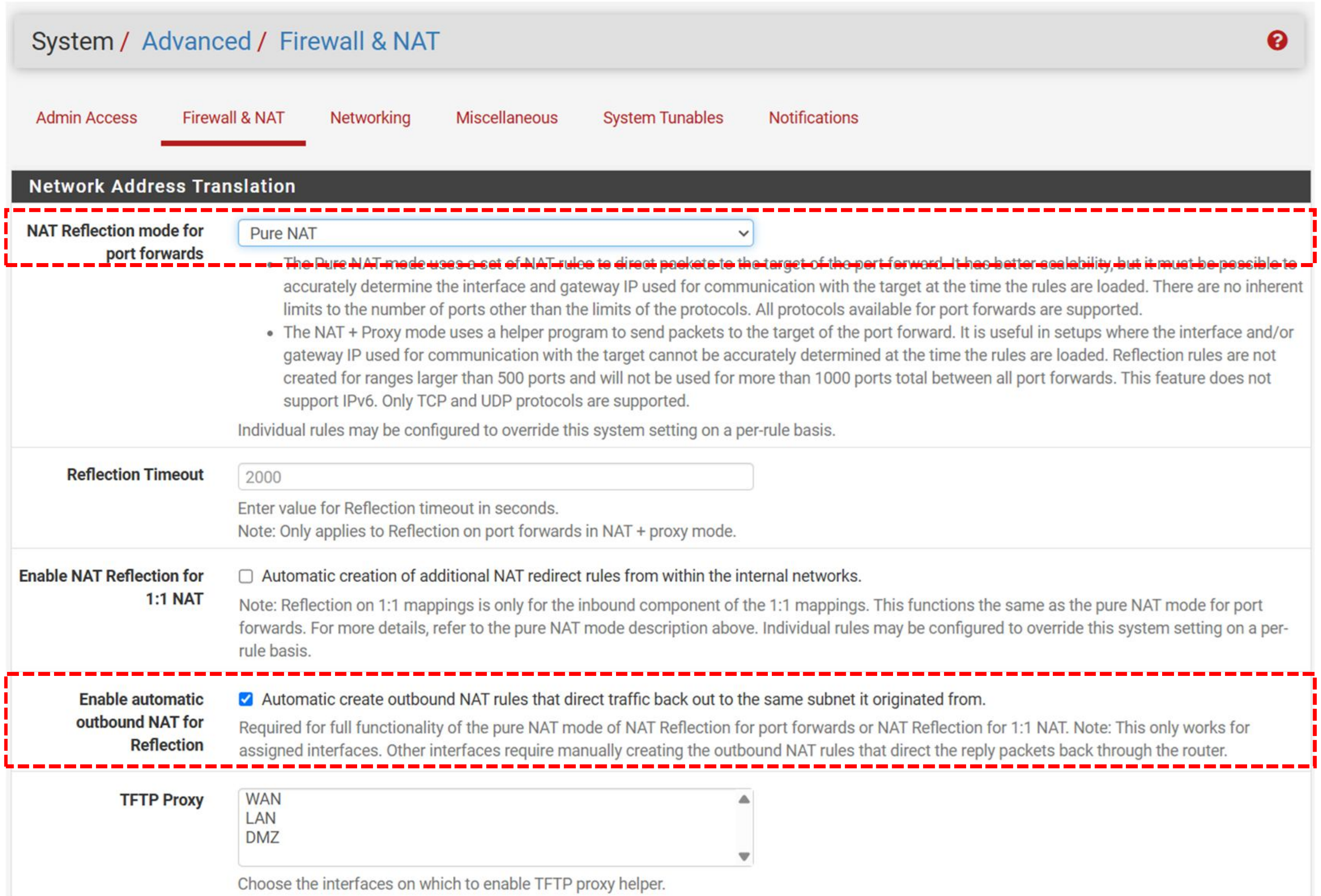
Server : 192.168.5.5(204.40.40.5)

《 NAT Reflection》

- 내부망 서버이지만 해당 서버의 IP가 공인IP인 경우, 인터넷 상의 라우터로 패킷을 보내서 내부망 서버로 접속하는 것이 일반적인 형태
- 이러한 과정을 걸치지 않고 NAT table에 목적지 서버가 내부망 서버로 확인이 되면 인터넷 망 게이트웨이를 경유하지 않고 바로 내부망 서버로 접속을 가능케 함

System > Advanced > Firewall & NAT > Network Address Translation >

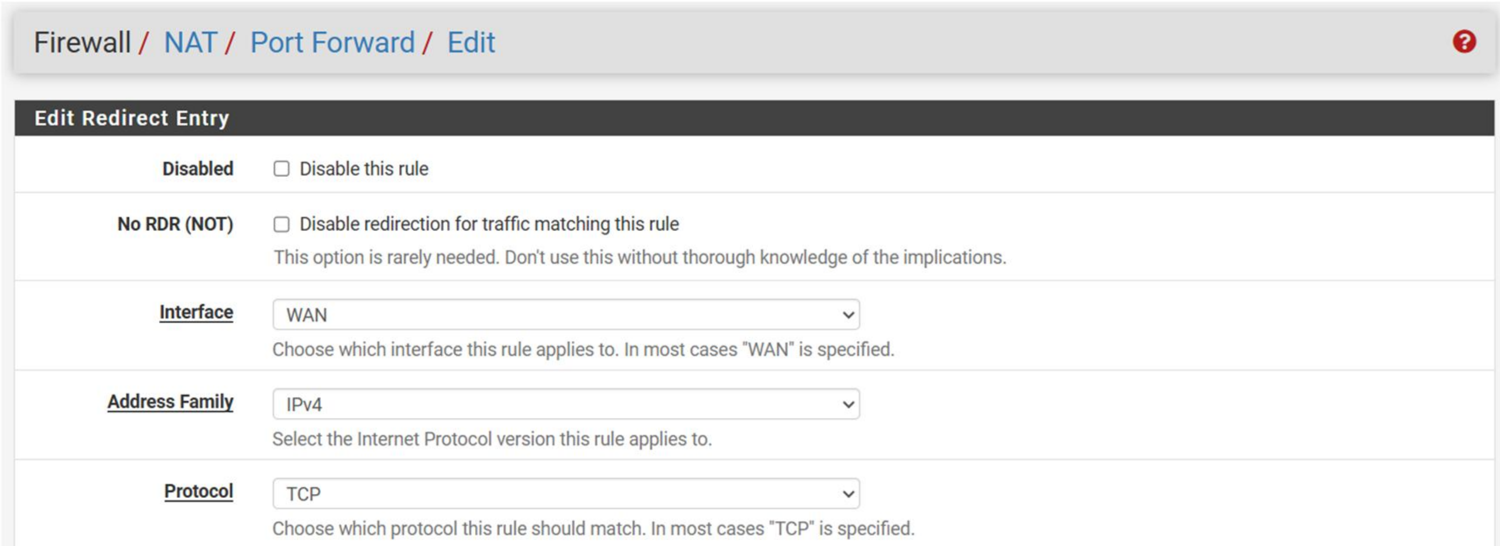
- NAT Reflection mode for port forward > Pure NAT로 변경
- Enable Automatic outbound NAT for Reflection 체크박스 활성화

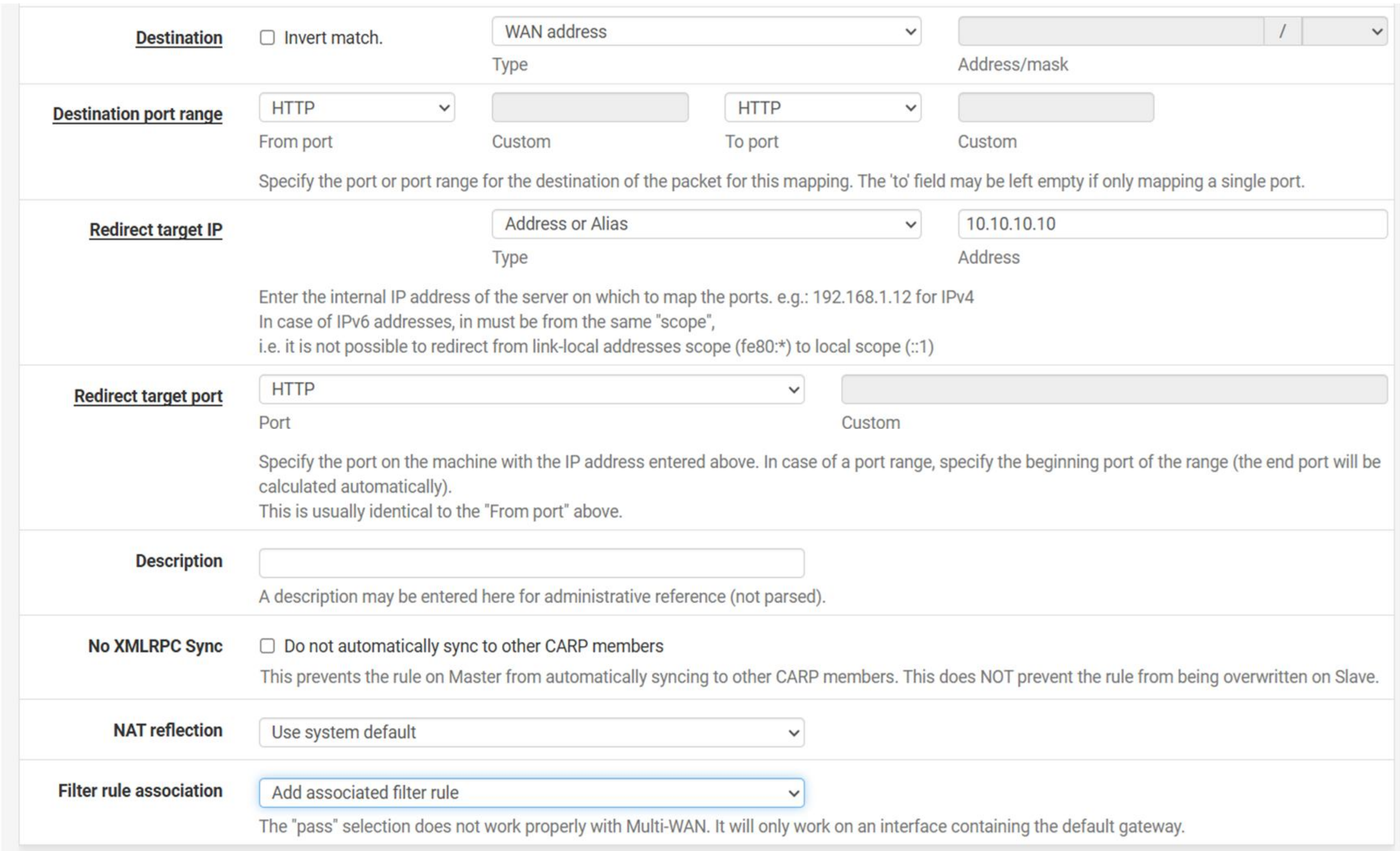


2단계. Port Forwarding 설정(1)

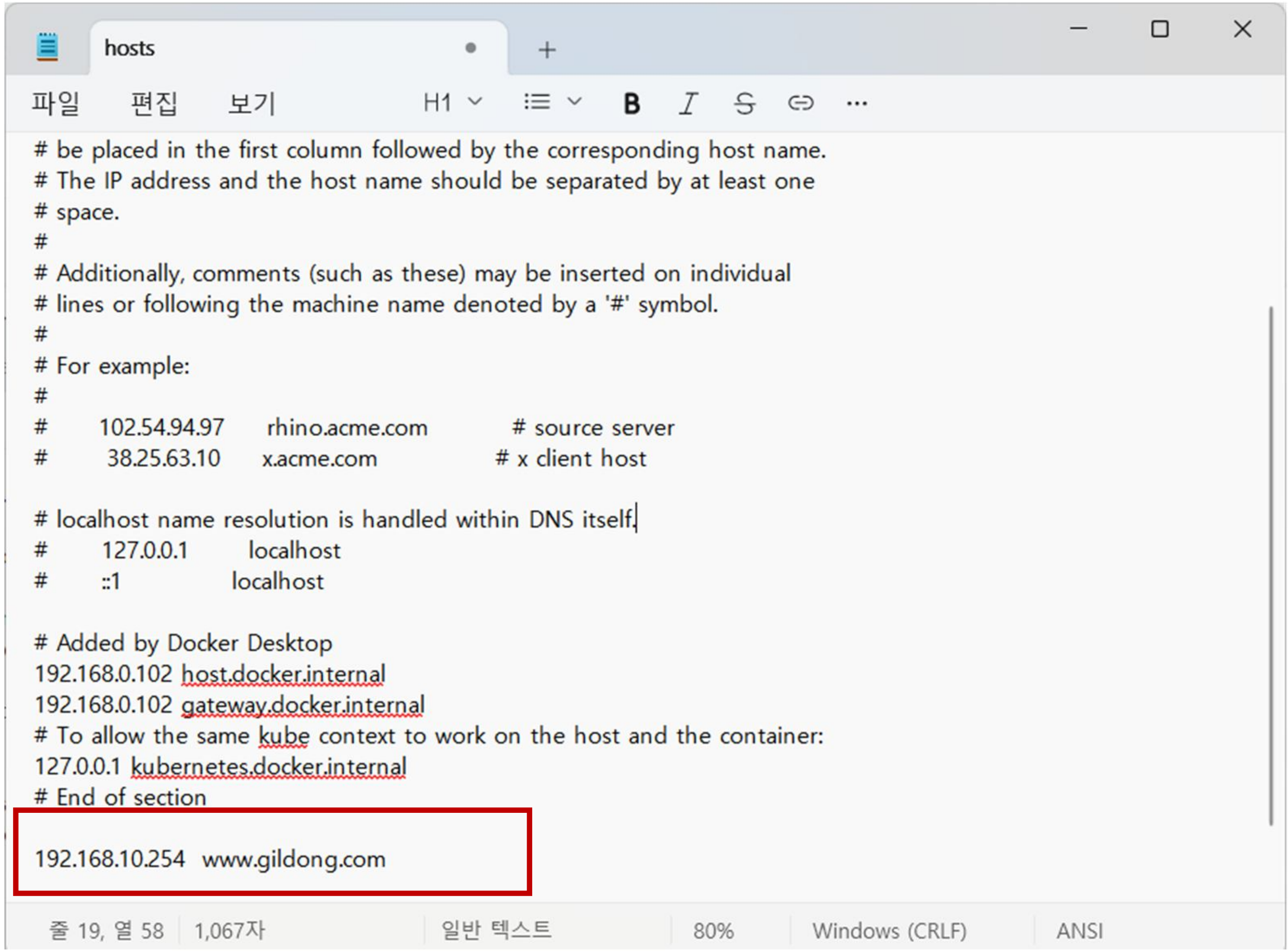
[Web Server(A) NAT 구성] Firewall > NAT > Port Forwarding

설정 필드	필드값
Interface	WAN
Address Family	IPv4
Interface	WAN
Destination	WAN address
Destination Port Range	HTTP
Redirect Target IP	Address or Alias 10.10.10.10
Redirect Target Port	HTTP
Filter rule association	Add associated filter rule


Firewall / NAT / Port Forward / Edit
Edit Redirect Entry
Disabled
Disable this rule
No RDR (NOT)
Disable redirection for traffic matching this rule
This option is rarely needed. Don't use this without thorough knowledge of the implications.
Interface
WAN
Choose which interface this rule applies to. In most cases "WAN" is specified.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which protocol this rule should match. In most cases "TCP" is specified.

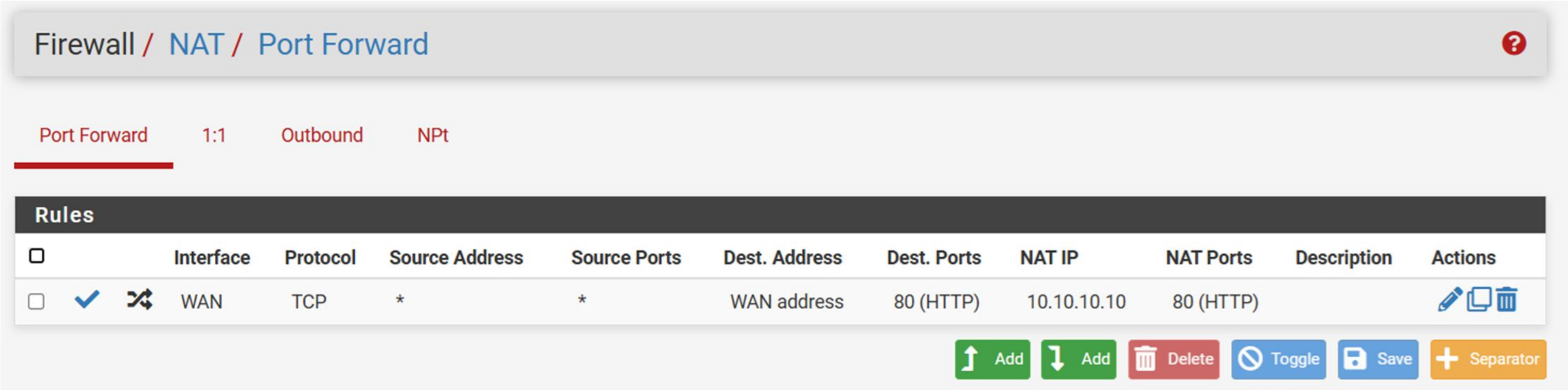

Destination
Invert match.
WAN address
Type
/
Address/mask
Destination port range
HTTP
From port
Custom
HTTP
To port
Custom
Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.
Redirect target IP
Address or Alias
Type
10.10.10.10
Address
Enter the internal IP address of the server on which to map the ports. e.g.: 192.168.1.12 for IPv4
In case of IPv6 addresses, in must be from the same "scope",
i.e. it is not possible to redirect from link-local addresses scope (fe80:*) to local scope (::1)
Redirect target port
HTTP
Port
Custom
Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).
This is usually identical to the "From port" above.
Description
A description may be entered here for administrative reference (not parsed).
No XMLRPC Sync
Do not automatically sync to other CARP members
This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.
NAT reflection
Use system default
Filter rule association
Add associated filter rule
The "pass" selection does not work properly with Multi-WAN. It will only work on an interface containing the default gateway.

외부접속자 PC 환경 변경

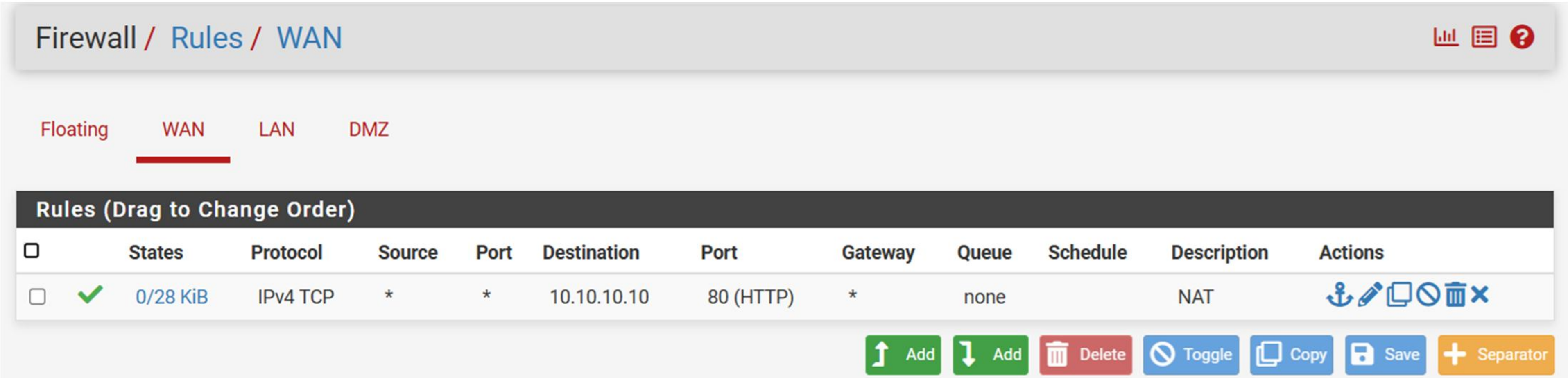


C:\Windows\System32\drivers\etc\host
*(관리자 권한으로 실행)

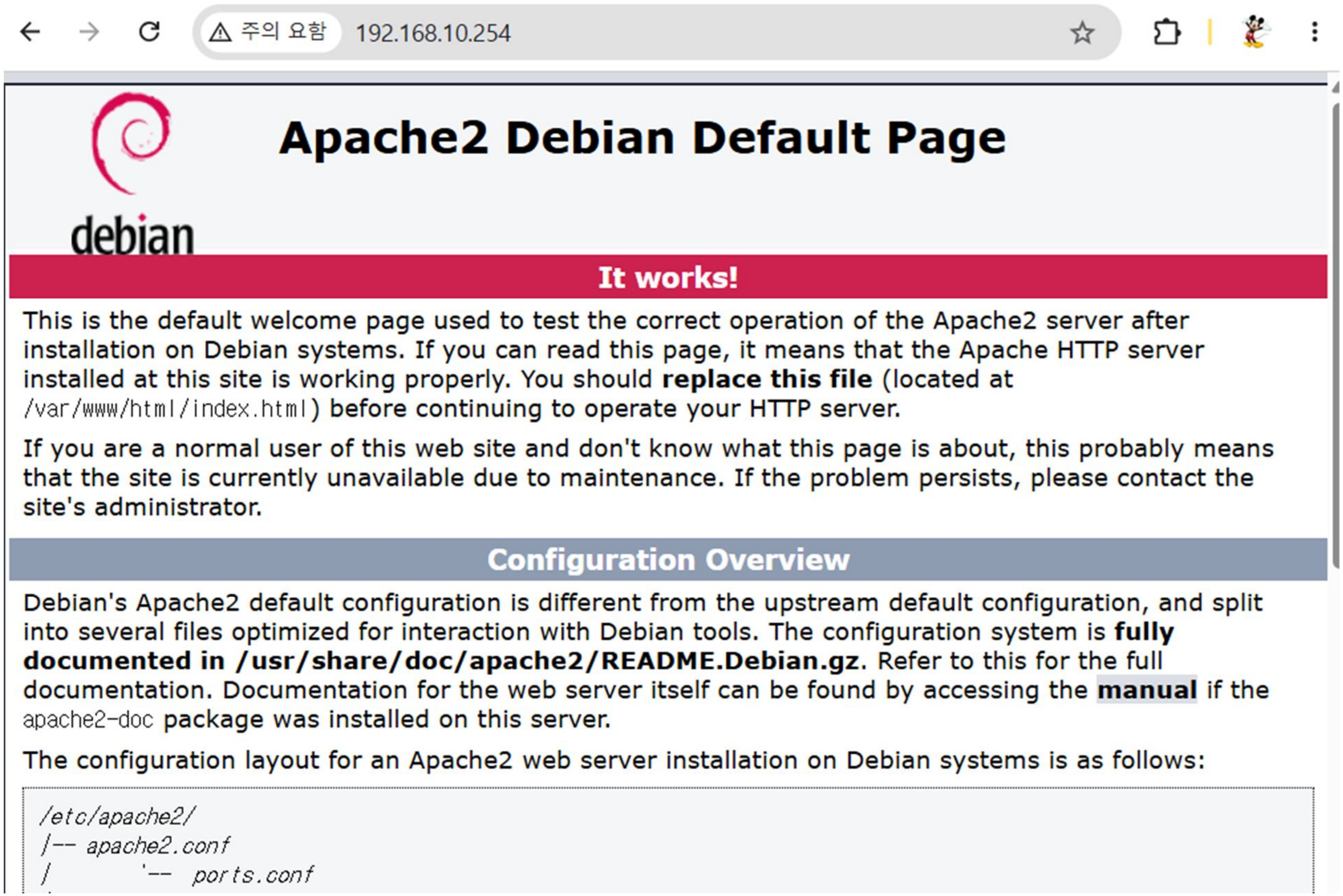
Port Forwarding 설정 후 WAN Rule 자동 등록



자동 등록됨



[크롬] http://www.gildong.com 또는 http://192.168.10.254



debian

Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

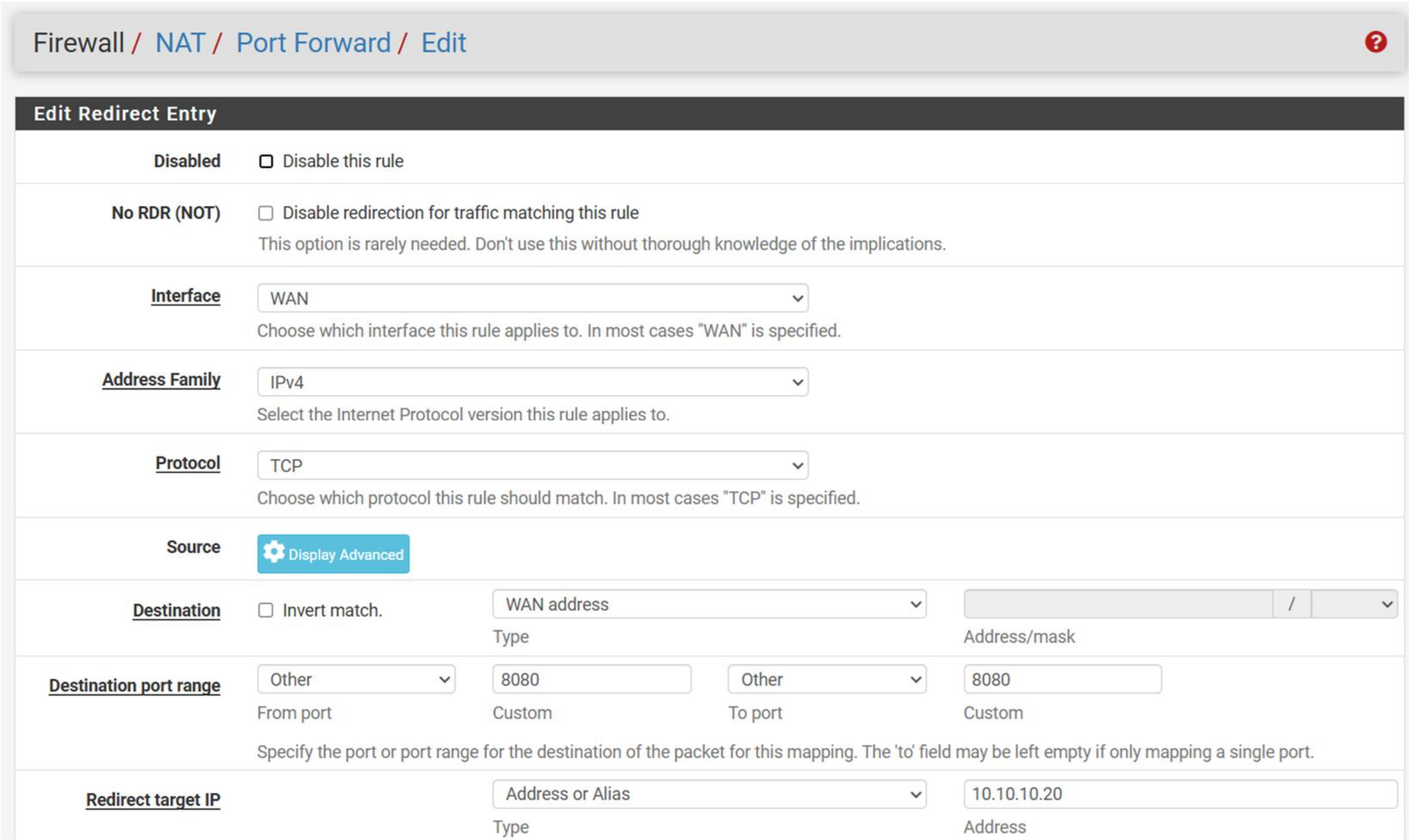
The configuration layout for an Apache2 web server installation on Debian systems is as follows:

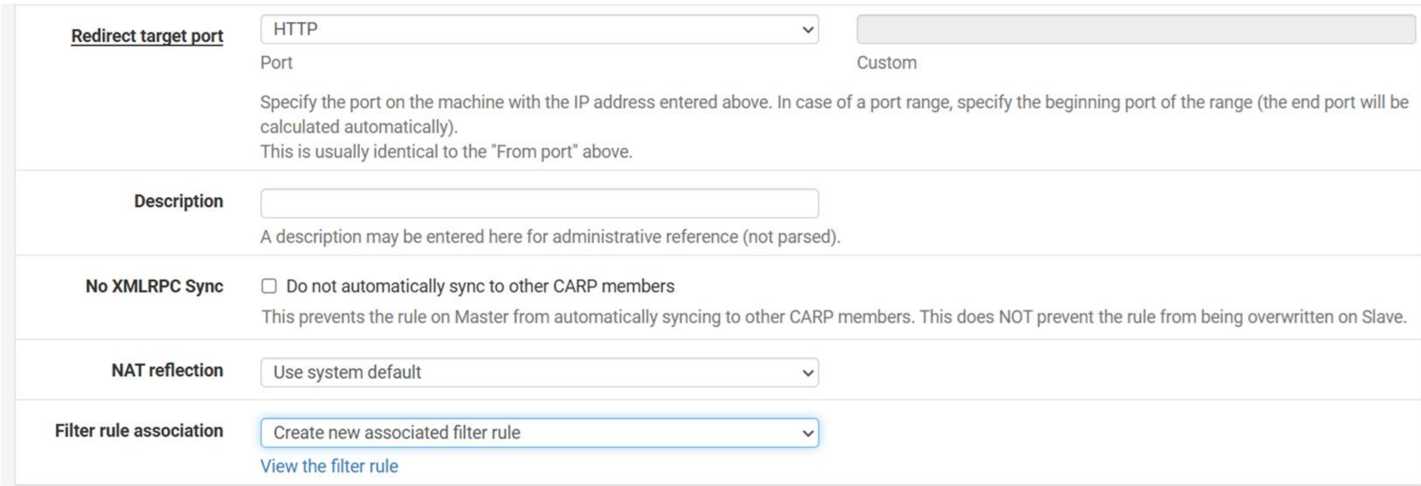
```
/etc/apache2/
|-- apache2.conf
|       `--  ports.conf
```

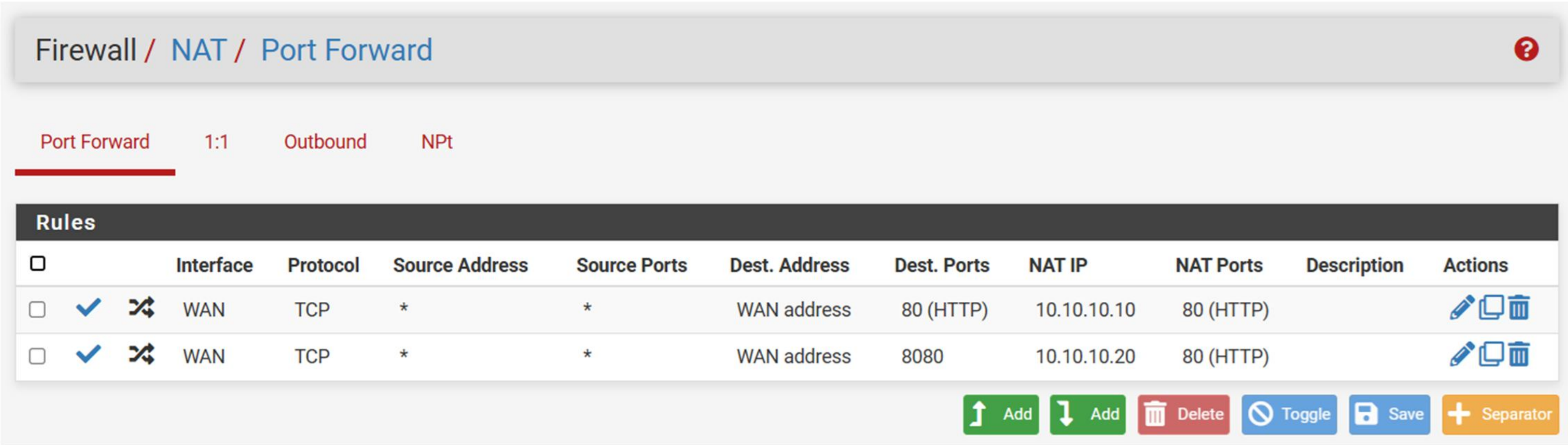
2단계. Port Forwarding 설정(2)

[Web Server(B) NAT 구성] Firewall > NAT > Port Forwarding

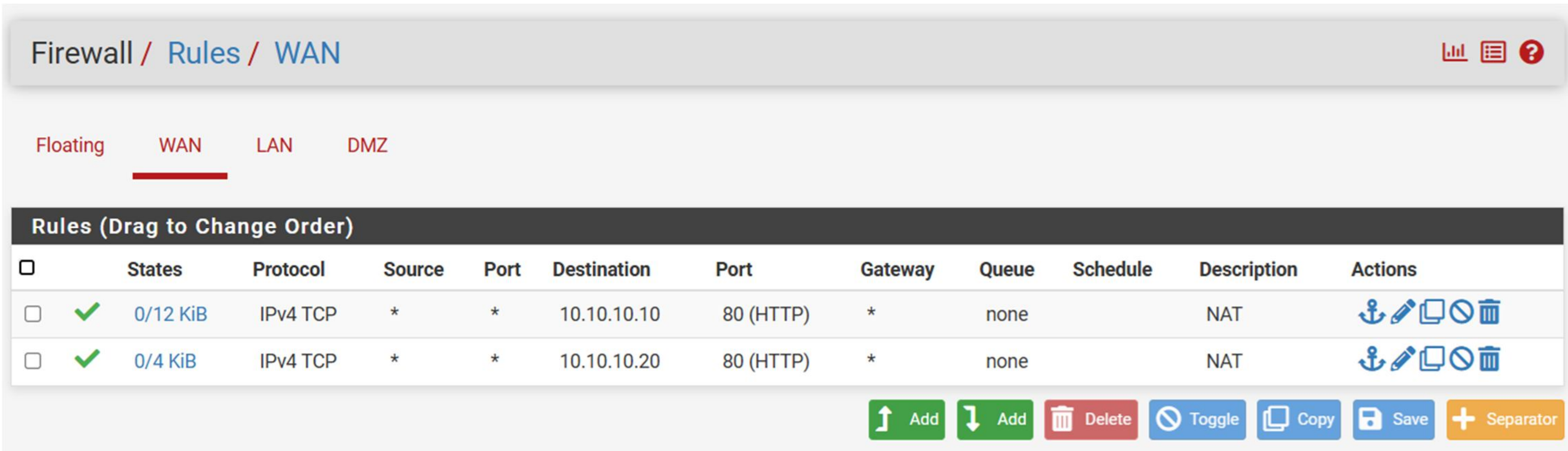
설정 필드	필드값
Interface	WAN
Address Family	IPv4
Protocol	TCP
Destination	WAN address
Destination Port Range	Other 8080
Redirect Target IP	Address or Alias 10.10.10.20
Redirect Target Port	HTTP
Filter rule association	Add associated filter rule


Firewall / NAT / Port Forward / Edit
Edit Redirect Entry
Disabled
Disable this rule
No RDR (NOT)
Disable redirection for traffic matching this rule
This option is rarely needed. Don't use this without thorough knowledge of the implications.
Interface
WAN
Choose which interface this rule applies to. In most cases "WAN" is specified.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which protocol this rule should match. In most cases "TCP" is specified.
Source
Display Advanced
Destination
Invert match.
WAN address
Type
Address/mask
Destination port range
Other
8080
Other
8080
From port
Custom
To port
Custom
Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.
Redirect target IP
Address or Alias
10.10.10.20
Type
Address

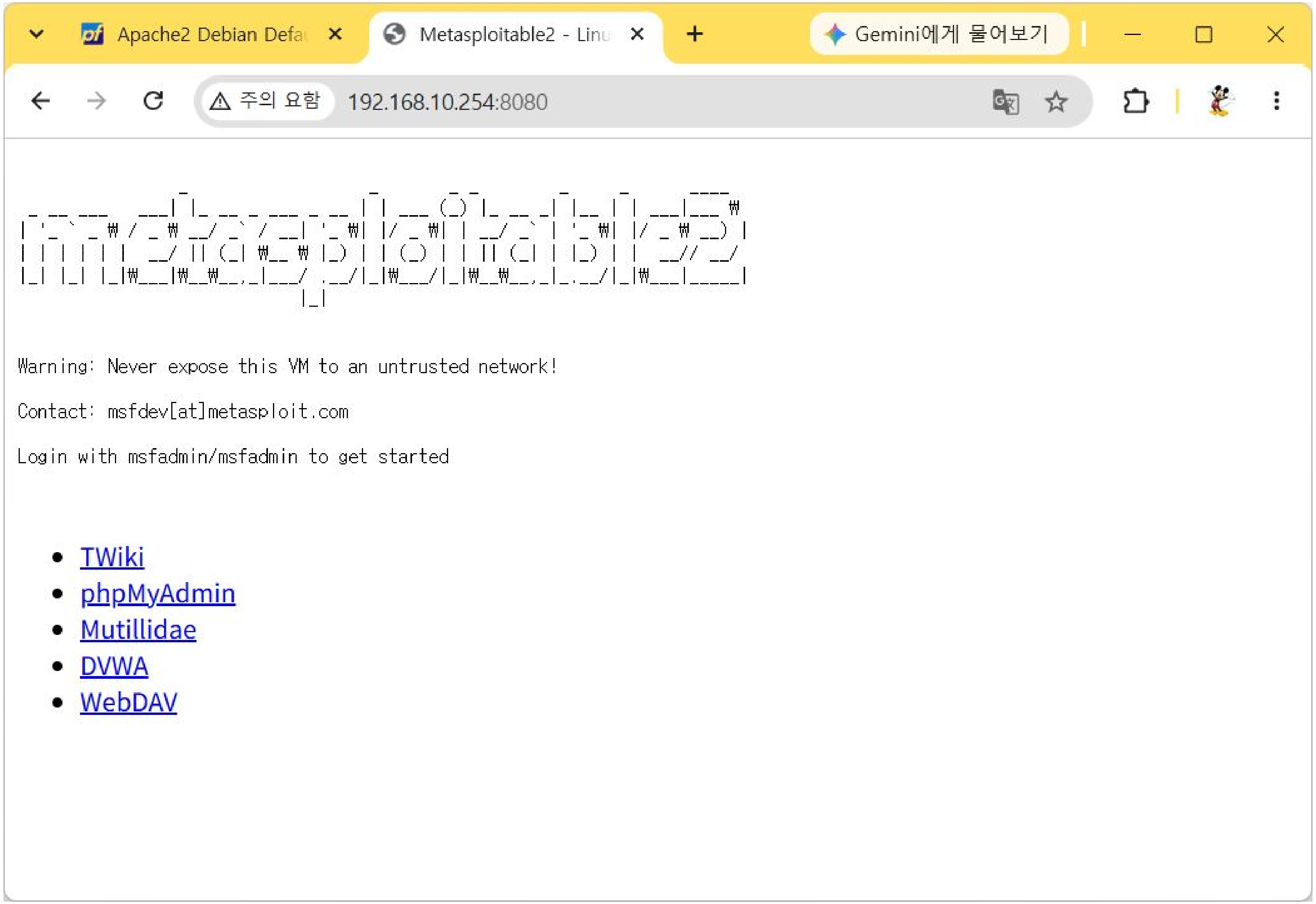

Redirect target port
HTTP
Port
Custom
Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).
This is usually identical to the "From port" above.
Description
A description may be entered here for administrative reference (not parsed).
No XMLRPC Sync
Do not automatically sync to other CARP members
This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.
NAT reflection
Use system default
Filter rule association
Create new associated filter rule
View the filter rule


Firewall / NAT / Port Forward
Port Forward
1:1
Outbound
NPt
Rules
Interface Protocol Source Address Source Ports Dest. Address Dest. Ports NAT IP NAT Ports Description Actions
WAN TCP * * WAN address 80 (HTTP) 10.10.10.10 80 (HTTP)
WAN TCP * * WAN address 8080 10.10.10.20 80 (HTTP)
Add
Add
Delete
Toggle
Save
Separator

자동 등록됨


Firewall / Rules / WAN
Floating
WAN
LAN
DMZ
Rules (Drag to Change Order)
States Protocol Source Port Destination Port Gateway Queue Schedule Description Actions
0/12 KiB IPv4 TCP * * 10.10.10.10 80 (HTTP) * none NAT
0/4 KiB IPv4 TCP * * 10.10.10.20 80 (HTTP) * none NAT
Add
Add
Delete
Toggle
Copy
Save
Separator

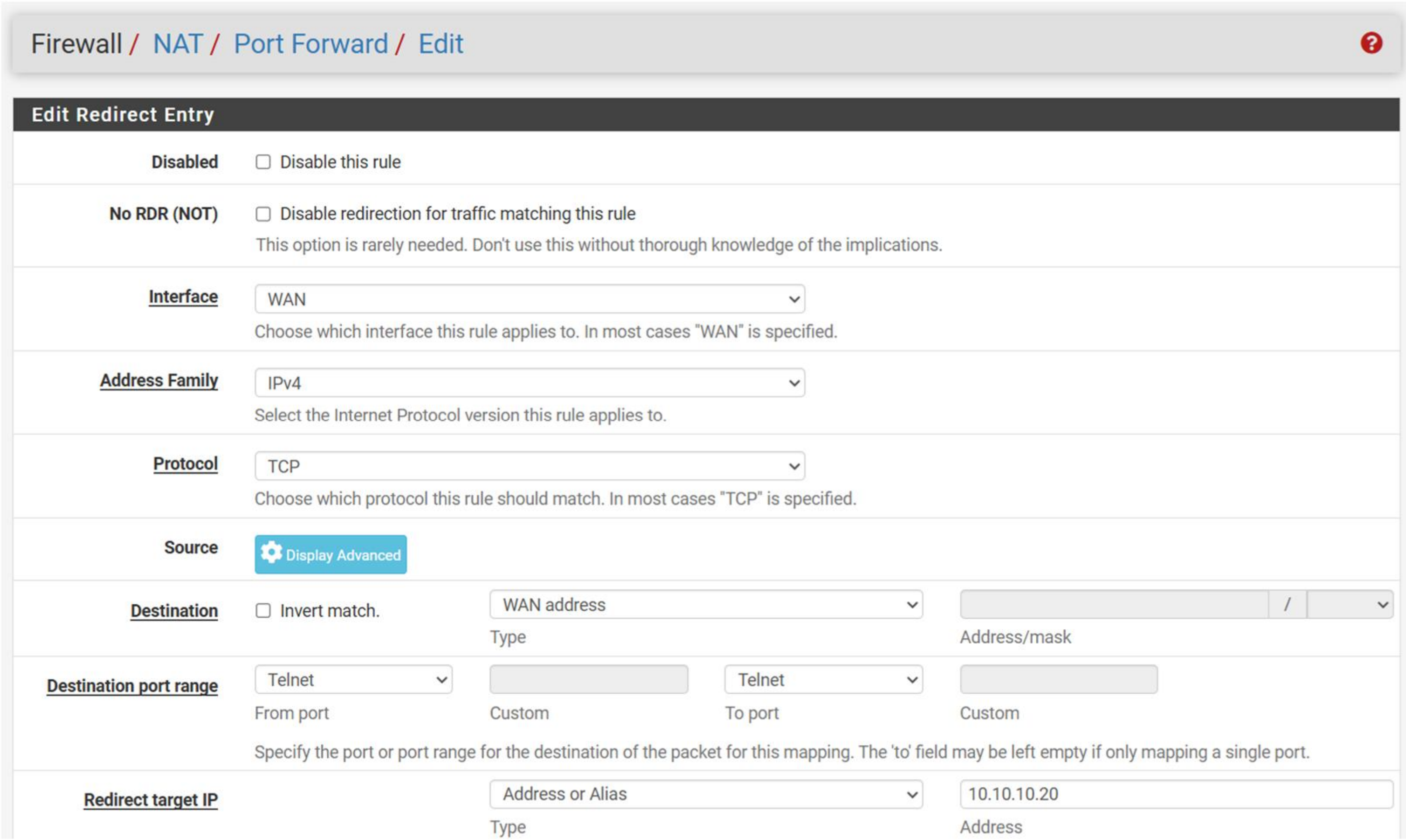
[크롬] http://www.gildong.com:8080 또는 http://192.168.10.254:8080

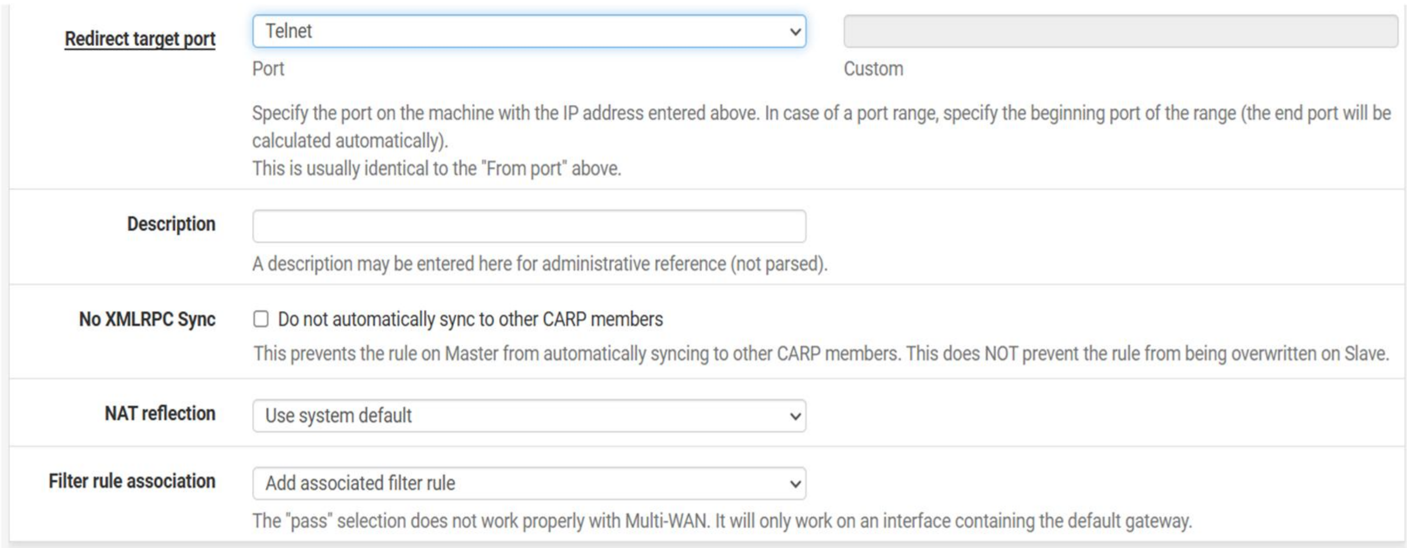


3단계. Port Forwarding 설정(3)

[Telnet Server NAT 구성] Firewall > NAT > Port Forwarding

설정 필드	필드값
Interface	WAN
Address Family	IPv4
Protocol	TCP
Destination	WAN address
Destination Port Range	Telnet
Redirect Target IP	Address or Alias 10.10.10.20
Redirect Target Port	Telnet
Filter rule association	Add associated filter rule

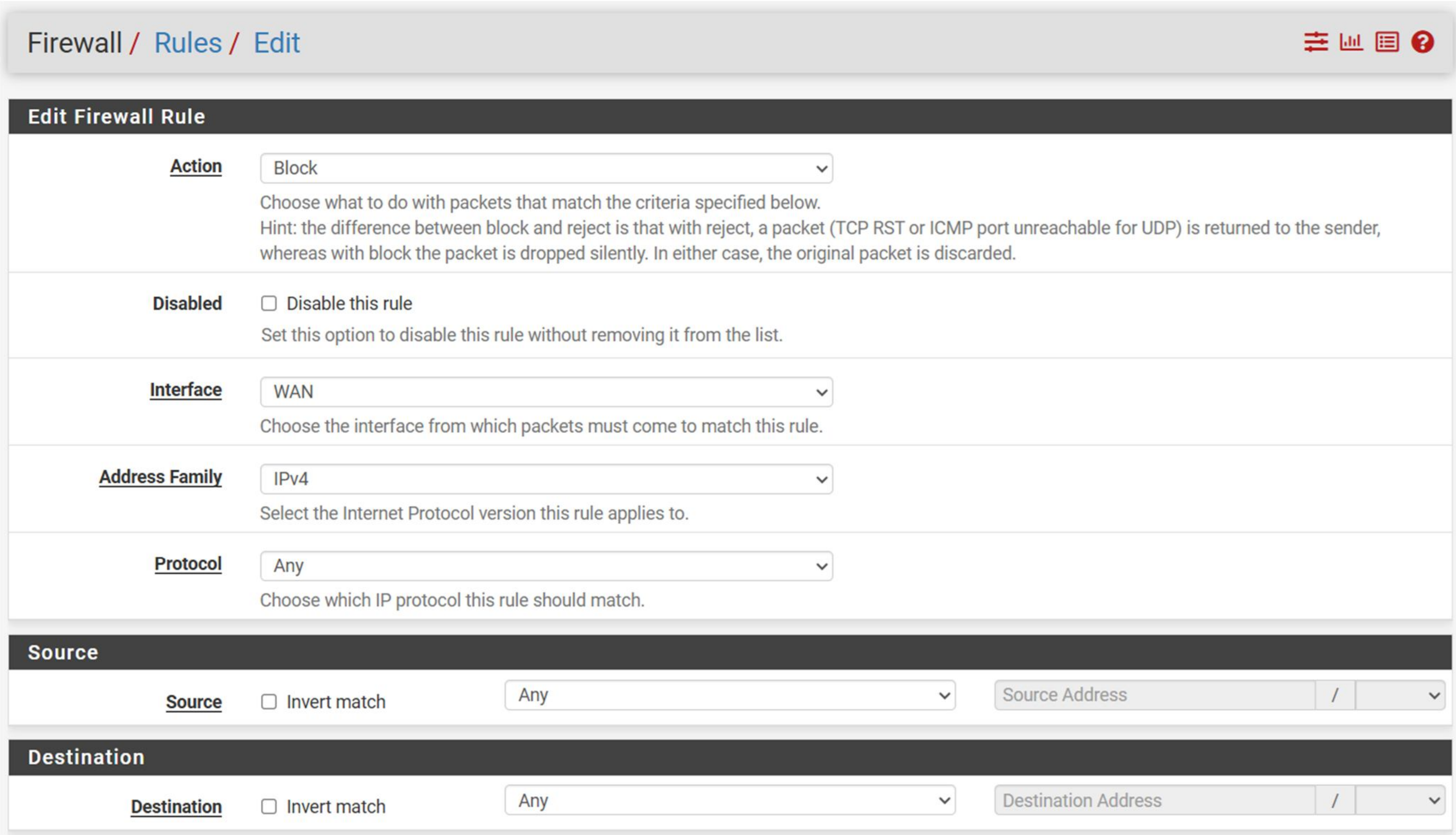

Firewall / NAT / Port Forward / Edit
Edit Redirect Entry
Disabled
Disable this rule
No RDR (NOT)
Disable redirection for traffic matching this rule
This option is rarely needed. Don't use this without thorough knowledge of the implications.
Interface
WAN
Choose which interface this rule applies to. In most cases "WAN" is specified.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which protocol this rule should match. In most cases "TCP" is specified.
Source
Display Advanced
Destination
Invert match.
WAN address
Type
Address/mask
Destination port range
Telnet
From port
Custom
Telnet
To port
Custom
Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.
Redirect target IP
Address or Alias
Type
10.10.10.20
Address


Redirect target port
Telnet
Port
Custom
Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).
This is usually identical to the "From port" above.
Description
A description may be entered here for administrative reference (not parsed).
No XMLRPC Sync
Do not automatically sync to other CARP members
This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.
NAT reflection
Use system default
Filter rule association
Add associated filter rule
The "pass" selection does not work properly with Multi-WAN. It will only work on an interface containing the default gateway.

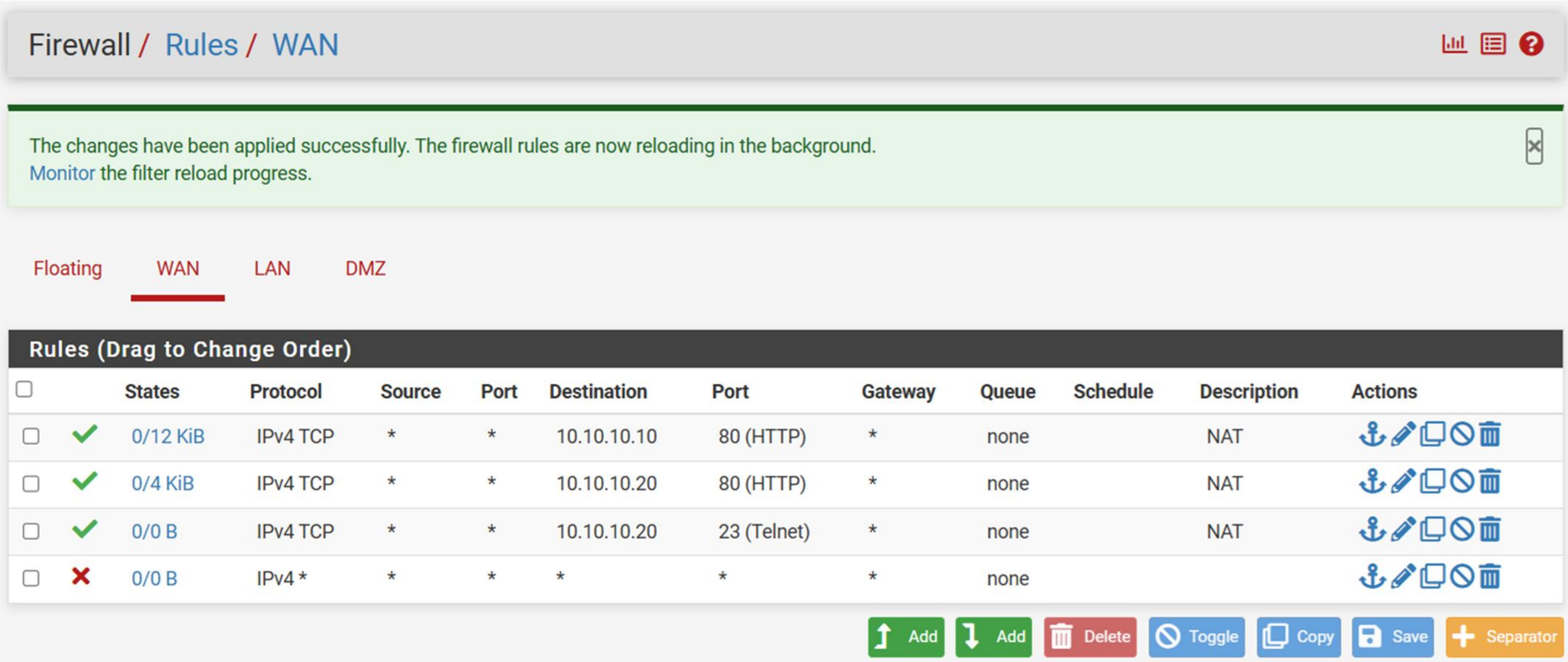
4단계. 그 외 모든 접근은 차단

- Firewall > Rules > WAN

설정 필드	필드값
Action	Block
Interface	WAN
Address Family	IPv4
Protocol	Any
Source	Any
Destination	Any


Firewall / Rules / Edit
Edit Firewall Rule
Action
Block
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
WAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
Any
Choose which IP protocol this rule should match.
Source
Source
Invert match
Any
Source Address
/
Destination
Destination
Invert match
Any
Destination Address
/

WAN Interface에 적용된 Rule Set


Firewall / Rules / WAN

The changes have been applied successfully. The firewall rules are now reloading in the background.
Monitor the filter reload progress.

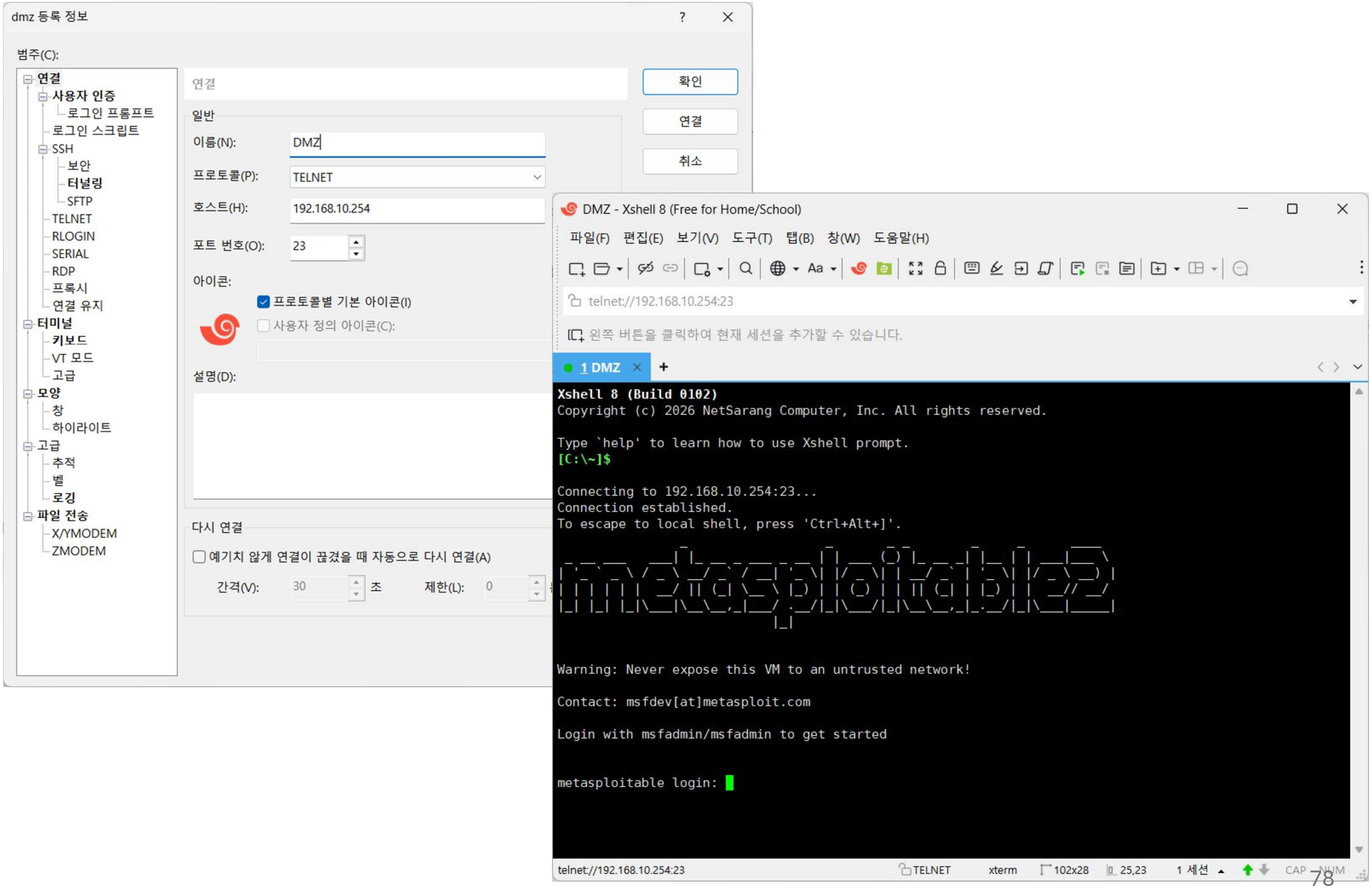
Floating WAN LAN DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✔	0/12 KiB	IPv4 TCP	*	*	10.10.10.10	80 (HTTP)	*	none		NAT	
✔	0/4 KiB	IPv4 TCP	*	*	10.10.10.20	80 (HTTP)	*	none		NAT	
✔	0/0 B	IPv4 TCP	*	*	10.10.10.20	23 (Telnet)	*	none		NAT	
✘	0/0 B	IPv4 *	*	*	*	*	*	none			

Add Add Delete Toggle Copy Save Separator

접근 테스트


dmz 등록 정보
범주(C):
연결
사용자 인증
로그인 프롬프트
로그인 스크립트
SSH
보안
터널링
SFTP
TELNET
RLOGIN
SERIAL
RDP
프록시
연결 유지
터미널
키보드
VT 모드
고급
모양
창
하이라이트
고급
추적
벨
로깅
파일 전송
X/YMODEM
ZMODEM
연결
일반
이름(N): DMZ
프로토콜(P): TELNET
호스트(H): 192.168.10.254
포트 번호(O): 23
아이콘:
프로토콜별 기본 아이콘(I)
사용자 정의 아이콘(C):
설명(D):
다시 연결
예기치 않게 연결이 끊겼을 때 자동으로 다시 연결(A)
간격(V): 30 초
제한(L): 0
확인
연결
취소
DMZ - Xshell 8 (Free for Home/School)
파일(F) 편집(E) 보기(V) 도구(T) 탭(B) 창(W) 도움말(H)
telnet://192.168.10.254:23
왼쪽 버튼을 클릭하여 현재 세션을 추가할 수 있습니다.
1 DMZ
Xshell 8 (Build 0102)
Copyright (c) 2026 NetSarang Computer, Inc. All rights reserved.
Type `help' to learn how to use Xshell prompt.
[C:\~]$
Connecting to 192.168.10.254:23...
Connection established.
To escape to local shell, press 'Ctrl+Alt+]'.
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login:
telnet://192.168.10.254:23
TELNET
xterm
102x28
25,23
1 세션
CAP NUM